

AIR POLLUTION CONTROL DIVISION
Field Inspection Report

COUNTY NUMBER: 013 **SOURCE NUMBER:** 0003 **DATE:** August 21, 2012
COMPANY: CEMEX, Inc. – Lyons Cement Plant **INSPECTOR:** Jason Long
SITE LOCATION: 5134 Ute Highway, Lyons **COUNTY:** Boulder
MAILING ADDRESS: P.O. Box 529, Lyons, CO, 80540
CONTACT PERSON: Denise Arthur, Environmental Manager **TIME:** 9:00am
TELEPHONE No: (303) 823-2115 **PERMIT No:** 95OPBO082
REPORT SUBMITTED: September 7, 2012

SOURCE CLASS: Major ☒ Syn Minor ☐ Minor ☐
INSPECTION TYPE: Full Compliance Evaluation ☒ Partial Compliance Evaluation ☐ Investigation ☐
HOURS: INSPECTION: 7.0 TRAVEL & PREP: 10.0 REPORT: 18.0 TOTAL: 35.0
COMPLIANCE ASSISTANCE [N] *Y/N **POLLUTION PREVENTION ASSISTANCE** [N] *Y/N

COMPLIANCE STATUS: ☒ In Compliance ☐ Out of Compliance

INSPECTION SUMMARY

The Air Pollution Control Division's (Division) Field Inspector Jason Long conducted multiple site visits to the Cemex Lyons Plant (CEMEX) which comprise the inspection for this source. CEMEX operates a portland cement manufacturing facility in Lyons Colorado, under Operating Permit No. 95OPBO082, Airs ID 013-0003. While onsite, Mr. Long was accompanied by Denise Arthur, Environmental Manager, and Melissa Mele, Environmental Engineer, who provided access to the facility and records to determine compliance.

Based upon observations made during the inspection and records provided by the source, CEMEX is in compliance with the terms and conditions of Operating Permit 95OPBO082.

COMPLIANCE ASSISTANCE

No compliance assistance was given to the source.

NSPS/MACT APPLICABILITY

NSPS Applicability:

Subpart OOO - Nonmetallic Mineral Processing;

Subpart F - Portland Cement Plants {On and after the MACT compliance date of June 10, 2002, any affected source subject to the provisions of 40 CFR Part 63, Subpart LLL is exempted from any otherwise applicable new source performance standard contained in 40 CFR Part 60, Subpart F. (63.1356)};

Subpart A - General Provisions.

MACT Applicability:

Subpart LLL - Portland Cement Manufacturing Industry

MALFUNCTION REPORT REVIEW

The source has submitted malfunction reports to the Division. The reports have been reviewed and will be discussed in the body of the report.

COMPLIANCE HISTORY

1. A 2000 stack test identified the source failed to operate the raw material dryer in compliance with the PM limit in the Operating Permit. Resolved with the issuance of a Compliance Advisory and an Order on Consent dated 10/02/00, civil penalty of \$1,400.00 and a SEP of \$8,400.00 to install video camera for control room staff to monitor plant conditions.

2. Violations identified during 2001 inspection include: failure to properly calculate and report emissions from a gasoline storage tank, failure to conduct Method 9 opacity observation as required, Failure to notify the Division of excessive opacity from an upset as required, and failure to submit annual compliance certifications which include statements of violations of conditions 1.3, 1.4, 3.2, 3.5, 3.5.7, 11.3, 11.6, 13.2, and 13.5. Resolved with a Compliance Order on Consent dated 07/25/02, SEP \$6,000.00.

3. Several violations were identified during 2003 inspections which were resolved with a Compliance Order on Consent dated 02/20/04, a civil penalty of \$37,460.00, pay at least \$149,840.00 towards a Supplemental Environmental Project, and pay \$94,839.00 for the economic benefit associated with not operating the control equipment required to control emissions from the A-Frame building.

4. Seven violations were identified during 2004 inspections and were resolved with a commitment to immediately implement procedures to control fugitive emissions, and install a new conditioning spray tower before the fall of 2005.

5. Numerous violations were identified during several 2005 and 2006 inspections and were addressed in Notice of Violations dated 9/23/05 and 6/8/06. Additional issues in 2006 were addressed in letters issued for Denial of Upset Reports dated 5/1/06 and 7/31/06 and a request for revision of Startup, Shutdown, and Malfunction Plan. All of these violations were resolved with a Compliance Order on Consent signed 12/22/06 and a penalty of \$1,500,000.00 to be paid as follows: \$300,000.00 cash penalty, \$50,000.00 non-compliance cash penalty, \$200,000.00 to operate a PM monitor for at least one year, \$450,000.00 donated to nonprofit organization for environmental projects, at least \$500,000.00 towards an Supplemental Environmental Project, and a moratorium on using tire-derived-fuels until at least January 1, 2008.

6. Violations identified in 2007 and 2008 inspections have been addressed in a Notice of Violation dated June 17, 2008. The case was resolved with a Compliance Order on Consent signed 2/11/09 with a total penalty assessed of \$528,325. Cemex agreed to pay a sum of \$105,665 in administrative penalties and perform Supplemental Environmental Projects (SEPs) totaling \$422,660. The SEPs included \$300,000 donated to the Boulder County Plug-In Hybrid and Vehicle to Grid Implementation Project and \$122,660 donated to the Governor's Energy Office (GEO) earmarked for energy efficiency and renewable energy projects for Lyons public schools.

7. A Warning Letter was issued for the 2011 inspection related to a stack test conducted on April 21, 2011 where Cemex exceeded the grain loading limit for Baghouse 625-14 (P009) found in the table in Condition 11.3. The test results indicated an emission rate of 0.038 gr/dscf in excess of the 0.03 gr/dscf limit found in the table. The baghouse was retested on May 25, 2011 indicating 0.01 gr/dscf emission rate and demonstrated compliance with the grain loading requirements.

INTRODUCTION

Cemex is permitted to emit air pollution into the atmosphere in accordance with Operating Permit 95OPBO082, which was first issued February 1, 2000 and renewed March 1, 2008. The facility is located near Lyons, 12 miles north of Boulder on Highway 66. There are no affected states within 50 miles of the plant. Rocky Mountain National Park, Rawah Wilderness Area and Eagle's Nest Wilderness Area are Federal Class I designated areas within 100 kilometers of the plant.

This facility manufactures portland cement. Limestone, shale, and other raw materials extracted from the quarry are processed through a primary crusher at the Dowe Flats quarry. The crushed material is transported to the plant on a 2.0 mile belt conveyor system and discharged to a stockpile. The stockpiled material is placed on a belt by means of a front end loader to be processed through a primary crusher, the dryer, and a secondary crusher. The material from the secondary crusher is stored in raw material storage silos. These storage silos contain silica and iron ore and various quarried raw materials. Material from these storage silos is discharged to weigh belts for the formulation of a desired product. The weigh belts discharge to the raw mill. The raw mill mixes and crushes the blended materials and delivers the homogenized material to storage silos. The homogenized material from the storage silos is delivered to the calciner portion of the kiln. Pulverized coal from the coal mill is fired at the bottom of the flash calciner. To improve the combustion characteristics, oxygen from an on-site oxygen generation plant, is injected through a swirl chamber directly below the first stage calciner (the oxygen plant is no longer in use and has been removed). The calcined material from the calciner then enters the rotary kiln, which is located at a slight incline along its horizontal axis. The material travels towards the clinker discharge end where additional pulverized coal is fired for the clinkering process. The clinker is discharged from the kiln onto the clinker cooler. The clinker is cooled by large amounts of air that is forced upwards through the clinker bed by undergrate fans. A large percentage of the cooling air is recovered for use as primary air in the kiln combustion process.

The cooled clinker is then moved to internal storage in an A-Frame building, or outside storage stockpiles. The stored clinker is the raw material for the finish mill. In the finish mill the clinker is combined with gypsum and ground to a fine material, passed through coolers and stored in the product silos. The material in the product silos can be loaded for bulk transport, or sent to a packaging system. From an over-all perspective, the manufacturing process may be viewed as two segments -- clinker production and cement production. The clinker storage allows the two processes to operate at different production rates. During periods of low demand for cement, clinker is accumulated. If cement is in high demand, the clinker production can be supplemented by purchase of clinker from other sources. The overall result is the clinker production can operate at a rather steady rate, while the cement production can operate in response to the current or projected demands.

This inspection also included a site visit to the sandstone quarry (previously AIRS 013/0124/001-004). The quarry has been added as point 013/0003/052 to the Cemex Lyons plant but has not been incorporated into the Operating Permit or new Construction Permit issued. The quarry was not operating at the time of inspection. Permit 02BO0176F demonstrates compliance with the fugitive particulate limits by limiting production of siliceous rock to 60,000 tons per year. Cemex loaded 33,813 tons of sandstone into trucks for the 12 month period ending June 2012. All sandstone mined at the quarry is hauled to Dowe Flats where it is crushed in the primary crusher before being sent to the plant.

2012 UPDATE

Since the last inspection, the Cemex facility has reduced the extended periods of planned outages due to market driven conditions. Since the last inspection, significant outages where the kiln was shut down have only occurred September 26 – October 17, 2011 and April 2 – April 11, 2012. During the inspection, the plant and kiln were operating normally on natural gas. According to plant personnel, due to the low cost of natural gas fuel, natural gas has been burned exclusively in the kiln since May 2012. A combination of natural gas and coal are being used in the precalciner tower. Also due to recent economic conditions, Cemex produces 3 different types of clinker comprising of 7 different types of cement to accommodate different markets.

Cemex has appointed Mr. Shane Wilson as the Plant Manager replacing Steve Goodrich at the facility. The plant has also hired a new Continuous Improvement Manager as part of a new management system being put into place. The management system follows the “Six Sigma” methodology, and Cemex has partnered with Caterpillar to implement the new management system that mirrors Caterpillar Production Systems.

Cemex submitted an application and APEN for a minor modification to change a pugmill (P007A) on March 7, 2012. The construction of the new pugmill was completed in May 2012 for loading CKD into trucks for disposal at C-Pit. In addition, a new 10,000 gallon calcium chloride tank has been added for use in the clinker process.

SECTION I – General Activities and Summary

Permitted Activities Summary

Until such time as this permit expires or is modified or revoked, the permittee is allowed to discharge air pollutants from this facility in accordance with the requirements, limitations, and conditions of this permit.

This Operating Permit incorporates the applicable requirements contained in the underlying construction permits, and does not affect those applicable requirements, except as modified during review of the application or as modified subsequent to permit issuance using the modification procedures found in Regulation No. 3, Part C. These Part C procedures meet the applicable substantive New Source Review requirements of Part B. Any revisions made using the provisions of Regulation No. 3, Part C, shall become new applicable requirements for purposes of this operating permit and shall survive reissuance. Any requirements that were designated in the compliance order on consent signed February 19, 2004 (No. 2002-124) as applicable requirements have been incorporated into this operating permit and shall survive reissuance as applicable requirements. This permit incorporates the applicable requirements (except as noted in Section II) from the following Construction Permit(s): P-10,225, P-10,535, 12BO444(1-2), P-10,298, P-10,284, P-10,266, P-10,292, 98BO0259, 10BO718, 93BO1414F, 94BO593, 98BO0292, 98BO0315 and 05BO0703.

All conditions in this permit are enforceable by the US Environmental Protection Agency, Colorado Air Pollution Control Division (hereinafter Division) and its agents, and citizens unless otherwise specified. State-only enforceable conditions are: Section II - Conditions 5.6 (Lead) and 10.11 (Lead), and Section IV - Conditions 3.d, 3.g (last paragraph), 14 (Odor) and 18 (Ozone Depleting Compounds). All information gathered pursuant to the requirements of this permit is subject to

the Recordkeeping and Reporting requirements listed under Condition 22 of the General Conditions in Section IV of this permit.

Prevention of Significant Deterioration/New Source Review (PSD/NSR)

This facility is categorized as a NANSR major stationary source (Potential to Emit of VOC and NO_x ≥ 100 tons/year). Future modifications at this facility resulting in a significant net emissions increase (see Regulation No. 3, Part D, Sections II.A.26 and 42) for VOC or NO_x or a modification which is major by itself (Potential to Emit ≥ 100 tons/year or either VOC or NO_x) may result in the application of the NANSR review requirements.

This source is categorized as a PSD major stationary source (Potential to Emit ≥ 100 tons/year) for PM, PM₁₀, SO₂, NO_x and CO. Future modifications at this facility resulting in a significant net emissions increase (see Regulation No. 3, Part D, Sections II.A.26 and 42) or a modification that is major by itself (Potential to Emit ≥ 100 tons/yr) for any pollutant listed in Regulation No. 3, Part D, Section II.A.42 for which the area is in attainment or attainment/maintenance may result in the application of the PSD review requirements.

There are no other Operating Permits associated with this facility for purposes of determining applicability of NANSR and PSD review regulations.

Accidental Release Program (112(r)):

The plant is not subject to the provisions of the Accidental Release Prevention Plan of Section 112 (r)(7) of the Clean Air Act.

Alternate Operating Scenarios:

No alternate operating scenarios have been specified for this facility.

Compliance Assurance Monitoring (CAM):

The following emission points at this facility use a control device to achieve compliance with an emission limitation or standard to which they are subject and have pre-control emissions that exceed or are equivalent to the major source threshold. They are therefore subject to the provisions of the CAM program as set forth in 40 CFR Part 64, as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV: **S005** – Raw Materials Drying; **S016** – Precalciner Kiln; **S018** – Clinker Cooler; **S032** – Bottom of A Frame Transfer; **S024** - #2 Clinker Silo; **S024 A** – Top of A Frame Transfer; **S031** – #6 Reclaim Feeder; **S036** – Finish Mill; **S037** – Finish Mill Auxillary Dust Collector; **S065** – Separator; **S043** – Cement Silos A10 and A13; **S044** – Cement Silo A7; **S046** – East Load Spout; **S047** – West Load Spout; **S049** – Packhouse; **S050** – Packhouse; **S001** – Waste Dust Silo. CAM requirements are set forth in Section II, Condition 24 of this permit.

COMPLIANCE WITH PERMIT CONDITIONS

Cemex is subject to the following conditions of Operating Permit #95OPBO082.

Inspection Report Guidance:

This inspection report follows the template of the Title V operating permit renewal issued March 1, 2008. **Inspector's comments are noted in bolded font.** Due to the large volume of records provided, recordkeeping to determine compliance is kept in the binder provided by Cemex labeled for the 2012 inspection.

Summary of Emission Units

The emissions units regulated by this permit are the following:

Process	Plant Identifier	AIRS Stack Number	Description	Size	Pollution Control Device	Construction Permit
Dowe Flats Quarry	P017	016	Drilling	3,200,000 tons/year	PM Emission Control Plan	93BO1414F
		017	Blasting			
		018	Truck Loading/Unloading			
		019	Haul Roads			
		120	Scraper Activities			
		021	Grading	800 tons/hour	Baghouse (8 each)	94BO593
		022	Bulldozing			
		023	Wind erosion of exposed areas			
		024	Wind erosion of stockpiles			
		026	Conveyor	800 tons/hour	Baghouse (8 each)	94BO593
		027	Primary Crusher (Quarry)	800 tons/hour	Baghouse	
Lyons Quarry	P018	028	Blasting, Grading, etc.	760,000 tons/year	PM Emission Control Plan	93BO1414F
Raw Material Storage and Handling at Plant Site	P004	024B	Discharge of Primary-Crushed Material onto Open Stockpile Front End Loader Activity	1,050,000 tons/year	PM Emission Control Plan	98BO0292
Primary Crusher	P001	001	S002 - Primary Crushing	300 tons/hour lime and shale 3 tons/hour iron oxide 15 tons/hour silica	Bagfilters	P-10,225 P-10,535
		027	S004 - Surge Silo			
Raw Materials Drying	P002	002	S005 - Raw Materials Drying	200 TPH Limestone and shale 3 TPH Iron Ore 15 TPH Silica	Baghouse	12BO444-1
Secondary Crushing	P003	003	Secondary Crushing and Screening	200 TPH Limestone and shale 3 TPH Iron Ore 15 TPH Silica	Baghouse (3 each)	Grandfather
			S003 - #4 Belt Transfer			P-10,298
Raw Material Silos	P004	004	S006 to S008 - Raw Material Storage Silos	200 TPH	Baghouse (3 each)	P-10,284
Raw Material Grinding	P005	005 031	S012 - Raw Mill Feeders	180 TPH Limestone and Shale 3 TPH Iron Ore 15 TPH Silica 150 TPH Raw Meal Product	Baghouse (4 each)	Grandfather
			S013 - Iron/Silica Silo			
			S010 - Raw Material Grinding			Grandfather
			S011 - Separator			
Homogenizing & Blending	P006	006	S014 - Homogenizing Silo	180 TPH Raw Meal	Baghouse (2 each)	Grandfather
			S015 - Kiln Feed Silo			Grandfather
Kiln Burning	P007	007	S016 - Precalciner Kiln	120 TPH (dry)	Baghouse	12BO444-2
Clinker Cooling	P008	008	S017 - Clinker Drag Chains S018 - Clinker Cooler	81 TPH	Baghouse (3 each)	
Clinker and Gypsum/Additive Silos and Weigh Feeders	P009	009	S026, S027, S029, S030, S031 - Weigh Feeders 1,2,4,5 and 6	600,000 tons/year (3) Silos, 800 tons each (2) 105 tons/hour weight-	Baghouse (12 each)	98BO0259
			S032 - Bottom of A Frame Transfer			

Process	Plant Identifier	AIRS Stack Number	Description	Size	Pollution Control Device	Construction Permit	
			S024 - #2 Clinker Silo	o-meters (1) 5 tons/hour weight-o-meter			
			S035 – Discharge of 629-3 Belt				
			S039 to S041 –Finish Mill Weigh Feeders				
			S038 - Surge Bin				
Sheltered (A-Frame) Clinker Storage and Reclaim	P010	010	S031 - #6 Reclaim Feeder	600,000 tons/year Maximum A-Frame capacity - 60,000 tons	Baghouse		
			S0?? – Top of A Frame Transfer		Baghouse		
Outdoor Clinker Storage and Handling	P015	030	Outdoor Clinker Storage Pile	120,000 tons storage	Emission Control Plan		
Cement Finish Mill and Auxiliaries	P011	011	Cement Finish Mill and Bucket Elevator/Auxiliaries S036 & S037 – Grinding and Limestone Handling	600,000 tons/year	Baghouse (6 each)		
	P012	031	S065 – Separator	Separator - 150 tons/hour fines produced			
Cement Silos/Packhouse /Loadout	P013	012 013	Cement Loadout Spouts S043 to S048 - Cement Silos S049 & S050 – Packhouse	(19) Silos, 34,330 tons total capacity	Baghouse (6 each)		
Clinker & Fuel Handling System	P014		S020 - Coal Silo S025 - Coal/Clinker Hopper	200 TPH Coal and Clinker			C-10,316
Haul Roads	P016	019	Haul roads				None
Fuel Storage Tank	T001		Fuel Storage Tank Breathing/Working Losses	8,000 gallons		None	
Pneumatic Conveyance of Materials	P007A	040	S066 Cement Silo A5		Bagfilters	98BO0315	
			S067 CKD Loading Spout S001Waste Dust Silo		Bagfilters		
Pug Mill Mixing, Pelletization and Truck Loading of CKD and Benefication Dust		041	Pug Mill	30 tons (dry) per hour			
Haulage and Disposal of Pelletized CKD and Benefication Dust		042	Truck Hauling to Lyons Quarry		PM Emission Control Measures		
Haul Roads	P016		Haul Roads			None	
Fuel Storage Tank	T001		T001 - Gasoline Storage Tank	3,000 gallons	Submerged Filling and	None	

Process	Plant Identifier	AIRS Stack Number	Description	Size	Pollution Control Device	Construction Permit
					Vapor Recovery	
Cement Rail Car Unloading System	P050	050	Cement Rail Car Unloading and Handling System – Hopper, screw conveyor and pneumatic transfer system		Bagfilter BH-825-8	05BO0703

SECTION II – Specific Permit Terms

1. P017- Dowe Flats Fugitive Dust Sources P018 - Lyons Quarry

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	1.1	3,200,000 tons/year 25,000 tons/day		Recordkeeping	Monthly
PM	1.2	Dowe Flats Quarry Operations - 134.2 tons/year Disturbed Areas @ Lyons Quarry - 19.0 tons/year	See Condition 1.2	Recordkeeping and Calculation	Annually
PM ₁₀		Dowe Flats Quarry Operations - 58.4 tons/year 916 lbs/day Disturbed Areas @ Lyons Quarry - 9.4 tons/year			
Fugitive Emission Control Plan	1.3			Inspection	Weekly
Restrictions on Lyons Quarry	1.4	Lyons - Mining Prohibited		Certification	Annually

Compliance Status: P017 - Dowe Flats Fugitive Dust Sources P018 - Lyons Quarry

Parameter		Limitations	Reported Data
Dowe Flats Quarry	Process Rates	3,200,000 ton/yr	2,161,770 tons/yr
		25,000 ton/day	19,213 tons/day (Oct-11)
	PM	134.2 ton/yr	104.4 tons/yr
	PM ₁₀	58.4 ton/yr	27.9 tons/yr
Lyons Quarry	PM	19 ton/yr	1.90 ton/yr
	PM ₁₀	9.4 ton/yr	0.0 ton/yr

Cemex provided the reported data above for the rolling 12-month period ending June 2012. The Lyons Quarry has been inactive for many years; however, particulate emissions from ongoing reclamation activities have been calculated. Source is in compliance with the above permit limits.

- 1.1 Total material (includes: topsoil, overburden, limestone, and waste rock) handled shall not exceed the limitations listed in the table above (Construction Permit 93BO1414F). The quantity of total material handled shall be monitored and recorded monthly. Monthly quantities of materials handled shall be used in a twelve month rolling total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months data.

The permittee shall maintain records of the number of days of operation during each month. Compliance with the daily limits shall be monitored by dividing the monthly handling rates by the number of days of operation for that month. Records of the number of days of operation, the daily, monthly and twelve month total handling rates and any other information used to determine the monthly quantities of materials handled shall be kept on site and made available for the Division upon request.

Cemex is tracking material handling on a daily basis and compiling monthly and yearly totals as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of material handling. Source is in compliance with this condition.

- 1.2 PM and PM₁₀ emissions shall not exceed the limits listed in the table above (Construction Permit 93BO1414F). In the absence of credible evidence to the contrary, compliance with the emission limitations is presumed provided the process limits set forth in Condition 1.1 are met and that the control measures set forth in Condition 1.3 have been followed.

For APEN reporting and fee purposes, the annual PM and PM₁₀ emissions shall be estimated using the actual annual processing rates recorded under Condition 1.1, appropriate emission factors and equations, and control measure efficiencies. A record of the annual emission calculations shall be maintained and made available to the Division for inspection upon request.

Cemex is calculating emissions on a monthly and yearly basis. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of particulate emissions. Source is in compliance with this condition.

- 1.3 Fugitive particulate matter emissions shall be controlled in accordance with the following (Construction Permit 93BO1414, Colorado Regulation No. 1, III.D.1.b).
 - 1.3.1 Mining Activities, including mined land reclamations - Visible emissions shall not exceed 20%, and no off-property transport of visible emissions. The 20% opacity and no off-property transport limitations are guidelines and not enforceable standards and no person shall be cited for violation thereof pursuant to C.R.S. 25-7-115.

Mining and blasting activities were observed during the inspection with no off property transport of emissions observed. As part of a Compliance Order on Consent (COC) signed February 20, 2004, Cemex agreed to implement additional weather forecast and control measures to minimize the risk of fugitive emissions during blasting activities. Cemex postpones the loading of explosives if the wind is forecasted to exceed 20 mph. Cemex

also videotapes each quarry blast to document opacity and direction of plume travel. Source is in compliance with this condition.

- 1.3.2 Haul Roads - No off-property transport of visible emissions shall apply to on-site haul roads, and the nuisance guidelines shall apply to off-site haul roads. The no off property transport and nuisance emission limitations are guidelines and not enforceable standards and no person shall be cited for violation thereof pursuant to C.R.S. 25-7-115.

No evidence of off-property transport was observed during the inspection.

- 1.3.3 Haul Trucks - No off-property transport of visible emissions except that when operating off the property of the owner or operator, the applicable guidelines shall be no off-vehicle transport of visible emissions. The no off-property transport limitations are guidelines and not enforceable standards and no person shall be cited for violation thereof pursuant to C.R.S. 25-7-115.

No evidence of off-property transport was observed during the inspection.

- 1.3.4 Adequate soil moisture must be maintained in topsoil and overburden to control emissions during removal.

Adequate soil moisture was observed during the inspection to control emissions from topsoil and overburden.

- 1.3.5 Topsoil and overburden stockpiles shall be reclaimed and revegetated in accordance with the Mined Land Reclamation Bureau (MLRB) permit conditions. Open acreage (mine pits and stockpiles) shall be minimized and in no circumstances shall they be in excess of MLRB or APCD permits, whichever is more restrictive. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit).

Disturbed areas of topsoil and overburden at Dowe Flats appeared to be minimized during the inspection and reclaimed areas appeared to have been re-vegetated as required.

- 1.3.6 Emissions from material handling (i.e. removal, loading, and hauling) shall be controlled by watering at all times, except during below-freezing temperatures, unless natural moisture is sufficient to control emissions. A sprinkler system shall be installed and operated to wet muck piles prior to loading, hauling and crushing. Spillages and accumulations of particulate matter shall be cleaned up and shall be managed to insure they do not contribute to fugitive emissions during operation. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit)

No fugitive emissions from material handling were observed during the inspections. Cemex employs a sprinkler system to wet the active disposal area and a water truck to control dust from haul roads. The haul roads were wet during the inspection and were sufficient in controlling fugitive dust. Cemex employs crews of laborers for cleaning up accumulated piles of dust on and around equipment at the plant. Source is in compliance with this condition.

- 1.3.7 Activities causing fugitive dust emissions shall be suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60-minute period. Only those activities affected by wind speed, and for which it is possible to “suspend operation” need be shut down (i.e., the permittee cannot “shut down” storage piles, thus this condition would not apply to storage piles). Activities may continue when the average wind speed drops below 30 m.p.h. (Construction Permit 93BO1414F, Attachment A, as modified per Section 1, Condition 1.3 of this permit)

Cemex is using a wind speed monitor mounted at the top of the primary crusher building to continuously record wind speeds, the control room operators reportedly evaluate the wind speed and suspend equipment operation when the wind exceeds 30 mph. The data acquisition system continuously monitors wind velocity and automatically notifies the control room operator when the wind speed exceeds 30 mph. The control room’s computer monitors have been modified to include a visual alarm to notify operators of high wind. Source is in compliance with this condition.

- 1.3.8 Vehicle speed on unpaved roads and disturbed areas shall not exceed a maximum of 35 m.p.h. Speed limit signs shall be posted.

Speed limit signs with a maximum of 35 mph were observed on the haul roads during the inspection. Source is in compliance with this condition.

- 1.3.9 Unpaved haul roads shall be treated with chemical dust suppressants to maintain a surface crust, and watered, as often as needed to control fugitive particulate emissions.

Haul roads at the facility appeared compacted and no visible emissions were observed from vehicle traffic. Chemical dust suppressant (calcium chloride) was applied at the quarry on 3/21/12 and 8/18/12 and water trucks are used judiciously to control fugitive dust. Records of water truck activities were reviewed during the inspection and appear to be sufficient to control fugitive emissions. Source is in compliance with this condition.

- 1.3.10 Drills shall be equipped with bag collectors to control emissions.

Drilling rigs were observed during previous inspections and were equipped with bag collectors. Source is considered to be in compliance with this condition.

- 1.3.11 Sequential blasting shall be employed.

Sequential blasting techniques are being employed. Cemex has contracted blasting activities out to a separate company for blasts at the quarry. Blasting activities are still being videotaped by the source to provide further evidence of this requirement. Source is in compliance with this condition.

- 1.3.12 Reclamation work and sequential extraction of material shall be initiated to keep the total disturbed areas at any one time to a minimum.

Reclamation following extraction is the operating procedure employed by Cemex. Ms. Arthur has employed extensive reclamation activities with native plants around the facility to revegetate disturbed areas. Source is in compliance with this condition.

- 1.3.13 The permittee shall maintain a copy of the facility’s Mining and Reclamation plan (as submitted and approved by the Colorado Department of Minerals and Geology -

Mine, Land, and Reclamation Division) on site for Division inspection upon request. (Construction Permit 93BO1414F, Attachment A, as modified for clarification per Section 1, Condition 1.3 of this permit)

The Mining and Reclamation plans are available on-site for review. Source is in compliance with this condition.

- 1.3.14 The permittee will monitor and record local weather forecast information for the current day prior to loading explosives in any drill hole. The forecasts at a minimum shall include anticipated wind speed and direction. Records of the forecasts shall be kept on file at the plant and made available upon request. (Compliance Order on Consent 2002-124, paragraph 38.a)

Records of local weather forecast information are maintained for each blast and were reviewed onsite. Source is in compliance with this condition.

- 1.3.15 The permittee will postpone the loading of explosives if the wind speed is forecasted to be greater than 20 miles per hour at the time of the planned blast. (Compliance Order on Consent 2002-124, paragraph 38.b)

Cemex postpones the loading of explosives if the wind is forecasted to exceed 20 mph. Cemex also videotapes each quarry blast to document opacity and direction of plume travel. The source has a blasting record for each day a blast occurs with a forecast report indicating wind speed and direction printed from a local weather station. In addition, the control room operator's log also includes entries of Dowe Flats and raw material handling equipment downtime during high wind events that correspond to postponed blasts. Source is in compliance with this condition.

- 1.3.16 The permittee will record the actual blast with a video camera, which is to be positioned on the large overburden hill at the south end of the quarry. Each recorded blast shall capture the detonation and the tracking of the ensuing dust plume until the plume's opacity dissipates to less than 5% opacity. The video record will be kept on site and made available upon request. (Compliance Order on Consent 2002-124, paragraph 38.c)

Cemex is recording each blast with video and continues to record until the opacity is dissipated below 5%. Source is in compliance with this condition.

Records of application of dust suppressants shall be maintained on site for inspection upon request.

Cemex is maintaining records of dust suppressant applications. The source applies calcium chloride periodically to control dust from unpaved roadways. Source is in compliance.

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when average wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h, averaged over a sixty minute period.

Cemex is using a wind speed monitor mounted at the top of the primary crusher building to continuously record wind speeds, the control room operators reportedly evaluate the wind speed and

suspend equipment operation when the wind exceeds 30 mph. The data acquisition system continuously monitors wind velocity and automatically notifies the control room operator when the wind speed exceeds 30 mph. The control room's computer monitors have been modified to include a visual alarm to notify operators of high wind. Records of wind speed exceedances and equipment shutdowns were reviewed onsite. Source is in compliance with this condition.

A weekly inspection of the site shall be conducted to ensure the emission control elements are in place and effective. The permittee shall maintain records of the weekly inspections and results. In addition, at any time when a fugitive dust problem is observed, the permittee shall take action to correct the problem. The permittee shall maintain records of the date and time of any fugitive dust problem observed, and the type and time of action taken to correct the problem. These records shall be maintained on site for inspection upon request.

Cemex is performing daily and weekly inspections of the control equipment and documenting them on the "Dowe Flats Weekly Fugitive Dust Inspection Log". The logs were reviewed onsite and appeared to be complete. Source is in compliance with this condition.

- 1.4 There shall be no mining of limestone/raw materials or overburden materials at the Lyons Quarry. Reclamation activities and cement kiln dust disposal (as permitted under Section II, Condition 13 of this permit) are allowed in the Lyons Quarry. (Construction Permit 93BO1414F)

The Lyons Quarry is no longer being mined. However, CKD disposal and reclamation activities still persist at the site. Particulate emissions calculated for the Lyons Quarry are associated with these activities. Source is in compliance with this condition.

2. P017- Dowe Flats Quarry

S055 - S064, Primary Crusher, Belt Conveyor, Radial Stacker to Stockpiles

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	2.1	1,050,000 tons/year		Recordkeeping	Monthly
PM	2.2	Crusher and Conveyor – 0.05 gram per dry standard cubic meter		Stack Test Baghouse Operation and Maintenance	Every Five (5) Years
	2.3	Conveyor – See Condition 2.3		Recordkeeping and Calculation	Monthly
	2.4	0.14 ton/year	Crusher: 0.020 lb/ton Conveyor: .00124 lb/ton (total for all transfer points)		
PM ₁₀	2.4	0.07 ton/year	Crusher: 0.009 lb/ton Conveyor: 0.00059 lb/ton (total for all transfer points)		

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Opacity – Crusher and Conveyor Transfer Points	2.2	Shall not exceed 7%		Inspection Method 9 Baghouse O&M	Daily Semiannually
Opacity	2.5	Shall not exceed 20%, except as provided for below			
		Certain Operating Conditions - Shall not exceed 30%			

**Compliance Status: P017- Dowe Flats Quarry
S055 - S064, Primary Crusher, Belt Conveyor, Radial Stacker to Stockpiles**

Parameter	Limitations	Reported Data
Process Rate	1,050,000 tons/yr	616,289 tons
PM	0.14 ton/yr	0.09 tons
PM ₁₀	0.07 ton/yr	0.04 tons

Cemex provided the reported data above for the rolling 12-month period ending June 2012. Source is in compliance with the above permit limits.

- 2.1 Processing and conveying of raw materials at the crusher and conveyor system shall not exceed the limitation listed in the table above (Construction Permit 94BO593). The quantity of raw materials processed and conveyed shall be monitored and recorded monthly. Monthly quantities of materials processed and conveyed shall be used in a twelve month rolling total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months data.

Records of the monthly and twelve month material processing rates and any information used to determine the monthly quantities of materials handled shall be kept on-site and made available for Division review upon request.

Cemex is recording monthly and annual process data as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of material processed. Source is in compliance with this condition.

- 2.2 The crusher and the conveyor are subject to the provisions of 40 CFR Part 60, Subpart OOO, Standards of Performance for Non-Metallic Mineral Processing Plants, as adopted by reference in Colorado Regulation No. 6, Part A, as set forth in Condition 20 of this permit.

Refer to inspector's comments in Condition 20.

- 2.3 During any consecutive sixty (60) minute period, PM emissions from the conveyor operation shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b and Colorado Regulation No. 6, Part B, II.C.2):

$$E = 17.31 (P)^{0.16} \quad \text{Where:} \quad E \text{ is the allowable particulate emissions in lbs/hr.}$$

P is the process weight rate in tons/hr

In the absence of credible evidence to the contrary compliance with the particulate matter limit is presumed provided the baghouse operating and maintenance requirements specified in Condition 20 are met.

The source is complying with the process rate limits in Condition 2.1 and the baghouse O&M requirements of Condition 20. Source is in compliance with this condition.

- 2.4 PM and PM₁₀ emissions shall not exceed the limits listed in the above table (Construction Permit 94BO593). Compliance with the emission limitations shall be monitored by calculating monthly emissions using the above emission factors in the following equation:

$$\text{Tons/mo} = \frac{\text{EF (lbs/ton)} \times \text{material conveyed or crushed (tons/mo)}}{2000 \text{ lbs/ton}}$$

Note that if the baghouses are operated and maintained in accordance with the requirements in Condition 21 a control efficiency of 98.6 % may be used in the above calculation for the enclosed conveyor and baghouses and the crusher baghouse.

Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions from the crusher and conveyor shall be summed together and be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month rolling total shall be calculated using the previous twelve months' data. Records of emission calculations shall be maintained and made available to the Division upon request.

Cemex is calculating emissions from material throughput on a monthly and annual basis as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of particulate emissions. No exceedances of the permit limits were noted in the records provided. Cemex conducted a stack test on June 18-19, 2008 on the primary crusher and a representative conveyor baghouse. The results of the test indicate compliance with the lb/ton PM emission factors in the table above and the grain loading limits in NSPS Subpart OOO (0.022 gr/dscf). However, this testing shows that Cemex is emitting 0.57 tons per year of controlled PM emissions from the primary crusher and eight conveyor baghouses when calculated using the lb/ton emission factor from the test and permitted annual throughput, in excess of the permitted limit of 0.14 tons per year. In contrast, compliance with the PM emission limit using the emission factors from the permit and actual throughput shows compliance with the limit. The discrepancy from the stack test results and permit emission factors represents an error in the calculation method that has been identified by Cemex and will be addressed with a future permit modification. Source is in compliance with this condition.

- 2.5 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Refer to inspector's comments in Condition 21.

3. P004 - Storage and Handling of Raw Materials at Plant Site
Discharge of Primary-Crushed Raw Materials onto Open Stockpile, S009 – Front End Loader Activity

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	3.1	Front End Loaders:		Recordkeeping	Monthly

		1,050,000 tons/year 4,365 tons/day* Iron Slag: 50,000 tons/year			
PM	3.2	15.5 tons/year	Front End Loader: .0282 lb PM/ton .0127 lb PM ₁₀ /ton	Recordkeeping and Calculation	Monthly
PM ₁₀		7.0 tons/year 53.00 lbs/day			
Opacity	3.3	Shall not exceed 20%, except as provided for below		Inspection Method 9	Daily Semiannually and As Needed
		Certain Operating Conditions - Shall not exceed 30%			
PM Emission Control Plan	3.4			Inspection	Weekly

*daily limit addresses all materials, including slag.

Compliance Status: P004 - Storage and Handling of Raw Materials at Plant Site
Discharge of Primary-Crushed Raw Materials onto Open Stockpile
S009 - Front End Loader Activity

Parameter	Limitations	Reported Data
Process Rate	1,050,000 tons/yr	658,008 tons
	4,365 tons/day	3,764 tons/day (Sept-11)
PM	15.5 tons/yr	9.3 tons
PM ₁₀	7.0 tons/yr	4.2 tons
	53 lbs/day	47.8 lbs/day (Sept-11)

Cemex provided the reported data above for the rolling 12-month period ending June 2012. Magnetite has replaced iron slag as the iron source for the kiln; therefore, no iron slag has been stored or handled. Source is in compliance with the above permit limits.

- 3.1 Process rates shall not exceed the rates listed in the table above (Construction Permit 98BO0292). The quantity of materials processed shall be monitored and recorded monthly. Monthly quantities of materials processed shall be used in a twelve month rolling total to monitor compliance with the annual limitations. Each month a new twelve month rolling total shall be calculated using the previous twelve months data.

The permittee shall maintain records of the number of days of operation during each month. Compliance with the daily limits shall be monitored by dividing the monthly limits by the number of days of operation for that month. Records of the number of days of operation, the daily, monthly and twelve month total processing rates and any other information used to determine the monthly quantities of materials processed shall be maintained and made available to the Division upon request.

Records provided by Cemex indicate compliance with the process rates and emissions limits stated above. Cemex is maintaining daily, monthly and yearly records as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of materials processed. Source is in compliance with this condition.

- 3.2 PM and PM₁₀ emissions shall not exceed the limits listed in the table above (Construction Permit 98BO0292). Compliance with the emission limitations shall be monitored by calculating monthly emissions using the above emission factors (from permit notes in Construction Permit 98BO0292, initial approval, modification 2, issued June 19, 2006) in the following equation:

$$\text{Tons/mo} = \frac{\text{EF (lbs/ton)} \times \text{material processed (tons/mo)}}{2000 \text{ lbs/ton}}$$

Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data. Records of emission calculations shall be maintained and made available to the Division upon request.

Cemex is maintaining and calculating daily, monthly and yearly emissions as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of particulate emissions. No exceedances of the permit limits have been noted. Source is in compliance with this condition.

- 3.3 These sources, except for Front End Loader Activity, are subject to the opacity limits set forth in Condition 21 of this permit.

Refer to inspector's comments in Condition 21.

- 3.4 The following particulate emission control measures shall be applied to the particulate emission producing sources (Construction Permit 98BO0292, Colorado Regulation No. 1, III.D.1.b).

- 3.4.1 Height of discharge from the stacker belt shall be adjusted to minimize the drop height. Water spray bars shall be used if the natural surface moisture is insufficient to limit opacity to less than 10 percent.

Cemex has installed a new stacker in 2009 equipped with a baghouse and a second foam system application at the transfer point to the new stacker from the two mile conveyor from Dowe Flats. Foam is first applied to material on the belt at Dowe Flats. The stacker was observed operating during the inspection and the drop distance was approximately 10-15 feet as it began a new pile of hi cal limestone. A dust event was noted from the stacker lasting approximately 20-30 seconds with opacity greater than 20%; however, the opacity curtailed below 5% continuously after the event. No Method 9 reading was taken. According to plant personnel, the dust was attributed to baghouse fines being dumped onto the belt when it started up for the new pile. The stacker is equipped with a "tattle-tail" sensor that maintains the stacker's height above the stock pile and has a water spray bar to control particulate emissions at the discharge. Source is in compliance with this condition.

- 3.4.2 Transfer points shall be enclosed and maintained under negative pressure.

The transfer points are shrouded and a dust collector capture fugitive dust and periodically release the collected material back onto the conveyor belt before the water spray nozzles. Source is in compliance with this condition.

- 3.4.3 Moisture content of the materials handled by front-end loaders shall be adequate to effectively control the emissions. (Construction Permit 98BO0292, as modified according to Section I, Condition 1.3 of this permit)

The material transported during the inspection contained adequate moisture to control emissions. Source is in compliance with this condition.

- 3.4.4 When feeding the primary crusher at the plant, material drop height from the front-end loaders shall be minimized. (Construction Permit 98BO0292, as modified per Section 1, Condition 1.3 of this permit)

A line is marked on the back of the primary crusher indicating the maximum drop height into the crusher. The equipment operator has been observed loading material into the hopper during previous inspections and the drop distance was minimized. Source is in compliance with this condition.

- 3.4.5 The stockpile work area on which the front-end loaders operate shall be treated with chemical dust suppressants and/or water to minimize the generation of fugitive emissions. The operation of front-end loaders and hauling activities in this area will be suspended whenever the wind speed reaches or exceeds 30 m.p.h., averaged over a 60-minute period. Activities may continue when the average wind speed drops below 30 m.p.h. (Construction Permit 98BO0292, as modified per Section 1, Condition 1.3 of this permit)

No visible emissions were observed from the operation of the loader equipment and the patio area was wet during the inspection. Source is in compliance with this condition.

- 3.4.6 Paved travel areas used by the front-end loader shall be regularly swept with a high efficiency vacuum sweeper to minimize material buildups. In addition, these areas will be watered as necessary and vehicle traffic suspended or rerouted to minimize fugitive emissions if fugitive emissions become a concern. (Construction Permit 98BO0292, as modified per Section 1, Condition 1.3 of this permit)

The vacuum sweeper vehicle was observed operating during the inspection and no visible emissions were observed from vehicle use on the paved areas. Source is in compliance with this condition.

- 3.4.7 Front end loader and hauling activities shall be suspended when the wind speed reaches or exceeds 30 m.p.h., averaged over a 60-minute period. Activities may continue when the average wind speed drops below 30 m.p.h. (Construction Permit 98BO0292, as modified per Section 1, Condition 1.3 of this permit)

Wind monitoring data and daily equipment operation logbooks were reviewed onsite during the inspection to verify equipment suspension during high wind events.

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h, averaged over a sixty minute period. The Annual Certification shall include a statement that the activities identified in Condition 3.5.7 (3.4.7) were suspended when average wind speeds reached or exceeded 30 m.p.h. threshold.

Cemex is using a wind speed monitor mounted at the top of the primary crusher building at Dowe Flats to continuously record wind speeds. The data acquisition system continuously monitors wind velocity and automatically notifies the control room operator when the wind speed exceeds 30 mph. The control room's computer monitors have been modified to include a visual alarm to notify operators of high wind. The anemometer was replaced in 2008 and calibrations were performed most recently on 2/14/12 and 8/11/12 (see attached). Routine calibrations are performed semiannually. Wind speed and equipment shut down records were reviewed onsite and appeared to indicate equipment and activities were shut down when the wind speed exceeded 30 mph. Source is in compliance with this condition.

A weekly inspection of the site shall be conducted to ensure the emission control elements are in place and effective. In addition, at any time when a fugitive dust problem is observed, the permittee shall take action to correct the problem. The permittee shall maintain records of the date and time of any fugitive dust problem observed, and the type and time of action taken to correct the problem. These records shall be maintained on site for inspection upon request. **Cemex is conducting weekly inspections of the site and recording the issues identified and any corrective actions implemented. Records were reviewed onsite. Source is in compliance with this condition.**

4. P001

S002 - Primary Crusher at Plant Site, S004 – Surge Silo

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Material Throughput	4.1			Recordkeeping	Annually
PM & PM ₁₀	4.2		PM & PM ₁₀ : Crusher: 0.001 lb/ton Surge Silo: 2.9 x 10 ⁻⁵ lb/ton	Recordkeeping and Calculation	Annually
PM	4.3	See Condition 4.3		Control Equipment Maintenance	Annual Certification
Opacity	4.4	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse Maintenance and Operation	Daily As Needed
		Certain Operating Conditions – Shall not exceed 30%, for a period or periods aggregating more than six (6) minutes in any 60 consecutive minutes			
CAM	4.5			See Condition 24	

Compliance Status: S002 - Primary Crusher at Plant Site
S004 - Surge Silo

Parameter	Limitations	Actual Data
Material Throughput	No limit	658,008 tons/yr

Crusher	PM	No limit	0.33 tons
	PM ₁₀	No limit	0.33 tons
Surge Silo	PM	No limit	0.01 tons
	PM ₁₀	No limit	0.01 tons

Cemex provided the reported data above for the rolling 12-month period ending June 2012. The primary crusher at the plant site is generally a pass through device with the installation of the crusher at Dowe Flats. The plant primary crusher is used occasionally during the winter to break up ice formed in the raw material. Source is in compliance with calculating and reporting of monthly and annual production throughput values.

- 4.1 Records of the annual amounts of raw materials throughput shall be maintained and made available to the Division for inspection upon request.

Cemex is recording monthly and annual raw material throughput as required.

- 4.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 4.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). These emission factors account for bagfilter control. Records of the calculations shall be maintained and made available for inspection upon request.

Cemex is calculating emissions using the raw material throughputs and the emission factors in the table above, as required.

- 4.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b):

$$E = 17.31 (P)^{0.16}$$

Where: E is the allowable particulate emissions in lbs/hr.
P is the process weight rate in tons/hr

In absence of evidence to the contrary, compliance with the particulate matter limit is presumed provided the baghouse operating and maintenance requirements specified in Condition 21 are met.

The source is complying with the baghouse O&M requirements, refer to Condition 21 for more detail.

- 4.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Refer to Condition 21 for more detail.

- 4.5 S002 is subject to the CAM requirements set forth in Condition 24 of this permit.

Refer to Condition 24 for more detail.

5. P002 - Raw Materials Drying (S005)

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Operating Hours	5.1	7,000 hours/year		Recordkeeping	Monthly
Dryer Feed		1,050,000 tons/year 160 tons/hour			
Coal	5.2	1.4 tons/hour			
Dryer Heat Input	5.3	210,000 million Btu/year		Fuel Sampling Recordkeeping	Semi-Annually Monthly
PM	5.4	22.8 tons/year	See Conditions 5.4 and 5.5.	Stack Test	From Annually to Every Five (5) Years (See Condition 5.4)
PM ₁₀		22.8 tons/year and 6.5 lbs/hour		Baghouse Maintenance and Operation Recordkeeping and Calculation	Monthly
SO ₂	5.5	36.7 tons/year		Stack Test Recordkeeping and Calculation	Every Five (5) years Monthly
NO _x		13.9 tons/year			
VOC		144.8 tons/year			
CO		57.3 tons/year			
Lead	5.5	1.6 tons/year			
	5.6	1.5 micrograms per standard cubic meter averaged over a one-month period (State-Only)		Modeling	See Condition 5.6
Opacity	5.7	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse Maintenance and Operation	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%			
RACT – VOC	5.8	Process Design		Compliance Certification	Annually
MACT Standard - Opacity	5.9	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every Five (5) years Monthly to Annually
CAM	5.10	See Condition 24			

Compliance Status: P002 - Raw Materials Drying (S005)

Parameter	Limitations	Reported Data
Operating Hours	7000 hrs	4,679 hrs
Dryer Feed	1,050,000 tons/yr	658,008 tons
	160 tons/hr	157 tons/hr (Sept-11)

Natural Gas Fuel	No limit	124,708 mscf
Coal Fuel	1.4 tons/hr	0.0 tons/hr
Dryer Heat Input	210,000 mmBtu/yr	131,461 mmBtu
PM	22.8 tons/yr	7.23 tons
PM ₁₀	22.8 tons/yr	4.02 tons
	6.5 lbs/hr	1.72 lbs/hr
SO ₂	36.7 tons/yr	1.74 tons
NO _x	13.9 tons/yr	4.46 tons
VOC	144.8 tons/yr	2.88 tons
CO	57.3 tons/yr	16.20 tons
Lead	1.6 tons/yr	0.94 tons

Cemex provided the reported data above for the rolling 12-month period ending June 2012. The highest hourly values from the annual records are reported. Source is in compliance with the above permit limits.

- 5.1 Annual operating hours and annual and hourly feed rates shall not exceed the limits listed in the above table. (Construction Permit 12BO444-1, revised in accordance with Section I, Condition 3.1 of this permit). Operating hours and the quantity of feed material to the dryer shall be monitored and recorded monthly. Monthly hours of operation and feed material to the dryer shall be used in twelve month rolling totals to monitor compliance with the annual limitations. Each month a new twelve month rolling total shall be calculated using the previous twelve months data.

Compliance with the hourly dryer feed limit shall be monitored by dividing the monthly quantities of dryer feed by the hours the dryer operated for that month. Records of the monthly and twelve month totals of operating hours and feed rates shall be maintained and made available to the Division for inspection upon request.

Cemex is recording operating hours, feed rates and fuel consumption as required. No exceedances of the permit limits were noted. Source is in compliance with this condition.

- 5.2 The terms and conditions of this permit are based on the dryer using natural gas as the primary fuel. Coal may be used as a backup fuel during emergencies and natural gas curtailments. The Division shall be notified, in writing, within seven (7) calendar days of the start of coal use. Records of the amounts of coal burned and the duration of the combustion must be maintained. The quantity of coal burned shall be included in monitoring compliance with the heat input limit as specified in Condition 5.3. Compliance with the hourly coal consumption limit shall be determined by dividing the monthly amount of coal burned by the number of hours coal was burned during the month. (Construction Permit 12BO444-1)

Cemex reported that natural gas was the only fuel supply used for the dryer during the previous year. Source is in compliance with this condition.

- 5.3 Dryer heat input shall not exceed 0.21×10^6 MMBtu per year (Construction Permit 12BO444-1). The quantity of fuel burned in the dryer shall be monitored and recorded monthly. Monthly quantities of fuel burned shall be converted to units of MMBtu based on the heat content for each fuel as determined below. Monthly heat input shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be determined using the previous twelve months' data. Records of the twelve month totals shall be maintained and made available to the Division for inspection upon request.

The heat content of the natural gas shall be determined semi-annually using ASTM Methods or other appropriate methods approved by the Division.

If coal is used as a fuel during any semi-annual period, it shall be sampled at least once during that semi-annual period to determine the heat content and weight percent sulfur. Prior to the first coal sampling event, a coal sampling plan shall be submitted to the Division for approval. The coal sampling plan shall at a minimum include the elements identified in Appendix G. The sulfur and heat content of the coal shall be determined by sampling and testing the coal in accordance with the Division approved coal sampling plan. Alternatively, vendor receipts, invoices, contracts or similar information may be used to provide the values for sulfur and heat content. Such information shall indicate that the coal sampling and analysis has been performed in accordance with ASTM procedures, or equivalent, identified in Appendix G.

Cemex is tracking the heat input to the dryer monthly and annually, the heat content of the fuel is provided quarterly, and records reviewed indicate no exceedances of the permit limits. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of dryer heat input. Source is in compliance with this condition.

- 5.4 PM and PM₁₀ emissions shall not exceed the limits listed in the table above (Construction Permit 12BO444-1). Compliance with the emission limits shall be monitored using the results of the stack tests and the baghouse maintenance and operating procedures required in this Condition 5.4.

Compliance with the annual emission limitations shall be monitored by calculating monthly emissions using the appropriate PM and PM₁₀ emission factors specified in the table below and hours of operation.

Pollutant	Fuel	Emission Factor	Source
PM ¹	Natural Gas	0.62 lb/hr	February 2001 stack test (natural gas used as fuel)
PM ₁₀ ¹	Natural Gas	0.62 lb/hr	
PM ²	Coal	2.64 lb/hr	1988 stack test (coal used as fuel)
PM ₁₀ ¹	Coal	2.64 lb/hr	

¹The emission factors in this table represent the emission factors from the most recent stack test. The permittee shall use emission factors from the most recent stack test to calculate emissions.

²These emission factors shall be used in the event that a stack test is not required for burning coal. If a stack test is conducted for coal burning, emission factors from that test shall be used in lieu of these factors.

Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

Compliance with the hourly emission limitation shall be monitored by dividing the monthly emissions by the number of hours the dryer operated for that month. Records of emission calculations shall be maintained and made available to the division upon request.

Cemex is calculating the particulate emissions using the operating hours and emission factors from the most recent stack test (6/21/2008) as required. PM emission factor = 3.09 lb/hr. PM-10 emission factor = 1.72 lb/hr. Source is in compliance with this condition.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 22 of this permit within 180 days of renewal permit issuance [March 1, 2008]. Frequency of testing, thereafter shall be annual except that: (1) if the first test required by this renewal permit or any subsequent test results indicate emissions are less than or equal to 50% of the emission limit, another test is required within five years; (2) if the first test required by this renewal permit or any subsequent test results indicate emissions are more than 50%, but less than or equal to 75% of the emission limit, another test is required within three years; (3) if the first test required by this renewal permit or any subsequent test results indicate emissions are greater than 75% of the emission limit, an annual test is required until the provisions of (1) or (2) are met.

Tests shall be performed using natural gas (the primary fuel). However, if coal is used for 45 days or more during any calendar year, stack testing shall be performed according to Condition 22 of this permit. This shall be a one-time test.

Cemex performed a stack test for PM on the raw materials dryer in June 2008 as required within the 180 days of renewal permit issuance. Results of the stack test performed using natural gas indicate PM emissions were less than 50% of the limit and therefore another stack test is required within 5 years. Source is in compliance with this condition.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouse shall be conducted in accordance with manufacturer's specifications and/or good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

The source is complying with the baghouse O&M requirements, refer to Condition 21 for more detail.

- 5.5 Emissions shall not exceed the limits listed in the table above (Construction Permit 12BO444-1). Compliance with the emission limits shall be monitored using the results of the stack test and calculating emissions as described below.

Compliance with the annual emission limitations shall be monitored by calculating monthly emissions using the appropriate emission factors specified in the table below and hours of operation.

Pollutant	Fuel	Emission Factor	Source
SO ₂ ¹	Natural Gas	0.3 lb/hr	August 2006 stack test (natural gas used as fuel)
NO _x ¹	Natural Gas	3.5 lb/hr	
CO ¹	Natural Gas	1.0 lb/hr	
VOC ¹	Natural Gas	2.4 lb/hr (0.7lb/hr)*	
Pb ¹	Natural Gas	2 x 10 ⁻⁴ lb/hr	
SO ₂ ²	Coal	1.04 lb/hr	1988 stack test (coal used as fuel)
NO _x ²	Coal	13.68 lb/hr	
CO ²	Coal	19.6 lb/hr	
VOC ²	Coal	2.4 lb/hr (0.7lb/hr)*	August 2006 stack test (natural gas used as fuel)
Pb ²	Coal	2 x 10 ⁻⁴ lb/hr	

¹The emission factors in this table represent the emission factors from the most recent stack test. The permittee shall use emission factors from the most recent stack test to calculate emissions.

²These emission factors shall be used in the event that a stack test is not required for burning coal. If a stack test is conducted for coal burning, emission factors from that test shall be used in lieu of these factors.

Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month rolling total shall be calculated using the previous twelve months' data.

Cemex is calculating emissions on a monthly and a rolling 12-month total basis as required. No exceedances were noted in the records provided during the inspection. *The table above indicates an emission factor of 2.4 lbs/hr which is in conflict of the actual emission factor derived from the August 2006 stack test of 0.7 lb/hr. Cemex is using the correct emission factor (0.7 lb/hr) from the stack test in their VOC emissions calculations through July 2011. Cemex performed another stack test for the raw materials dryer on 7/6/11 establishing a new VOC emission factor (1.3 lbs/hr) that will be used in future calculations. Source is in compliance with this condition.

Stack Testing

Performance testing for lead, sulfur dioxide, nitrogen oxides, carbon monoxide, and volatile organic compounds shall be performed within one year of issuance of the original permit [February 1, 2000] and every five years thereafter, in accordance with the requirements and procedures set forth in the following EPA Test Methods: Method 12 (lead), Method 6-6C (SO₂), Method 10 (CO), Method 7E (NO_x) and Method 25A (VOC).

Tests shall be performed under natural gas conditions, however, tests shall be performed under coal burning conditions if a test as described under Condition 5.4 is required (a one-time coal test is required, if applicable).

Test Protocol, Notification and Report Submittal Requirements: A stack testing protocol shall be submitted for Division approval at least thirty (30) calendar days prior to performance of the test required under this condition. No stack test required herein shall be performed without prior written approval of the protocol by the Division. The Division reserves the right to witness the test. In order to facilitate the Division's ability to make plans to witness the test, notice of the date(s) for the stack tests shall be submitted to the Division at least thirty (30) calendar days prior to the test. The Division may for good cause shown, waive this thirty (30) day notice requirement. In instances when a scheduling conflict is presented, the division shall immediately contact the permittee in order to explore the possibility of making modifications to the stack test schedule. The required number of copies of the compliance test results shall be submitted to the Division within forty-five (45) calendar days of the completion of the test unless a longer period is approved by the Division.

The first stack test on the raw materials dryer after issuance of the original permit was conducted in April 2000 and met the requirements outlined in this condition. A stack test was conducted in August 2006 (as referenced in the table above) also demonstrated compliance with the permitted limits and met the requirements established by EPA's test methodologies. Cemex performed the most recent stack test on the raw materials dryer on July 6, 2011 for the above pollutants demonstrating compliance with the permitted limits. Source is in compliance with this condition.

- 5.6 Emissions of lead shall not be such that emissions result in an ambient lead concentration exceeding 1.5 micrograms per standard cubic meter averaged over a one-month period (Colorado Regulation No. 8, Part C, Section I.B). The permittee submitted modeling results which indicate compliance with this applicable requirement. A copy of the modeling analysis and results shall be maintained and made available for inspection upon request.

The emission rate of lead was determined from the stack test data (7/2011) and demonstrates compliance with the regulation and modeling analysis previously reviewed. The test results indicate lead emissions of 1.8×10^{-4} lb/hr. Source is in compliance with this condition.

- 5.7 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

The raw materials dryer is meeting the opacity requirements outlined in Condition 21, refer to Condition 21 for more detail.

- 5.8 This source shall utilize Reasonably Available Control Technology (RACT) for VOC emissions (Colorado Regulation No. 7, II.C). Operation of this dryer as designed represents RACT. Any modification of the design shall require a new RACT determination and modification or reopening of this permit.

No modifications have been made to the dryer's operation or design. Source is in compliance with this condition.

- 5.9 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 23 of this permit.

Refer to Condition 23 for more detail.

- 5.10 This source is subject to the CAM requirements set forth in Condition 24 of this permit.

Refer to Condition 24 for more detail.

6. P003 - Secondary Crushing Crusher and Screen, S003 - #4 Belt Transfer

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	6.1			Recordkeeping	Annually
PM & PM ₁₀	6.2		PM & PM ₁₀ : Screening and Crushing: 0.00031 lb/ton Silo and Belt Transfer: 2.9×10^{-5} lb/ton (each bagfilter)	Recordkeeping and Calculation	Annually
PM	6.3	See Condition 6.3		Control Equipment Maintenance	Annual Certification

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Opacity	6.4	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse O&M	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%, for a period or periods aggregating more than six (6) minutes in any 60 consecutive minutes			

Compliance Status: **P003 - Secondary Crushing
Crusher and Screen
S001 - Secondary Crushing Dust Silo
S003 - #4 Belt Transfer**

Parameter		Limitations	Reported Data
Material Throughput		No limit	658,008 tons
Screening & Crushing	PM	No limit	0.10 tons
	PM ₁₀	No limit	0.10 tons
Silo & Belt Transfer	PM	No limit	0.01 tons
	PM ₁₀	No limit	0.01 tons

Cemex provided the reported data above for the rolling 12-month period ending June 2012. The emissions calculations for PM and PM₁₀ are based on the emission factors stated in the permit and the actual material throughput.

6.1 Records of the annual amount of raw material throughput shall be maintained and made available to the Division for inspection upon request.

Cemex is tracking raw material throughput on a monthly and annual basis, as required.

6.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 6.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). These factors take the control equipment efficiency into account. Records of the calculations shall be maintained and made available for inspection upon request.

Cemex is using the correct emission factor to calculate PM and PM₁₀ emission from the Secondary Crusher Activities on a monthly and annual basis as required. Source is in compliance with this condition.

6.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b):

$$E = 17.31 (P)^{0.16}$$

Where: E is the allowable particulate emissions in lbs/hr.
 P is the process weight rate in tons/hr

In absence of credible evidence to the contrary, compliance with the particulate matter limit is presumed provided the baghouse operating and maintenance requirements specified in Condition 21 are met.

The source is complying with the baghouse O&M requirements, refer to Condition 21 for more detail.

6.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Wind speed and equipment shut down records were reviewed onsite. Cemex appears to be meeting the opacity requirements outlined in Condition 21, refer to Condition 21 for more detail.

**7. P004 - Raw Materials Storage
S006 through S008 - Raw Materials Storage Silos**

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	7.1			Recordkeeping	Annually
PM & PM ₁₀	7.2		PM & PM ₁₀ : 2.9 x 10 ⁻⁵ lb/ton (for each baghouse stack)	Recordkeeping and Calculation	Annually
PM	7.3	See Condition 7.3		Control Equipment Maintenance	Annual Certification
Opacity	7.4	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse Maintenance and Operation	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%, for a period or periods aggregating more than six (6) minutes in any 60 consecutive minutes			
MACT Standard - Opacity	7.5	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every five (5) years Monthly to Annually

**Compliance Status: P004 - Raw Materials Storage
S006 through S008 - Raw Materials Storage Silos**

Parameter		Limitations	Reported Data
Material Throughput		No limit	658,008 tons
Raw Material Storage (S006, S007, S008)	PM	No limit	0.003 tons
	PM ₁₀	No limit	0.003 tons

Cemex provided the reported material throughput data above for the rolling 12-month period ending June 2012. Particulate emissions are calculated on an annual basis as required under Condition 7.2. The emissions calculations for PM and PM₁₀ are based on the emission factors stated in the permit and the actual material throughput.

7.1 Records of the annual amounts of raw material throughputs shall be maintained and made available to the Division for inspection upon request.

Cemex is tracking raw material throughput on a monthly and annual basis, as required.

7.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 7.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). This factor accounts for the control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.

Cemex is using the correct emission factor to calculate PM and PM₁₀ emission from the Raw Material Storage areas on a monthly and annual basis as required. Source is in compliance with this condition.

7.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b):

$$E = 17.31 (P)^{0.16}$$

Where: E is the allowable particulate emissions in lbs/hr.
P is the process weight rate in tons/hr

In absence of credible evidence to the contrary, compliance with the particulate matter limit is presumed provided the baghouse operating and maintenance requirements specified in Condition 21 are met.

The source is following the baghouse O&M requirements, refer to Condition 21 for more detail.

7.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Cemex is meeting the opacity requirements outlined in Condition 21, refer to Condition 21 for more detail.

7.5 These sources are subject to the MACT requirements, as set forth in Conditions 23 of this permit.

Refer to Condition 23 for more detail.

**8. P005 - Raw Mill Grinding
S010 - Raw Mill, S011 – Raw Mill Auxiliary (Separator), S012 – Raw Mill Weigh Feeder,
S013 - Iron/Silica Silo**

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	8.1			Recordkeeping	Annually
PM & PM ₁₀	8.2		PM & PM ₁₀ : S010 - 0.012 lb/ton S011 - 0.032 lb/ton S012 - 0.019 lb/ton	Recordkeeping and Calculation	Annually

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
			S013 - 0.0031 lb/ton		
PM	8.3	See Condition 8.3		Control Equipment Maintenance	Annual Certification
Opacity	8.4	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse Maintenance and Operation	Daily As Needed
		Certain Operating Conditions – Shall not exceed 30%			
MACT Standard – Opacity	8.5	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every five (5) years Daily

Compliance Status: **P005 - Raw Mill Grinding**
S010 - Raw Mill
S011 - Raw Mill Auxiliary (Separator)
S012 - Raw Mill Weigh Feeder
S013 - Iron/Silica Silo

Parameter		Limit	Reported Data
Process Rate		No limit	650,543 tons
S010	PM	No limit	3.90 tons
	PM ₁₀	No limit	3.90 tons
S011	PM	No limit	10.41 tons
	PM ₁₀	No limit	10.41 tons
S012	PM	No limit	6.18 tons
	PM ₁₀	No limit	6.18 tons
S013	PM	No limit	0.08 tons
	PM ₁₀	No limit	0.08 tons

Cemex provided the Reported Data above for the rolling 12-month period ending June 2012. The emissions calculations for PM and PM₁₀ are based on the emission factors stated in the permit and the actual material throughput.

8.1 Records of the annual amounts of raw material throughputs shall be maintained and made available to the Division for inspection upon request.

Cemex is tracking raw material throughput on a monthly and annual basis, as required.

8.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual raw material throughputs recorded under Condition 8.1, and the emission factors listed in the table above (AP-42, Section 11.6, January, 1995). These factors account for the control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.

Cemex is using the correct emission factor to calculate PM and PM₁₀ emission from the Raw Mill point sources S010, S011, S012, and S013 on a monthly and annual basis as required. Source is in compliance with this condition.

- 8.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b):

$$E = 17.31 (P)^{0.16}$$

Where: E is the allowable particulate emissions in lbs/hr.
P is the process weight rate in tons/hr

In absence of credible evidence to the contrary, compliance with the particulate matter limit is presumed provided the baghouse operating and maintenance requirements specified in Condition 21 are met.

The source is following the baghouse O&M requirements, refer to Condition 21 for more detail.

- 8.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Cemex is meeting the opacity requirements outlined in Condition 21, refer to Condition 21 for more detail.

- 8.5 The raw mill is subject to the MACT standard, as set forth in Condition 23 of this permit.

Refer to Condition 23 for more detail.

9. P006 - Homogenizing and Blending S014 - Homogenizing Silo, S015 Kiln Feed Silo

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	9.1			Recordkeeping	Annually
PM & PM ₁₀	9.2		PM & PM ₁₀ : 2.9 x 10 ⁻⁵ lb/ton (for each baghouse stack)	Recordkeeping and Calculation	Annually
PM	9.3	See Condition 9.3		Control Equipment Maintenance	Annual Certification
Opacity	9.4	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse Maintenance and Operation	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%, for a period or periods aggregating more than six (6) minutes in any 60 consecutive minutes			
MACT Standard – Opacity	9.5	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every five (5) years Daily

Compliance Status: **P006 - Homogenizing and Blending**
 S014 - Homogenizing Silo
 S015 - Kiln Feed Silo

Parameter		Limit	Reported Data
Process Rate		No limit	650,543 tons
S014	PM	No limit	0.009 tons
	PM ₁₀	No limit	0.009 tons
S015	PM	No limit	0.009 tons
	PM ₁₀	No limit	0.009 tons

Cemex provided the Reported Data above for the rolling 12-month period ending June 2012. The emissions calculations for PM and PM₁₀ are based on the emission factors stated in the permit and the actual material throughput.

9.1 Records of the annual amounts of material throughput shall be maintained and made available to the Division for inspection upon request.

Cemex is tracking raw material throughput on a monthly and annual basis, as required.

9.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual material throughputs recorded under Condition 9.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). This factor accounts for control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.

Cemex is using the correct emission factor to calculate PM and PM₁₀ emission from the Homogenizing and Blending silos on a monthly and annual basis as required. Source is in compliance with this condition.

9.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b):

$$E = 17.31 (P)^{0.16}$$

Where: E is the allowable particulate emissions in lbs/hr.
 P is the process weight rate in tons/hr

In absence of credible evidence to the contrary, compliance with the particulate matter limit is presumed provided the baghouse operating and maintenance requirements specified in Condition 21 are met.

The source is following the baghouse O&M requirements, refer to Condition 21 for more detail.

9.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Cemex is meeting the opacity requirements outlined in Condition 21, refer to Condition 21 for more detail.

9.5 The raw mill is subject to the MACT standard, as set forth in Condition 23 of this permit.

Refer to Condition 23 for more detail.

10. P007- Kiln, Calciner (S016)

P008 – Clinker Drag Chains (S017), Clinker Cooler (S018)

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Kiln Feed Rate & Clinker Production Rate	10.1			Recordkeeping	Daily
Operating Hours	10.2	8,064 hours/year		Recordkeeping	Daily and Monthly
Kiln Feed Rate	10.3	120 tons/hour 967,680 tons/year (dry basis)		Recordkeeping	Daily and Monthly
Kiln Fuel		Natural Gas: 2,438 MMSCF/yr Coal: 113,945 tons/yr Whole or Shredded Tires: 18,400 tons/yr Petroleum Coke/Coal Blend: 10,000 tons/yr Blend not to exceed 10% petroleum coke and petroleum coke not to exceed 2% sulfur by weight The use of any other fuel requires Division approval		Recordkeeping	Monthly
Opacity	10.5	Shall not exceed 20%, except as provided for in 10.6		Continuous Opacity Monitoring System Method 9	Continuously
	10.6	Certain Operating Conditions - Shall not exceed 30%			During Each Spray Tower Blasting Event
PM & PM ₁₀	10.7	133 tons/year	See Condition 10.7	Stack Test Recordkeeping and Calculation	Annually to Every Five (5) Years (See Condition 10.4) Monthly
PM & PM ₁₀ – Kiln	10.4	0.275 pound/ton of feed (dry basis)	N/A	Stack Test	Annually to Every Five (5) Years (See Condition 10.4)

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
NO _x	10.7	2649.0 tons/year		Continuous Emission Monitoring Systems	Continuously
CO		396.0 tons/year			
SO ₂		1340 tons/year			
	10.8	Facility Wide Limit: 7 lbs/ton of material		See Condition 10.8	Daily
VOC	10.7	138 tons/year	Stack Test	Stack Test Recordkeeping and Calculation	Annually Monthly
HCl	10.9		0.0381 lb/ton clinker	Recordkeeping and Calculation	Annually
Lead	10.10	4.4 tons/year	See Condition 10.10	Stack Test Recordkeeping and Calculation	Every Five (5) Years Monthly
	10.11	1.5 micrograms per standard cubic meter averaged over a one-month period (State-Only)		Modeling	See Condition 10.11
RACT VOC –	10.12	Process Design		Compliance Certification	Annually
MACT Standards	10.13	Kiln: 0.30 lb PM/ton feed (dry) 20% opacity D/F emission limit Clinker Cooler: 0.10 lb PM/ton feed (dry) 10% opacity		Stack Test COM (opacity), Temperature (D/F) O & M Plan Kiln: PM CEMS	PM: Every Five (5) Years D/F: Every thirty (30) months* Continuously (Deferred)
CAM	10.14	See Condition 24			

*In addition, for periodic monitoring purposes (Colorado Regulation No. 3, Part C, Section V.C.5.b), testing for D/F shall be conducted 45 days after burning tire derived fuel.

Compliance Status: P007- Kiln, Calciner (S016)
P008 - Clinker Cooler (S018)

Parameter		Limitations	Reported Data
Clinker Production		No limit	390,132 tons
Operating Hours		8,064 hrs/yr	6,598.5 hrs
Kiln Feed Rate		120 tons/hr	107 tons/hr (Oct-11)
		967,680 tons/yr	648,369 tons
Fuel	Natural Gas	2,438 Mmscf/yr	425.77 Mmscf
	Coal	113,945 tons/yr	71,113 tons
PM from Kiln		133 tons/yr	22.43 tons
PM ₁₀ from Kiln		133 tons/yr	22.43 tons
NO _x		2649.0 tons/yr	900.84 tons
CO		396.0 tons/yr	366.03 tons
SO ₂		1340.0 tons/yr	19.0 tons
		7.0 lbs/ton fuel	0.049 lbs/ton (Jun-12)

VOC	138.0 tons/yr	2.2 tons/yr
HCl	No limit	7.43 tons/yr
Lead	4.4 tons/yr	0.0 tons/yr

Cemex provided the Reported Data above for the rolling 12-month period ending June 2012. The emissions calculations are based on the continuous emission monitoring systems and emission factors derived from stack tests and stated in the permit.

- 10.1 The permittee shall record the daily production rates and kiln feed rates (Construction Permit 12BO444-2). In addition, the permittee shall maintain a record of daily raw material consumption and clinker production rates (Construction Permit 12BO444-2).

Cemex is recording the daily production rates, kiln feed rates, raw material consumption and clinker production rates. The daily production rates are compiled into monthly and annual records as required.

- 10.2 Annual operating hours shall not exceed 8,064 (Construction Permit 12BO444-2). The permittee shall monitor and record hours of operation daily (Construction Permit 12BO444-2). Daily hours of operation shall be summed to determine monthly hours of operation. Monthly hours of operation shall be used to monitor compliance with the annual limitation. Records of daily, monthly and annual operating hours shall be kept on-site and made available for inspection upon request.

Cemex is tracking operating hours daily and maintaining monthly and annual records as required.

- 10.3 The Construction Permit was issued based on permitted fuels consisting of natural gas, coal, and/or used tires. The use of petroleum coke is incorporated directly into this operating permit according to Section 1, Condition 1.3 of this permit. No other fuels shall be used without prior approval from the Division.

Kiln fuel consumption shall not be exceed the limitations listed in the above table (Construction Permit 12BO444-2). Records of the amount of each type of fuel shall be monitored and recorded daily. (Construction Permit 12BO444-2) Daily quantities of each type of fuel shall be summed to determine monthly quantities of fuel. Monthly quantities of each type of fuel shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month new twelve month totals shall be calculated using the previous twelve months' data.

Cemex uses both coal and natural gas as fuel sources in the kiln and has considered incorporating used tires as an alternative fuel source. Cemex agreed to suspend all tire-derived fuels (TDF) for the kiln until at least 12/31/07, per the Compliance Order on Consent signed 12/22/06. No TDF or pet coke has been burned in the kiln since the issuance of the Operating Permit renewal on March 1, 2008. Cemex tracks kiln fuel consumption and kiln feed rates on an hourly, daily, monthly and yearly basis as required. Beginning with the issuance of the renewal permit, Cemex is maintaining a rolling 12-month total of kiln fuel consumption.

The permittee shall provide the Division written notice at least sixty (60) days prior to the commencement of burning tire derived fuel in the kiln.

A petroleum coke/coal blend containing no more than 10% petroleum coke may be used. The sulfur content of the petroleum coke used shall not exceed 2% by weight. The sulfur content of the petroleum coke used in the blend shall be determined by sampling and analyzing each shipment of petroleum coke, using the appropriate ASTM methods of equivalent, if approved in

advance by the Division. In lieu of sampling of vendor data may be used to determine the weight percent sulfur provided sampling and analysis was performed using the appropriate ASTM methods.

No tire derived fuels (TDF) or petroleum coke/coal blends have been used since the last inspection.

Kiln feed rate shall not exceed 120 tons/hour and 967,680 tons/year (dry basis). (Construction Permit 12BO444-2, as modified under the provisions of Section I, Condition 1.3 to increase the annual limitation.) Daily quantities of the kiln feed shall be summed to determine monthly quantities of kiln feed. Monthly quantities of kiln feed shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month rolling total shall be calculated using the previous twelve months' data.

Compliance with the hourly kiln feed rate shall be determined by dividing the daily kiln feed rate, as recorded under Condition 10.1, by the daily hours of operation, as recorded under Condition 10.2. Records of the hourly, daily and twelve month total kiln feed rates shall be maintained and made available for inspection upon request.

Records of daily kiln feed and hours of operation were reviewed onsite and no exceedances of the hourly kiln feed rate were reported since the last inspection. Daily silo measurements are still used by the source to verify kiln feed weigh scales to determine compliance with the 120 tons/hr limit. Cemex maintains that exceeding the hourly kiln feed rate is an impossibility due to the physical limitations of the kiln. During the inspection, the kiln feed rate was 74.2 tons per hour as recorded in the control room. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of kiln feed. Source is in compliance with this condition.

- 10.4 Particulate matter emissions from the kiln shall not exceed 0.275 pound per ton of feed (dry basis). (Incorporated directly into this operating permit through a combined construction/operating permit process, to make the annual PM emission limits enforceable.)

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 22 of this permit. Frequency of testing shall be annual, except that: (1) if the first annual test or any subsequent test results indicate emissions are less than or equal to 50% of the emission limit, another test is required within five years; (2) if the first annual or any subsequent test results indicate emissions are more than 50%, but less than or equal to 75% of the emission limit, another test is required within three years; (3) if the first annual or any subsequent test results indicate emissions are greater than 75% of the emission limit, an annual test is required until the provisions of (1) or (2) are met.

Note that the last performance test conducted for the kiln was performed in May 2006. The results of the test indicates that PM emissions from the kiln were greater than 50% but less than 75% of the limitation (results were 0.142 lb/ton feed, which is 51.6% of the standard), therefore, the next test is required within three years of the May 2006 test.

Tests shall be performed using the primary fuel that will be used during the annual, three year or five year period, whichever is applicable, however, a stack test shall be performed according to

Condition 22 of this permit, to determine compliance when any other fuel is used, if any other fuel is used for 45 days or more during any calendar year.

Alternatively, the permittee may test using the worst case particulate matter emitting fuel, as approved by the Division, and shall then use this emission rate to estimate PM emissions from all fuels.

The emission rate of particulate matter shall be computed for each run using the procedures and equation set forth in 40 CFR Part 63, Subpart LLL, § 63.1349.

In addition, stack testing shall be performed within forty five (45) days of commencing burning of tire derived fuel in the kiln. If the burning of tire derived fuel does not occur for 45 days or more during a rolling twelve month period, no stack testing is required. The 45 days is the total number of days that tires are burned in the kiln. If tire derived fuel is burned in the kiln only part of a day, that day counts towards the 45 day total. If tires are burned for more than 20% of the total plant operating hours during the five year term of this permit, an additional stack test shall be required during the term of the renewal permit, subject to the 45 day criteria set forth above.

Cemex conducted a stack test for particulate matter emissions from the kiln on 6/19/08 and demonstrated an emission rate of 0.0692 lbs of PM per ton of feed material. Cemex conducted a stack test for particulate matter emissions from the clinker cooler on 4/6/11 and demonstrated an emission rate of 0.03 lbs of PM per ton of feed material, which is less than the MACT limit of 0.10 lbs PM per ton feed. Since results of the stack tests indicate that emissions are less than 50% of the standard, another test is required within five years.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

The permittee shall maintain records of the occurrence and duration of any startup, shutdown, or malfunction in the operation of the kiln and clinker cooler; any malfunction of the air pollution control equipment.

Records of the baghouse operational and maintenance procedures were reviewed during previous inspections. The differential pressure change across the baghouses (kiln and clinker cooler) is monitored continually from the control room. The pressure drop across the kiln and Hart baghouses was 4.8" w.c. with the clinker cooler baghouse at 3.1" w.c. The alkali bypass baghouse is no longer being used by the source. Cemex conducts weekly inspections of the baghouses and all maintenance activities are tracked by work order requests. Source is in compliance with this condition.

- 10.5 Except as provided in Condition 10.6, below, no owner or operator of a source shall allow or cause the emission into the atmosphere of any air pollutant which is in excess of 20% opacity. This standard is based on 24 consecutive opacity readings taken at 15-second intervals for six minutes. The approved reference test method for visible emissions measurement is EPA Method 9 (40 CFR Part 60, Appendix A (July, 1992)) in all subsections of Section II.A of Regulation No. 1. (Colorado Regulation No. 1, II.A.1).

Compliance with this opacity limit shall be monitored using the continuous opacity monitor as set forth in Condition 23.

In addition, during each spray tower blasting event, an EPA Reference Method 9 visual opacity observation (in accordance with 40 CFR Part 60, Appendix A, as adopted by reference in Colorado Regulation No. 6, Part A) shall be performed. If any such observation indicates an exceedance of the limit, additional observations shall be performed. Subject to the provisions of C.R.S. 25-7-123.1 and in the absence of credible evidence to the contrary, exceedance of the limit shall be considered to exist from the time a Method 9 reading is taken that shows an exceedance of the opacity limit until a Method 9 reading is taken that shows the opacity is less than the opacity limit. All EPA Reference Method 9 readings shall be conducted by a certified observer.

Records of the date, time and length of each blasting event shall be maintained and made available for inspection upon request. Records of the results of Method 9 readings and a copy of the Method 9 reader's certification shall be kept on site and made available to the Division upon request. Copies of the blasting event date and time information, and observations shall be submitted on a quarterly basis, based on the operating permit reporting dates. A copy of the COM data for each blasting event shall also be submitted with the quarterly report.

During the inspection, both the kiln and the clinker cooler were operating normally and no visible emissions were observed, which is consistent with the continuous opacity monitor readings observed in the control room (2.87% kiln; 0.04% cooler). No spray tower blasting with dynamite has occurred since issuance of the renewal permit. A straightening vane added to the spray tower has changed the airflow inside the tower reducing buildup and the need for blasting. All cleaning of the spray tower was conducted using cardox or spray cannons as reported in quarterly reports received by the Division. Beginning in 2008, Cemex has begun submitting quarterly CEMS reports, as required, that is more frequent than the semiannual reports previously submitted. Since the last inspection, opacity exceedances from the kiln related to an upset/malfunction were recorded by the COMS during fourth quarter of 2011. The malfunctions occurred on 11/12/2011, 11/13-14/2011, and 12/10/2011 related to bags blown out in single compartments of the kiln/hart baghouses. In order to troubleshoot the cause of the high opacity, the kiln remained running at a lower speed and feed rate during each event per SOP to identify the problem baghouse compartments. Once the compartments were identified, the compartment was taken out of service and the bags were repaired/replaced. The 12/10/2011 event occurred from bringing the kiln up after an outage where bags in four separate compartments had been replaced including the compartment where the malfunction occurred. Each of the events were reported to the Division and appear to meet the criteria of a valid upsets at this time. Cemex has instituted a "Beyond Compliance" program set to maintain kiln opacities below 5% and includes aggressive baghouse maintenance and bag replacement schedule. Monitor downtime has been less than 3.0% including downtime for calibrations since the last inspection. See table below.

COMS DATA	Operating Time	Excess Emissions	Monitor Downtime
Kiln	Hours	Hours	Hours
2010 1 st Qtr	2034	0.2	13.7 (0.6%)
2010 2 nd Qtr	961	0	6.0 (0.6%)
2010 3 rd Qtr	1317	0.6	7.2 (0.5%)
2010 4 th Qtr	1329	0	5.7 (0.4%)
2011 1 st Qtr	1104	0	4.9 (0.4%)
2011 2 nd Qtr	1228	2.2	15 (1.2%)
2011 3 rd Qtr	2096	0.1	32.6 (1.6%)
2011 4 th Qtr	1613	1.3	24.3 (1.5%)

2012 1 st Qtr	1425	0	34.3 (2.4%)
2012 2 nd Qtr	1869	0	28.5 (1.5%)
Clinker Cooler			
2010 1 st Qtr	2034	0	12.4 (0.6%)
2010 2 nd Qtr	961	0	4.3 (0.4%)
2010 3 rd Qtr	1317	0	6.3 (0.4%)
2010 4 th Qtr	1328	0	5.5 (0.4%)
2011 1 st Qtr	1104	0	4.8 (0.4%)
2011 2 nd Qtr	1228	0	5.3 (0.4%)
2011 3 rd Qtr	2096	0	41.4 (2.0%)
2011 4 th Qtr	1613	0.6	13.2 (0.8%)
2012 1 st Qtr	1425	0	8.6 (0.6%)
2012 2 nd Qtr	1869	0	13.9 (0.7%)

- 10.6 No owner or operator of a source shall allow or cause to be emitted into the atmosphere any air pollutant resulting from the building of a new fire, cleaning of fire boxes, soot blowing, start-up, any process modification, or adjustment or occasional cleaning of control equipment, which is in excess of 30% opacity for a period or periods aggregating more than six minutes in any sixty consecutive minutes (Colorado Regulation No. 1, Section II.A.4).

Compliance with this opacity limit shall be demonstrated using the continuous opacity monitor as set forth in Condition 23.

Cemex is monitoring opacity during startup, shutdown, process modifications and control equipment cleaning with the COMS, as required. A malfunction reported for 9/7/2011 resulted in opacity exceeding the 30% limit for six minutes after repairing a screw conveyor servicing an offline compartment of the kiln baghouse. The compartment was immediately taken back out of service and a man door was found open on the clean side of the compartment after the repair. The door was closed and residual material was cleaned out before the compartment was returned to service. Source is considered in compliance with this condition.

- 10.7 Emissions of PM, PM₁₀, NO_x, SO₂, CO and VOC shall not exceed the limits listed in the above table. (Construction Permit 12BO444-2, revised according to Section I, Condition 1.3, to revise the NO_x, CO, and VOC emission limits).

Emissions of NO_x, SO₂, and CO emissions shall be determined in units of pounds per hour using the continuous emission monitoring systems (CEMS). Hourly emissions from the CEMS shall be summed to determine monthly emissions. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual, tons per year, limitations. Each month a new twelve month rolling total shall be calculated using the previous twelve months' data.

Proper installation, certification and operation of the CEMS shall include annual certification and quarterly audits of the calibration gases. (Construction Permit 12BO444-2). The CEMS shall be installed, operated and certified in accordance with a Division-approved QA/QC plan. When the CEMS are unable to provide quality assured data the following applies: When data is not available for 48 hours or less, the highest recorded hourly emission rate from the previous 24 hours shall be used. When data is not available for more than 48 hours, the maximum daily emission rate, based on the previous 30 days, shall be used.

CEMEX operates a continuous emission monitor (CEM) to monitor and track emissions of NO_x, SO₂, and CO from the kiln and clinker cooler exhaust stream. Cemex does not have short-term emission limits for NO_x, SO₂,

and CO; therefore, no excess emissions were reported. Cemex reported that the gaseous continuous monitoring system (CMS) was down due to quality assurance calibrations, malfunctions, and other causes for less than 5.0% of operating time during the last four quarters.

Beginning in June 2007, CEMEX has installed and been using a new DAS manufactured by Nexus Solutions, Inc. to monitor compliance with its emission limits. Rather than using a default moisture correction value as with the previous system, the Nexus system uses real-time moisture concentration measurements taken by the CEMS and combined with wet flow measurements and dry pollutant concentrations in the DAS to compute actual pollutant mass emission rates. The Division has obtained a copy of the Configuration Report for the Nexus DAS showing the origin of every data input and the calculations used to compute all output. Additionally, CEMEX installed a new flow monitor in December 2007 to replace the old flow monitor installed in 2001. CEMEX performed a flow RATA to certify the new flow monitor on December 26, 2007. CEMEX agreed to perform another RATA including all gaseous pollutants with flow in lbs/hr to fulfill the request outlined in a letter dated September 27, 2007 by the Division requiring CEMEX to recertify all components of the CEMS and DAS. Cemex performed a RATA on 5/6/08, certifying the CEMS and DAS with the new flow monitor. Following the Consent Order signed 2/11/09, CEMEX has performed semiannual RATAs for two years (2009 and 2010) to ensure the quality of data from the CEMS. CEMEX has subsequently gone back to the annual RATA schedule with cylinder gas audits (CGA) in between.

Compliance with the PM and PM₁₀ emission limits shall be monitored using the stack test results and baghouse maintenance procedures specified in Condition 10.3. Compliance with the annual emission limits shall be monitored by calculating emissions monthly using the PM and PM₁₀ emission factors (in lbs/ton feed) indicated in the table below and the monthly quantity of feed to the kiln.

Pollutant	Emission Factor	Source
PM ¹	0.142 lbs/ton feed	May 2006 Stack Test
PM ₁₀ ¹	0.142 lbs/ton feed	
Lead ¹	2.52 x 10 ⁻⁵ lbs/ton feed	

¹The emission factors in this table represent the emission factors from the most recent stack test. The permittee shall use emission factors from the most recent stack test to calculate emissions.

Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

PM emissions from the kiln and clinker cooler shall be calculated in the manner prescribed in 40 CFR Part 63, Subpart LLL, Section 63.1349 for all records, calculations and reports. (Compliance Order on Consent 2002-124, Paragraph 37.b, revised to require procedure in Part 63 instead of Part 60)

Cemex is calculating PM and PM₁₀ emissions using the emission factors in the table above. Cemex performed another stack test for particulate emissions on the kiln on June 19, 2008, establishing a 0.069 lb/ton of feed emission factor. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of kiln particulate emissions.

Compliance with the VOC emission limit shall be monitored through stack tests and emission calculations as discussed below.

Stack Tests

Performance testing for VOC shall be performed once during each calendar year, in accordance with the requirements and procedures set forth in EPA Test Method 25A forth in 40 CFR Part 60, Appendix A. The length of time between each test shall be at least six months. Test results shall be converted to units of lbs/ton feed, for use in subsequent emission calculations. The initial test shall be conducted within 180 days of renewal permit issuance [March 1, 2008].

Testing shall be performed for each proposed fuel type, except natural gas. No testing is required if natural gas is the only fuel used during the calendar year. Alternatively, the permittee may test using the worst case VOC emitting fuel, and shall then use this emission rate to estimate VOC emissions from all fuels for that year.

In addition, stack testing shall be performed within forty five (45) days of commencing burning of tire derived fuel in the kiln. If the burning of tire derived fuel does not occur for 45 days or more during a rolling twelve month period, no stack testing is required. The 45 days is the total number of days that tires are burned in the kiln. If tire derived fuel is burned in the kiln only part of a day, that day counts towards the 45 day total. If tires are burned for more than 20% of the total plant operating hours during the five year term of this permit, an additional stack test shall be required during the term of the renewal permit, subject to the 45 day criteria set forth above.

The stack testing is subject to the "Test Protocol, Notification and Report Submittal Requirements" specified in Condition 5.5 of this permit.

Until recently, this emission point was equipped with a VOC continuous emission monitor system (CEMS) to measure actual VOC emissions. The requirement to operate the VOC CEMS was removed in this version of the Title V permit renewal, and replaced with a requirement to perform annual stack testing to determine VOC emissions. Cemex must use the results of this testing to calculate VOC emissions from the main kiln stack using the actual measured kiln feedrate in tons per hour and a VOC emission factor of pounds emitted per ton of kiln feedrate. Cemex performed the initial stack test for VOC emissions using coal on 4/21/08 establishing an emission rate of 0.0051 lb/ton of feed. The next stack test for VOC was scheduled for October 2009 to coincide with the semiannual RATA; however, the VOC testing was performed and passed on 1/27/2010 due to kiln downtime for market conditions. Current VOC emissions are being calculated with the 8/23/2011 stack test derived emission factor. The most recent annual test was conducted on 7/19/2012 with an average 0.004lb VOC/ton of kiln feed. Future VOC emissions from the kiln shall be calculated using this emission factor. No TDF as fuel in the kiln has been proposed for use by Cemex, to date.

Emission Calculations

Compliance with the annual VOC emission limits shall be monitored by calculating monthly emissions using the emission factor determined from the stack test (in units of lbs/ton feed) and the monthly quantity of feed to the kiln. The emission factor from the most recent stack test shall be used to calculate emissions, when more than one stack test has been conducted. Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual emission limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data. Prior to conducting the initial stack test, emissions shall be calculated using an emission factor of 41.4 lbs/hr and monthly hours of operation.

Cemex is using the results of the 8/23/2011 stack test to calculate VOC emissions from the main kiln stack using the actual measured kiln feedrate in tons per hour and a VOC emission factor of pounds emitted per ton of kiln feed rate derived from the test. The results of the 7/19/2012 test will be used going forward in the VOC emission calculations.

- 10.8 **Facility Wide Limit** Sulfur dioxide emissions shall not exceed 7 pounds per ton of material (including fuel) processed. This emission limit shall be calculated over each 24-hour period that commences at midnight. If the source does not operate for the entire 24-hour period, the actual hours of operation shall be used as the averaging time. At no time shall the averaging time be greater than 24 hours (Construction Permit 12BO444-2, Colorado Regulation No. 1, Section VI.A.3.f.). Compliance with the facility wide limit shall be monitored using the CEMS data obtained under Condition 10.7, and actual material throughputs recorded under Conditions 10.1 and 10.3, the relevant information recorded for the dryer, and any other information necessary from any other sources emitting sulfur dioxide at this facility.

Cemex is calculating the SO₂ emissions based on the CEM emissions data and the tons of material processed as required. Records provided indicate that the average SO₂ emissions during the previous 12 months were 0.03 pounds of SO₂ per ton of material processed. Source is in compliance with this condition.

- 10.9 For APEN reporting and fee purposes, annual HCl emissions shall be estimated using actual clinker production and the emission factor listed in the table above (Portland Cement Association Report, September, 1996). Records of emission calculations shall be maintained for inspection upon request.

Cemex is calculating HCl emissions using the emission factor specified in the permit (0.0381 lbs HCl/ton clinker). For APEN purposes and fee reporting, Cemex correctly reported the HCl emissions (6.55 tons) using the correct emission factor and actual clinker production in the APEN received 5/13/2011. Source is in compliance with this condition.

- 10.10 Emissions of lead shall not exceed the limits shown in the table above. Compliance with the annual limitations shall be monitored using the results of the stack tests and the baghouse maintenance and operating procedures required in this condition 10.10.

Compliance with the annual limitations shall be monitoring by calculating monthly emissions using the emission factor (in lbs/ton feed) indicated in the table in Condition 10.7 and the monthly quantity of feed to the kiln. Monthly emissions shall be calculated by the end of the subsequent month. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data. Records of emission calculations shall be maintained and made available to the Division upon request.

Emission calculations are not required for any twelve month period for which only natural gas was used as fuel for the kiln. In these cases, compliance with the annual limitations is presumed, in the absence of credible evidence to the contrary.

Cemex is using the emission factor represented in the table above from a May 2006 stack test (2.52×10^{-5} lbs/ton feed) to calculate monthly and rolling 12-month total emissions for reporting and emissions fees as required. Source is in compliance with this condition.

Stack Testing

Performance testing for lead shall be performed every five years in accordance with the requirements and procedures set forth in EPA Test Method 12 forth in 40 CFR Part 60, Appendix A.

Testing shall be performed for each proposed fuel type, except natural gas. No testing is required if natural gas is the only fuel used during the permit term. A stack test shall be performed to determine compliance when any other fuel is used, if any other fuel is used for 45 days or more during any calendar year. Alternatively, the permittee may test using the worst case lead emitting fuel, and shall then use this emission rate to estimate lead emissions from all fuels except natural gas if natural gas is the only fuel used for any twelve month rolling period.

The stack testing is subject to the "Test Protocol, Notification and Report Submittal Requirements" specified in Condition 5.5 of this permit.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

Cemex has performed performance testing for lead as described above. The previous stack test was performed May 11, 2006. Records of the baghouses' operational maintenance procedures were reviewed during previous inspections. Information about the differential pressure change across the baghouses (kiln and clinker cooler) is monitored continually from the control room. Cemex conducts weekly inspections of the baghouses and all maintenance activities are tracked by work order requests. Source is in compliance with this condition.

10.11 Emissions of lead shall not be such that emissions result in an ambient lead concentration exceeding 1.5 micrograms per standard cubic meter averaged over a one-month period (Colorado Regulation No. 8, Part C, Section I.B). The permittee submitted modeling results, which indicate compliance with this applicable requirement. A copy of the modeling analysis and results shall be maintained and made available for inspection upon request.

The 6/04/02 stack test determined lead emissions rates that were used to model the ambient lead emissions analysis. Source is in compliance with this condition.

10.12 This source shall utilize Reasonably Available Control Technology (RACT) for VOC emissions (Colorado Regulation No. 7, II.C). Operation of this kiln and clinker cooler as designed represents RACT. Any modification of the design shall require a new RACT determination and modification or reopening of this permit.

No modifications have been made to the kiln or clinker cooler operation. Source is in compliance with this condition.

10.13 On and after the MACT compliance date of June 10, 2002, these sources are subject to the MACT standard, as set forth in Conditions 23 of this permit. Note that as specified in Condition 23, under "Performance Testing Requirements" for nonopacity standards a test for dioxin/furans

(D/F) is required within 45 days of burning tire-derived fuel in the kiln (Colorado Regulation No. 3, Part C, Section V.C.5.b).

Refer to condition 23 for more detail.

10.14 These sources are subject to the CAM requirements set forth in Condition 24 of this permit.

Refer to condition 24 for more detail.

- 11. P009 - Clinker/Gypsum Silos and Weigh Feeders**
S026, S027, S029, S030, S031 – Weigh Feeders 1,2,4, 5 and 6, S024 - #2 Clinker Silo, S038 – Surge Bin, S039 - S041 – Finish Mill Weigh Feeders, S035 – Discharge of 629-3 Belt, S032 – Bottom of A Frame Transfer, P010 - Sheltered Clinker Storage Stockpile with Underground Reclamation, S024A – Transfer from top of A-Frame Building, S031 - #6 Reclaim Feeder, P015 - Outdoor Clinker Storage, and Handling of Clinker with Front End Loaders, P011 - Cement Finish Mill and Bucket Elevator/Auxiliaries, S036 & S037 – Grinding and Limestone Handling, P012 – S065 - High Efficiency Separator (Classifier) and Baghouse Collector, P013 - Cement Storage Silos/Packhouse/Loadout, S043 to S048 - Cement Silos, S049 & S050 – Packhouse

Parameter	Permit Condition Number	Limitations		Emission Factors	Monitoring	
					Method	Interval
Process Limits	11.1	P009	Clinker Handled: 600,000 tons/year 4,000 tons/day		Recordkeeping	Monthly
		P010	Clinker Handled: 600,000 tons/year 5,500 tons/day			
		P015	Maximum Clinker Stored: 120,000 tons Clinker Handled: 180,000 tons/year 5,500 tons/day			
		P011	Overall Fresh Feed to Mill: 631,600 tons/year 4,500 tons/day 15,000 tons/year limestone			
		P012	Cement Produced: 631,600 tons/year 4,500 tons/day			

Parameter	Permit Condition Number	Limitations		Emission Factors	Monitoring	
					Method	Interval
				P013	Cement Handled: 681,600 tons/year (includes 50,000 tons/yr imported cement) 4,500 tons/day	
Operating Hours	11.2	8,064 hours/year			Recordkeeping	Monthly
PM	11.3	P009	7.82 tons/year	Based on Grain Loading Limitation – See Table in Condition 11.3	Stack Test	Every Five (5) Years Monthly
		P010	21.96 tons/year		Recordkeeping and Calculation	
		P011	17.05 tons/year		Stack Test	Every Five (5) Years
		P012	21.52 tons/year			
		P013	12.3 tons/year			
		P015	2.05 tons/year	3.8 lb/vmt	Recordkeeping and Calculation Emission Control Plan	Monthly
PM ₁₀	11.3	P009	3.91 tons/year 52 lbs/day	50% of PM	Stack Test	Every Five (5) Years Monthly
		P010	10.98 tons/year 201 lbs/day	50% of PM	Recordkeeping and Calculation	
		P011	8.65 tons/year 48 lbs/day	50% of PM		
		P012	10.76 tons/year 107 lbs/day	50% of PM		
		P013	6.2 tons/year 43 lbs/day	50% of PM		
		P015	0.92 ton/year 78 lbs/day	0.3 mile one way PM ₁₀ – 1.7 lb/vmt and 80% control	Recordkeeping Emission Control Plan	Monthly
Opacity	11.4	Shall not exceed 20%, except as provided for below			Inspection Method 9	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%				
Fugitive Particulate Emissions	11.5	Emission Control Plan			Inspection	Weekly

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
MACT Standard – Opacity	11.6	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every Five (5) Years Monthly to Annually
CAM	11.7	Condition 24			

Compliance Status: P009 - Clinker/Gypsum Silos and Weigh Feeders

P010 - Sheltered Clinker Storage Stockpile with Underground Reclamation

P015 - Outdoor Clinker Storage, and Handling of Clinker with Front End Loaders (formerly 10A)

P011 - Cement Finish Mill and Bucket Elevator/Auxiliaries

P012 - High Efficiency Separator (Classifier) and Baghouse Collector

P013 - Cement Storage Silos/Packhouse/Loadout

Parameter	Source	Limitations	Reported Data
Process Limits	P009	600,000 tons/yr	416,654 tons
		4,000 tons/day	1,900 tons/day (Jan-12)
	P010	600,000 tons/yr	390,132 tons
		5,500 tons/day	1,543 tons/day (Oct-11)
	P015	180,000 tons/yr	39,013 tons
		5,500 tons/day	154 ton/day (Oct-11)
	P011	631,600 tons/yr	416,954 tons
		4,500 tons/day	1,900 tons/day (Jan-12)
		15,000 tons/yr (limestone)	1,247 tons
	P012	631,600 tons/yr	446,683 tons
		4,500 ton/mo	2,038 tons/day (Jan-12)
	P013	681,600 tons/yr	448,896 tons
		4,500 ton/day	2,022 tons/day (May-12)
Operating Hours		8,064 hrs/yr	6,672 hrs (278 days-P013)

Parameter	Source	Limitations	Reported Data
PM	P009	7.82 tons/yr	1.85 tons
	P010	21.96 tons/yr	2.13 tons
	P011	17.05 tons/yr	2.49 tons
	P012	21.52 tons/yr	2.55 tons
	P013	12.3 tons/yr	0.15 tons
	P015	2.05 tons/yr	1.48 tons
PM ₁₀	P009	3.91 tons/yr	2.22 tons

	P010	52 lbs/day	19.43 lbs/day (Jul-11)
		10.98 tons/yr	2.13 tons
	P011	201 lbs/day	25.38 lbs/day (Oct-11)
		8.65 tons/yr	1.24 tons
	P012	48 lbs/day	10.75 lbs/day (Jan-12)
		10.76 tons/yr	0.00 tons
	P013	107 lbs/day	0.00 lbs/day
		6.2 tons/yr	0.00 tons
	P015	43 lbs/day	0.00 lbs/day
		0.92 tons/yr	0.13 tons
		78 lbs/day	1.049 lbs/day (Oct-11)

Cemex provided the Reported Data above for the rolling 12-month period ending June 2012. The emissions are calculated based on daily, monthly and yearly material throughputs. The highest monthly and daily values are reported. P012 and P013 PM₁₀ emissions are based upon a 4/18-19/2011 stack tests with a 0.000 gr/dscf average. Source is considered to be in compliance with this condition.

- 11.1 The amount of clinker and cement handled shall not exceed the limits listed in the table above (Construction Permit 98BO0259). The quantity of materials handled through each emission group shall be monitored and recorded monthly. Monthly quantities of material handled shall be used in a twelve month rolling total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

The permittee shall maintain records of the number of days of operation for each month. Note that if any unit within an emission group operates during a day, that day counts as a day of operation. Compliance with the daily limits shall be monitored by dividing the monthly handling rates by the monthly number of days of operation. Records of the number of days of operation, the daily, monthly and twelve month total handling rates and any other information used to determine the monthly quantities of material handled shall be maintained and made available upon request.

Cemex is tracking the amount of clinker handled daily, monthly, and yearly as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of materials handled.

- 11.2 Annual hours of operation shall not exceed 8,064 (Construction Permit 98BO0259). Hours of operation shall be monitored and recorded monthly. Monthly hours of operation shall be used in a twelve month rolling total to monitor compliance with the annual limitation. Each month a new twelve month total shall be calculated based on the previous twelve months' data. Records of monthly and twelve month totals of operating hours shall be kept on-site and made available for inspection upon request.

Cemex is tracking the hours of operation on a daily, monthly, and yearly as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of hours of operation.

- 11.3 PM and PM₁₀ emission rates shall not exceed the limits listed in the table above (Construction Permit 98BO0259). For all sources except P015, compliance with the emission limits shall be monitored using the results of the stack tests and the baghouse maintenance and operating procedures required in this Condition 11.3.

For all sources except P015, compliance with the annual emission limitations shall be monitored by calculating monthly emissions using the PM emission factor determined from the stack test (in gr/dscf), hours of operation (as required by Condition 11.2) and the maximum design flow rate of the control devices (see table below). PM₁₀ emissions shall be presumed to be 50% of PM.

The emission factor for any baghouse within an emission group that has been performance tested shall be the results of the most recent test. In lieu of using individual test results for a baghouse, the most conservative test result from the most recent performance test for a baghouse within that same emission group may be used. The emission factor for any baghouse within an emission group that has not been performance tested, shall be the most conservative emission factor from the most recent performance test for any baghouse within that emission group that has been performance tested.

Monthly emissions of PM and PM₁₀ shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

Emission Group	Baghouse ID	Baghouse Grain Loading (gr/dscf)	Design Flow Rate (ACFM)	Baghouse ID	Baghouse Grain Loading (gr/dscf)	Design Flow Rate (ACFM)
P009	BH 625-1	0.03	2,000	BH 625-9	0.03	1,000
	BH 725-5	0.03	1,000	BH 625-14	0.03	1,000
	BH 625-4	0.03	1,000	BH 725-6	0.03	1,000
	BH 625-5	0.03	1,000	BH 725-8	0.03	1,000
	BH 625-7	0.03	1,000	BH 725-8	0.03	1,000
	BH 625-8	0.03	1,000	BH 625-10	0.03	2,000
P010	BH 625-15	0.02	45,000	BH 525-17	0.02	10,000
P011	BH 725-2	0.03	18,200	BH 725-3	0.03	1,300
P012/031*		0.006	147,060			
P013	BH 825-1	0.03	4,400	BH 824-5	0.03	2,540
	BH 825-2	0.03	4,400	BH 825-6	0.03	1,280
	BH 825-3	0.03	4,400	BH925-1	0.03	9,200
	BH 825-4	0.03	1,640	BH 925-2	0.03	9,200

*identified in Construction Permit 95BO0259 as AIRS pt 031.

Compliance with the daily PM₁₀ emission limitations shall be monitored by dividing the monthly PM₁₀ emissions by the number of days the emission group operated during that month. Note that if any unit within the emission group operates during a day, that day counts as a day of operation.

For P015, compliance with the emission limits shall be demonstrated by meeting the production limits set forth in Condition 11.1, and by applying the Fugitive Particulate Emission Control Plan set forth in Condition 11.5. For purposes of APEN reporting and fees, annual PM and PM₁₀ emissions shall be calculated using the actual throughput rates recorded under Condition 11.1 and the appropriate emissions factors, equations and control efficiencies. Records of the emission calculations shall be maintained and made available for inspection upon request.

Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of particulate emissions from these sources. Cemex conducted stack tests in April and May 2011 establishing PM and PM₁₀ emission rates for each activity group also in compliance with the limits. Cemex is using the design flow rate from the table above along with the grain loading emission factors from May 2006 and April-May 2011 stack tests to calculate particulate emissions. No exceedances of the permit limits have been noted. Source is in compliance with this condition.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 22 of this permit. Frequency of testing is every five years. A test shall be performed for a representative bagfilter for each activity group (P009, P010, P011, P012, and P013) to monitor compliance with the grain loading (gr/scf) requirements.

At least one activity group will be tested within one year of initial permit issuance (February 1, 2000) to include a PM₁₀ fraction analysis to verify the assumption that PM₁₀ emissions are 50% of total PM. (Construction Permit 98BO0259). Subsequent tests shall be conducted every five years, thereafter to verify the assumption that PM₁₀ emissions are 50% of PM. If the permittee elects to perform additional analyses due to the failure of a test to verify that PM₁₀ emissions are less than 50% of PM, approval and the schedule for, performing such additional tests shall be obtained from the Division. The permittee has relied upon PM₁₀ emissions being 50% or less of PM, in order to avoid major source non-attainment new source review (NSR) requirements. Failure to meet the 50% requirement and/or to exceed the permitted PM₁₀ emission limitations may result in the application of major source non-attainment area NSR requirements.

Cemex performed a stack test for PM and PM₁₀ in September 1999 observed by the Division. Results of the PM₁₀ were shown to be less than 50% of PM and under the previous version of this permit in effect for 2007 no additional testing for PM₁₀ was required. Cemex conducted stack tests in May 2006 establishing PM and PM₁₀ emission rates for each activity group in compliance with the permit limits. However, results of the 2006 stack test indicate that PM₁₀ emissions were above 50% of PM for AIRS Point P011 and P012. The PM₁₀ fraction was greater than the 50% level stated in the permit, however it was still well below the permit limit, with total particulate emissions still less than the permitted PM₁₀ limit. The results of the stack test show that actual PM is significantly less than PM assumed in modeling, and the actual PM₁₀ is less than 50% of the assumed PM (the modeled PM₁₀ level).

Cemex was issued a completeness letter April 4, 2011 for a minor modification to Operating Permit 96OPBO082 allowing grain loading standards from the table above for each baghouse to be used if a stack test has not yet been conducted on that baghouse and removing the assumption of PM₁₀ emissions are 50% of PM. Stack tests are now required every five years on a different baghouse for each activity group until all baghouses are tested within the activity group. Cemex performed PM/PM₁₀ stack testing for representative baghouses within each activity group in April-May 2011. Baghouse 625-14 (P009) showed PM emission rates of 0.038 gr/dscf in excess of the limit 0.03 gr/dscf represented in the table above during the April 21, 2011 test and was subsequently retested on May 25, 2011 achieving 0.01 gr/dscf. Cemex is currently using the emission factor from the May retest to calculate PM emissions. Source is in compliance with this condition.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouse shall be conducted in accordance with manufacturer's specifications and good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be

documented and maintained to be made available to the Division upon request. (Construction Permit 98BO0259)

Cemex's operation and maintenance plan for the bagfilter equipment was reviewed during a previous inspection. Cemex conducts and documents weekly inspections, maintenance items identified, corrective actions, and preventative maintenance activities performed, as required.

11.4 **Except for P015**, these sources are subject to the opacity limits set forth in Condition 21 of this permit.

Cemex is meeting the opacity requirements outlined in Condition 21, refer to Condition 22 for more detail.

11.5 The following particulate emission control measures shall be applied to particulate emission producing sources (Construction Permit 98BO0259).

11.5.1 If, at any time, the opacity of particulate emissions from the pile(s) exceeds ten (10) percent (as required under 40 CFR Part 63, Subpart LLL), then the pile(s) shall be watered at least once per day until a crust forms on the surface.

Roads and outdoor clinker storage piles are watered to control fugitive emissions. There was no outdoor storage of clinker during the inspection and no dust was observed from any stockpiles or roadways. A sweeper truck was observed operating at the plant site during the inspection. Source is in compliance with this condition.

11.5.2 The permittee shall operate a water truck during the day shift for 12 hours a day, 7 days a week to minimize fugitive particulate emissions from haul roads, loading areas, and edges of piles. Watering shall occur according to this schedule excluding periods of freezing conditions, snow/ice covered roads, rain or a shutdown of the kiln and crushing/drying system for greater than 24 hours. Additionally, haul roads shall be treated with chemical dust suppressants, as often as required, to maintain a surface crust. Such controls shall achieve a minimum control efficiency of 80%. Records of such application of dust suppressants and watering shall be maintained on site.

Cemex reported deviations in the SARs received 1/31/2012 and 7/30/2012 where the water truck was not operated for 12 hours per day on 5 days since the last inspection. Cemex operated a water truck at the plant following the above requirements almost 100% of the time with only the previously mentioned exceptions. Water truck records were reviewed onsite and indicated 12 hours of operation each day except when there was precipitation or freezing or when repairs on the vehicles was required (deviations). Records reviewed also indicate Cemex treats unpaved haul roads with calcium chloride multiple times per year. Source is in compliance with this condition.

11.5.3 Traffic on and around storage pile(s) shall be minimized.

Traffic on and around the storage piles appeared to be minimized during the inspection. Source is in compliance with this condition.

11.5.4 Height of fall material shall be minimized. Dust extractor used shall be in close proximity to the emission source.

The height of fall material appeared to be minimized during the inspection. Source is in compliance with this condition.

11.5.5 Vehicle traffic on unpaved surfaces shall be restricted to established roadways.

Plant vehicles and the front end loader were the only vehicles observed on unpaved roadways during the inspection. Source is in compliance with this condition.

11.5.6 Clinker shall be reclaimed from the storage pile(s) as soon as possible.

According to Cemex personnel, clinker is no longer routinely stored outdoors. No clinker was observed stored outside during the inspection. Source is in compliance with this condition.

11.5.7 Paved areas shall be kept clean using high efficiency sweeper(s) (vacuum trucks).

Water trucks and the vacuum sweeper vehicle were observed during the inspection and no visible emissions were observed from the paved areas. Cemex has two sweepers onsite with the old sweeper used as a backup. A truck wash system was installed as a pollution control device following a Memorandum of Understanding signed August 13, 2001 by Boulder County Health Department and Cemex. Cemex submitted a revised Fugitive Dust Control Plan to the Division on August 23, 2006 incorporating the use of the truck wash. Trucks are now channeled through the truck wash with barricades. Trucks were observed using the truck wash during the inspections and no evidence of vehicle carry out was observed. Source is in compliance with this condition.

11.5.8 Activities causing fugitive particulate matter emissions shall be suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60-minute period. Only those activities affected by wind speed, and for which it is possible to “suspend operation” need be shut down (i.e., the permittee cannot “shut down” storage piles, thus this condition would not apply to storage piles). Activities may continue when the average wind speed drops below 30 m.p.h. (Construction Permit 98BO0259, as modified per Section 1, Condition 1.3 of this permit)

Cemex monitors wind speed continuously. Records of suspended operations during high wind events are maintained. Records of suspended operations during times of dust complaints were reviewed during the inspection and Cemex has maintained that activities were discontinued during high wind events. Source is in compliance with this condition.

11.5.9 Spillages and other particulate matter accumulations shall be cleaned up with the least delay. The permittee shall operate a powered sweeper during day shift for 12 hours a day, 7 days a week to control accumulations on paved areas. Sweeping shall occur according to this schedule except under the following circumstances: wet pavement, snow/ice covered pavement, or shutdown of the kiln and crushing/drying system for greater than 24 hours.

The vacuum sweeper vehicle was observed on several occasions during the inspection and no visible emissions were observed from the paved areas. Cemex has two sweepers onsite with the old sweeper used as a backup. Vacuum sweeper truck records were reviewed onsite and indicated 12 hours of operation each day except when there was precipitation or freezing or when the sweeper was down for maintenance (6/27/2012 deviation). Generally, during sweeper truck downtime, the water truck is used to wet the paved areas; however, this was not the case during this deviation. Cemex affirms that fugitive dust controls were not compromised during the deviation. No corresponding complaints were received during that time. Cemex also has labor crews that manually clean up accumulated piles of material around the plant. Source is considered in compliance with this condition.

11.5.10 During the night shift, the Outdoor Clinker Discharge area shall be swept or watered at least every two hours while diverting clinker to the pit. Sweeping and watering will occur according to this schedule except under the following circumstances: Wet pavement, snow/ice covered pavement, or during a shutdown of the crushing/drying

and kiln system for greater than 24 hours. (Compliance Order on Consent 2002-124, Paragraph 39.b and c)

Cemex is sweeping and watering the outdoor clinker discharge area at night as required except during periods of precipitation or freezing or when the water truck and sweepers were down for maintenance on 8/1/11, 4/24/12, 5/21/12, and 6/22/12 (deviations reported in SAR). Cemex affirms that fugitive dust controls were not compromised during the deviations. Source is in compliance with this condition.

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when average wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h, averaged over a sixty minute period.

A wind speed monitor mounted to the top of the primary crusher building continuously records wind speeds and alerts the control room operators with an auditory alarm when the wind speed exceeds 30 mph. Records of suspended operations are maintained during high wind events. Source is in compliance with this condition.

A weekly inspection of the site shall be conducted to ensure the emission control elements are in place and effective. In addition, at any time when a fugitive dust problem is observed, the permittee shall take action to correct the problem. The permittee shall maintain records of the date and time of any fugitive dust problem observed, and the type and time of action taken to correct the problem. These records shall be maintained on site for inspection upon request.

Records of daily and weekly inspections were reviewed during the inspection. Cemex documents the operation of emission control equipment, the presence of visible emissions, and any corrective measures implemented. Source is in compliance with this condition.

11.6 These sources are subject to the MACT standard, as set forth in Conditions 23 of this permit.

Refer to Condition 23 for more detail.

11.7 The following sources are subject to the CAM requirements set forth in Condition 24 of this permit: S032, S024, S024A, S031, S065, S043, S044, S046, S047, S049 and S050.

Refer to Condition 24 for more detail.

12. P014 - Clinker and Fuel Handling S020 - Coal Silo, S025 - Coal/Clinker Hopper

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	12.1			Recordkeeping	Annually
PM & PM ₁₀	12.2		PM & PM ₁₀ : 2.9 x 10 ⁻⁵ lb/ton (for each baghouse stack)	Recordkeeping and Calculation	Annually
PM	12.3	See Condition 12.3		Control Equipment Maintenance	Annual Certification

Opacity	12.4	Shall not exceed 20%, except as provided for below		Inspection Method 9	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%, for a period or periods aggregating more than six (6) minutes in any 60 consecutive minutes		Baghouse Maintenance and Operation	
MACT Standard – Opacity	12.5	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every Five (5) Years Monthly to Annually

Compliance Status: **P014 - Clinker and Fuel Handling**
 S020 - Coal Silo
 S025 - Coal/Clinker Hopper

Parameter		Limitations	Reported Data
S020	Coal	No limit	65,769 tons
	PM	No limit	0.001 tons
	PM ₁₀	No limit	0.001 tons
S025	Clinker Hopper	No limit	65,769 tons
	PM	No limit	0.001 tons
	PM ₁₀	No limit	0.001 tons

Cemex provided the Reported Data above for the rolling 12-month period ending June 2012. The emissions are calculated based on daily, monthly and yearly material throughputs. Source is in compliance.

12.1 Records of the annual amounts of material throughput shall be maintained and made available to the Division for inspection upon request.

Cemex is tracking emissions on a monthly and annual basis, as required.

12.2 For APEN reporting and fee purposes, annual emissions shall be estimated using the actual annual material throughputs recorded under Condition 12.1, and the emission factor listed in the table above (AP-42, Section 11.6, January, 1995). This factor accounts for control equipment efficiency. Records of the calculations shall be maintained and made available for inspection upon request.

Cemex is calculating emissions using the raw material throughputs and the emission factor below, as required.

12.3 During any consecutive sixty (60) minute period, PM emissions shall not exceed the following limitation (Colorado Regulation No. 1, III.C.1.b):

$$E = 17.31 (P)^{0.16}$$

Where: E is the allowable particulate emissions in lbs/hr.
 P is the process weight rate in tons/hr

In absence of credible evidence to the contrary, compliance with the particulate matter limitation is presumed provided the baghouse operating and maintenance requirements of Condition 21 are met.

Cemex is following the baghouse operating & maintenance requirements, refer to Condition 21 for more detail.

12.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Cemex is meeting the opacity requirements outlined in Condition 21, refer to Condition 21 for more detail.

12.5 On and after the MACT compliance date of June 10, 2002, the clinker hopper is subject to the MACT standard, as set forth in Conditions 23 of this permit.

Refer to Condition 23 for more detail.

13. P007A - Handling and Processing of CKD And Raw Material Waste Dust

S001 – Waste Dust Silo, S066 – Cement Silo A5, 041 - Pug Mill, 042 - Truck Hauling to Lyons Quarry, S067 – CKD Loading Spout

Parameter	Permit Condition Number	Limitations		Emission Factors	Monitoring	
					Method	Interval
Process Rate	13.1	S001, S066, S067, 041	Total Quantity of Materials conveyed, CKD and benification dust, together, on a dry basis: 133,000 tons/year 600 tons/day		Recordkeeping	Monthly
		042	Total Quantity of material Hauled and Disposed, on Wet Basis: 173,403 tons/year 800 tons/day			
PM	13.2	S001, S066, S067	15.39 tons/year	See Condition 13.2	Stack Test S066 – Pressure Drop	Every Five (5) Years Daily
		041	2.66 tons/year	0.8 lb/ton	Recordkeeping and Calculation	Monthly
		042	5.50 tons/year		Recordkeeping and Calculation Emission Control Plan	Monthly

Parameter	Permit Condition Number	Limitations		Emission Factors	Monitoring	
					Method	Interval
PM ₁₀	13.2	S001, S066, S067	7.70 tons/year 69.5 lbs/day	50% of PM	Recordkeeping and Calculation	Monthly
		041	2.66 tons/year 24.00 lbs/day	0.2 lb/ton		
		042	2.50 tons/year 23 lbs/day		Recordkeeping and Emission Control Plan	Monthly
Hours of Operation	13.3	S001: 8064 hours/year S066 and S067: 3000 hours/year			Recordkeeping	Monthly
MACT Standard – Opacity	13.4	Shall not exceed 10%			Method 9 Method 22 O&M Plan	Every Five (5) Years Monthly to Annually
Opacity	13.5	Shall not exceed 20%, except as provided for below			Inspection Method 9 Baghouse O&M S066 – Pressure Drop	Daily As Needed
		Certain Operating Conditions - Shall not exceed 30%				Daily
Fugitive PM Emissions	13.6	Emission Control Plan			Inspection	Twice per Shift
Pit Restriction	13.7	Pit C Only			Compliance Certification	Annually
CAM	13.8	See Condition 24				

Compliance Status: **P007A - Handling and Processing of CKD And Raw Material Waste Dust**
S001 - Waste Dust Silo
S066 - Cement Silo A5
S067 - CKD Loading Spout
041 - Pug Mill
042 - Truck Hauling to Lyons Quarry

Parameter		Limitations	Reported Data
Process Rate	S001, S066, S067	133,000 tons/yr	48,037 tons
		600 tons/day	266 tons/day (Jan-12)
	041	133,000 tons/yr	35,610 tons
		600 tons/day	182.90 tons/day (Jan-12)
	042	173,403 tons/yr	44,513 tons
		800 tons/day	228.62 tons/day (Jan-12)
PM	S001, S066, S067	15.39 tons/yr	0.39 tons
	041	2.66 tons/yr	0.71 tons
	042	5.50 tons/yr	0.05 tons
PM ₁₀	S001, S066, S067	7.70 tons/yr	0.00 tons
		69.5 lbs/day	0.00 lbs/day
	041	2.66 tons/yr	0.71 tons
		24.00 lbs/day	7.32 lbs/day (Jan-12)

	042	2.50 tons/yr	0.02 tons
		23 lbs/day	0.23 lbs/day (Jan-12)
Hours of Operation	S001	8064 hours/yr	6,599 hours
	S066, S067	3000 hours/yr	82.85 hours

Cemex provided the Reported Data above for the rolling 12-month period ending June 2012. The emissions are calculated based on daily, monthly and yearly material throughputs. Source is in compliance with the above permit limits.

- 13.1 Process rates shall not exceed the maximum rates specified in the table above (Construction Permit 98BO0315). The quantity of materials handled shall be monitored and recorded monthly. Monthly quantities of material handled shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

The permittee shall maintain records of the number of operating days for each month. Note that if any unit in an emission group operates during a day, that day counts as a day of operation. Compliance with the daily process rates shall be monitored by dividing the monthly process rate by the number of days of operation. Records of the number of days of operation, the monthly and twelve month total handling rates and any other information used to determine the monthly quantities of material handled shall be maintained and made available for inspection upon request.

Cemex is maintaining the number of operating days per month and the process rates on a daily, monthly, and yearly basis, as required. Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of process rates from these sources. No exceedances of the permit limits are noted. Source is in compliance with this permit condition.

- 13.2 PM and PM₁₀ emission rates shall not exceed the limits listed in the table above. (Construction Permit 98BO0315).

For 041 compliance with the emission limitations shall be monitored by calculating PM and PM₁₀ emissions monthly using the above emission factors and the monthly quantity of materials processed. Control efficiencies of 95% for PM and 80% for PM₁₀ may be applied to the monthly calculations provided the pug mill and pelletizing machine are operated and maintained in accordance with manufacturer's recommendations and good engineering practices to provide a minimum moisture content of 20% water by weight.

Cemex is using the above emission factor and control efficiencies to calculate PM and PM₁₀ emissions from the pug mill and pelletizing machines which appeared to be operated and maintained according to manufacturer's recommendations and good engineering practices. Source is in compliance with this permit condition.

For S001, S066, and S067 compliance with the emission limits shall be monitored using the results of the stack test and the baghouse maintenance and operating procedures and pressure drop monitoring for S006 required in this Condition 13.2. Compliance with the emission limitations shall be monitored by calculating monthly emissions using the PM emission factor determined from the stack test (in gr/dscf), hours of operation (as required by Condition 13.7) and the maximum design flow rate of the control devices (see table below). PM₁₀ emissions shall be presumed to 50% of PM.

The emission factor for any baghouse within an emission group that has been performance tested shall be the results of the most recent test. In lieu of using individual test results for a baghouse, the most conservative test result from the most recent performance test for a baghouse within that same emission group may be used. The emission factor for any baghouse within an emission group that has not been performance tested, shall be the most conservative emission factor from the most recent performance test for any baghouse within that emission group that has been performance tested.

Cemex is using the maximum design flow rate of the control devices and emission rates determined from a stack test conducted on 4/7/11 to calculate the emissions and demonstrate compliance with the permit limits. Source is in compliance with this permit condition.

For S001, S066, S067 and 041, monthly emissions of PM and PM₁₀ shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of particulate emissions from these sources.

Baghouse ID	Baghouse Grain Loading (gr/dscf)	Design Flow Rate (ACFM)
BH 225-3	0.01	16,100
BH 525-28	0.01	3,800
BH 825-7	0.01	1,400

Compliance with the daily PM₁₀ limitations shall be monitored by dividing the monthly PM₁₀ emissions by the number of days the unit was operated. Note that if any unit within the emission group operates during a day, that day counts as a day of operation.

Cemex is maintaining the number of operating days each month and are used calculate daily PM₁₀ emissions, as required.

For 042, compliance with the emission limits shall be monitored by meeting the production limits set forth in Condition 13.1 and by applying the Fugitive Particulate Emission Control Plan set forth in Condition 13.6. For purposes of APEN reporting and payment of fees, annual PM and PM₁₀ emissions shall be calculated using the actual throughput rates recorded under Condition 13.1, appropriate emission factors and equations, and control efficiencies shall be used to estimate emissions. Records of the emissions calculations shall be maintained and made available for inspection upon request.

Cemex is calculating PM and PM₁₀ emissions from truck hauling using the appropriate emission factors and control efficiencies. Cemex is applying the measures of the Fugitive Particulate Emission Control Plan as required.

Stack Testing

Stack testing for PM from S001, S066 and S067 shall be performed as set forth in Condition 22 of this permit. Frequency of testing shall be every five years. Performance testing shall be conducted at a representative baghouse to demonstrate compliance with the grain loading (gr/dscf) requirements. Notwithstanding the above, a stack test shall be conducted to demonstrate compliance with the grain loading (gr/dscf) requirements for baghouses BH 525-28 and 225-3 within 180 days of renewal permit issuance [March 1, 2008]. The stack test shall be performed in accordance with the provisions set forth in Condition 22.

The first test (to performed within one year of initial permit issuance February 1, 2000) shall include a PM₁₀ fraction analysis to verify the assumption that PM₁₀ emissions are 50% of total PM. (Construction Permit 98BO0315) Subsequent tests shall be conducted every five years, thereafter to verify the assumption that PM₁₀ emissions are 50% of PM. If the permittee elects to perform additional analyses due to the failure of a test to verify that PM₁₀ emissions are less than 50% of PM, approval and the schedule for, performing such additional tests shall be obtained from the Division. The permittee has relied upon PM₁₀ emissions being 50% or less of PM, in order to avoid major source non-attainment area new source review (NSR) requirements. Failure to meet the 50% requirement and/or to exceed the permitted PM₁₀ emission limitations may result in the application of major source non-attainment area NSR requirements.

Cemex performed a stack test on 1/21/00 that demonstrated compliance with the permitted emission limits for PM and verified that PM₁₀ emissions were no more than 50% of the total particulate. Cemex performed another stack test on 5/16/06 of AIRS Pt 040; however, results of the 2006 stack test indicate that PM₁₀ emissions were above 50% of PM. The results of the stack test show that actual PM is significantly less than PM assumed in modeling, and the actual PM₁₀ is less than 50% of the assumed PM (the modeled PM₁₀ level). Stack tests on baghouses BH 525-28 and BH 225-3 were performed in June 2008 and demonstrate compliance with the grain loading (gr/dscf) requirements. A stack test was performed on BH 225-3 (waste dust silo) on April 7, 2011 demonstrating compliance with the PM grain loading (gr/dscf) limits and show PM-10 less than 50% of PM. Cemex was issued a completeness letter April 4, 2011 for a minor modification to Operating Permit 96OPBO082 allowing grain loading standards for each baghouse to be used if a stack test has not yet been conducted on that baghouse and removing the assumption of PM₁₀ emissions are 50% of PM. Stack tests are required every five years on a different baghouse for each activity group until all baghouses are tested within the activity group. Source is in compliance with this condition.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request. (Construction Permit 98BO0315)

Baghouses are inspected weekly and the differential pressure recorded to determine proper operation and ensure compliance with the emissions limit. Baghouses on the Waste Dust silo, and A5 cement silo were replaced during an outage in August 2007. Cemex has installed additional pressure monitor equipment on the control equipment and programmed the dust conveyance system to automatically shut down if positive pressure is detected. Source is in compliance with this condition.

S066 – Cement Silo A5 Pressure Drop Monitoring

The pressure drop across the inlet and outlet of the baghouse shall be recorded daily, when the silo is operating. Results of the daily reading will be recorded in a log book and made available for Division inspection upon request. A reading outside of the manufacturer's recommendation shall trigger the source to investigate the baghouse performance and make any repairs or adjustments necessary. A log of any repairs shall be maintained and made available upon request. The manufacturer's recommended pressure drop shall be maintained for Division inspection upon request. Note that the recording of the pressure drop readings is not required on days when the cement silo is not operating.

Differential pressures are recorded daily to determine proper operation and ensure compliance with the emissions limit. Source is in compliance with this condition.

- 13.3 Annual hours of operation shall not exceed the above limitations (Construction Permit 98BO0315, as modified under the provisions of Section I, Condition 1.3 to include the 3,000 hrs/yr limit for the A5 silo (S066) and CKD loadout spout (S067)). Hours of operation shall be monitored and recorded monthly. Monthly hours of operation shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data. Records of monthly and twelve month totals of operation hours shall be kept on-site and made available for inspection upon request.

Beginning with the issuance of the Operating Permit renewal on March 1, 2008, Cemex is maintaining a rolling 12-month total of hours of operation from these sources.

- 13.4 These sources are subject to the requirements of 40 CFR Part 63, Subpart LLL, National Emission Standards for Hazardous Air Pollutants from the Portland Cement Manufacturing Industry, as set forth in Condition 23 of this permit.

Refer to Condition 23 for more detail.

- 13.5 These sources, except for hauling and disposal, are subject to the opacity limits set forth in Condition 21 of this permit.

Source is in compliance with the opacity limits in Condition 21. Refer to Condition 21 for more detail.

- 13.6 The following particulate emission control measures shall be applied to particulate emission producing sources (Construction Permit 98BO0315 and Compliance Order on Consent 2002-124)

- 13.6.1 Transfer points shall be enclosed.

All transfer points appeared to be enclosed during the inspection.

- 13.6.2 Moisture content of the materials prior to transfer to pug mill shall be adequate to effectively control emissions.

Transfer of CKD from the pugmill was not observed during the inspection; however, no evidence of fugitive emissions was observed in the area around the transfer points. Source is in compliance with this condition.

- 13.6.3 The permittee shall operate a water truck during day shift for 12 hours a day, 7 days a week to minimize fugitive particulate emissions from haul roads, loading areas, and edges of piles. Watering shall occur according to this schedule excluding periods of freezing conditions, snow/ice covered roads, rain or a shutdown of the kiln and crushing/drying system for greater than 24 hours. Additionally, haul roads shall be treated with chemical dust suppressants, as often as required, to maintain a surface crust. Such controls shall achieve a minimum control efficiency of 80%. Records of such application of dust suppressants and watering shall be maintained at the site.

The haul road into the Lyon's Quarry disposal site (C Pit) appeared compacted and a water truck was operating. No fugitive dust issues from vehicle traffic were noted. Water truck logs and dust suppressant

application records were reviewed during the inspection and appeared to be complete. Source is in compliance with this condition.

- 13.6.4 At the disposal pit, the material shall be compacted and stabilized to minimize emissions.

During the inspection the material in the disposal site appeared compacted and stabilized. Source is in compliance with this condition.

- 13.6.5 Disturbed and exposed areas at the disposal site shall be minimized, and no more than seven (7) acres shall be used for disposal at a time. Fugitive particulate matter emissions from areas not being actively used shall be minimized using a shale cover.

Cemex has reduced the size of the active disposal to less than approximately one acre and the remaining area in the quarry was capped with material (waste shale) to control fugitive emissions. Source is in compliance with this condition.

- 13.6.6 A sprinkler system shall be in operation as much as necessary to effectively control emissions at the disposal area, and at a minimum be in operation when material is being dumped, and continue operating until emissions are suppressed.

The sprinkler system is set to water for 10 minutes every two hours. The sprinklers appear to be effectively controlling emissions from the disposal site. The disposal area was wet and no fugitive dust was observed during the inspection. Source is in compliance with this condition.

- 13.6.7 Haul trucks of 95 tons capacity shall be used to minimize the vehicle-miles traveled. Spillage and exposure to wind shall be minimized by restricting the material load to 75 percent of the volume capacity of the trucks.

Large haul trucks ("triple 7") were observed in the CKD loading area that are used to transport CKD into the disposal site. Minimization of vehicle-miles traveled occurs by reductions in the amount of material disposed. No material transport was observed during the inspection. Source is considered in compliance with this condition.

- 13.6.8 Spillages and other particulate matter accumulations shall be cleaned up with the least delay. The permittee shall operate a powered sweeper during the day shift for 12 hours a day, 7 days a week to control accumulations on paved areas. Sweeping shall occur according to this schedule except under the following circumstances: wet pavement, snow/ice covered pavement, or shutdown of the kiln and crushing/drying system for greater than 24 hours.

No evidence of material spillage was observed during the inspection. Sweeper trucks were observed in operation at the facility during the inspection. Source is in compliance with this permit condition.

- 13.6.9 Compliance inspections shall be conducted on a daily basis. Records of such inspections shall be maintained at the site.

Cemex conducts daily opacity inspection throughout the facility and documents observations as required. Source is in compliance with this condition.

- 13.6.10 Activities causing fugitive dust emissions shall be suspended when wind speeds reach or exceed 30 miles per hour, averaged over a 60-minute period. Only those activities affected by wind speed, and for which it is possible to “suspend operation@ need be shut down (i.e., the permittee cannot “shut down@ storage piles, thus this condition would not apply to storage piles). Activities may continue when the average wind speed drops below 30 m.p.h. (Incorporated directly into this operating permit per Section 1, Condition 1.3 of this permit)

Cemex monitors wind speed continuously. Records of suspended operations during high wind events are maintained. Records of suspended operations during times of dust complaints were reviewed during the inspection and Cemex has maintained that activities were discontinued during high wind events. Source is in compliance with this condition.

- 13.6.11 Operate an automated sprinkler system to water the active CKD disposal site.

13.6.11.1 Sprinklers will be set for 10 minutes or longer on each station. Cycle times will be set for at least one cycle every two hours, except CEMEX may reduce watering if the CKD disposal area is too wet for equipment operations.

13.6.11.2 The sprinklers will be positioned to cover 100% of the active CKD disposal area.

13.6.11.3 The sprinkler system will be in service from mid-April through mid-October each year, except during rain, snow, or freezing condition.

Cemex operates an automated sprinkler system set to water for 10 minutes every two hours. The sprinklers appear to be effectively controlling emissions from the disposal site. No fugitive dust was observed during the inspection. Source is in compliance with this condition.

- 13.6.12 Water trucks will be used to water the active CKD disposal area as follows:

13.6.12.1 The access road will be watered at least every three hours during the day, and as needed at night to minimize fugitive emissions. Watering may be reduced or suspended during cold weather if the road is ice covered and such ice cover is sufficient to minimize fugitive emissions.

13.6.12.2 When the sprinklers are not in service, water trucks will be used to water the active disposal area at least every 3 hours during the day, and as needed at night to minimize fugitive emissions.

13.6.12.3 Water truck operation as previously described will occur except in the following circumstances: freezing conditions, rain, or snow. As used here, the term “freezing conditions” means weather conditions severe enough to clog the water truck due to freezing. CEMEX shall take reasonable precautions, including but not limited to storing the water truck in a heated garage at night, to prevent such freezing conditions.

Cemex operates a water truck at the plant following the above requirements. Water truck records were reviewed onsite and indicated 12 hours of operation each day except when there was precipitation or freezing or

when repairs were needed. Water trucks have been observed watering the active disposal site during inspections when sprinklers were not in use. Source is in compliance with this condition.

13.6.13 CEMEX agrees to limit the active disposal or working area of the CKD storage pit to 3 acres at any time.

13.6.13.1 Inactive or unused portions of the pit shall be covered with rock or treated with hygroscopic materials to minimize fugitive emissions.

13.6.13.2 Signage or berms shall be used to delineate the 3 acre active disposal area.

Records of dust suppressant use shall be maintained and made available upon request.

Cemex has reduced the size of the active disposal to less than approximately one acre and the remaining area in the quarry was capped with material (waste shale) to control fugitive emissions. Cemex treats unpaved haul roads with calcium chloride approximately twice per year. Source is in compliance with this condition.

The permittee shall install, calibrate, and operate a wind speed instrument which will be used to alert personnel when average wind speeds reach or exceed 30 m.p.h. The permittee shall maintain records of those dates and times when wind speed reaches or exceeds 30 m.p.h, averaged over a sixty minute period.

A wind speed monitor mounted to the top of the primary crusher building continuously records wind speeds and alerts the control room operators with an auditory alarm when the wind speed exceeds 30 mph. Records of suspended operations are maintained during high wind events. Source is in compliance with this condition.

13.7 Only Pit "C" shall be used for disposal of CKD and Benefication Dust. The one-way haul distance is 0.38 mile. (Construction Permit 98BO0315).

Only C Pit is used for CKD and Benefication Dust disposal.

13.8 The following sources are subject to the CAM requirements set forth in Condition 24 of this permit: S001.

Refer to Condition 24 for more detail.

14. P016 - Haul Roads

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Haul Road Activity Information	14.1			Recordkeeping	Annually
PM &PM ₁₀	14.2			Calculation	Annually

14.1 Records of the annual amount of material hauled, and all other information necessary to estimate emissions, shall be maintained and made available to the Division for inspection upon request.

Records provided by Cemex show the source is tracking the number of vehicle miles traveled per year for both empty and loaded trucks, and calculating the tons of particulate matter using emission factors and an 80% control efficiency for the application of water as a dust suppressant. Source is in compliance with this permit condition.

- 14.2 For APEN reporting purposes, annual PM and PM₁₀ emissions shall be estimated using the records obtained under Condition 14.1, and appropriate emissions factors and/or equations and control efficiencies. Records of the calculations shall be kept on site for Division inspection upon request.

Cemex is calculating emissions on an annual basis as required. Source is in compliance with this permit condition.

NOTE: Measures to minimize fugitive emissions from the haul roads are included in the Particulate Matter Emission Control Plans listed in other Conditions of this Operating Permit.

- 14.3 The permittee shall treat haul roads with a magnesium chloride or calcium chloride solution as often as necessary to maintain a surface crust, as required in Section II, Condition 13.5.3 of this permit. Such solutions shall be applied to the haul road to the CKD disposal pit at least every six months. The solution shall have a concentration of at least 32 percent magnesium chloride or calcium chloride by weight. As an alternative, the permittee may substitute another hygroscopic material, as long as such treatment is at least as effective as the solution described in this condition in controlling fugitive emissions. (Compliance Order on Consent 2002-124, Paragraph 43)

Cemex treats unpaved haul roads with calcium chloride semiannually. Records reviewed during previous inspections indicate the dust suppressant solution has a calcium chloride concentration of 28-45% and treatments were applied twice per year. Calcium chloride was applied on the plant side on 3/21/2012 and 8/18/2012. Source is in compliance with this condition.

15. T001 - Gasoline Storage Tank, 3,000 Gallon Capacity

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Throughput Rate	15.1			Recordkeeping	Annually
VOC	15.2		13 lbs/1000 gallons	Calculation	Annually
RACT VOC	15.3	Submerged Fill, etc.		Compliance Certification	Annually

Compliance Status: T001 - Gasoline Storage Tank 3,000-Gallon Capacity

Parameter	Limitations	Reported Data
Throughput Rate	No limit	8,987 gallons
VOC	No limit	0.06 tons

Cemex is recording fuel throughput and calculating emissions on a monthly and rolling 12-month total basis. Cemex provided the reported data in the table above for the rolling 12-month period ending June 2012. Source is in compliance with this permit condition.

- 15.1 The permittee shall maintain records of the annual amount of gasoline throughput. Records shall be made available for inspection upon request.

Cemex is tracking fuel throughput and maintaining records on an annual basis, as required.

- 15.2 For APEN reporting and fee purposes, annual VOC emissions shall be estimated using the actual throughput rate and the emission factor listed above (AP-42, Table 5.2-7). Records of the emission calculations shall be made available for inspection upon request.

Cemex is calculating VOC emissions based on AP-42 emission factor of 13 lbs of VOC per 1000 gallons of fuel throughput, and reported on an annual basis as required. Source is in compliance with this permit condition.

- 15.3 This source is subject to the Reasonably Available Control Technology (RACT) requirements set forth in Colorado Regulation No. 7, VI.B.3, as follows.

- 15.3.1 The permittee shall not allow the transfer of petroleum liquid from any delivery vessel into this tank unless the tank is equipped with a submerged fill pipe and the vapors displaced from the storage tank during filling are processed by a vapor control system. A vapor balance system shall be deemed "approved" if its design and operation are in accordance with the applicable requirements of Appendices A and B of Colorado Regulation No. 7. Tanks equipped with a submerged fill pipe shall meet the specifications of Appendix A of Colorado Regulation No. 7.

- 15.3.2 The vapor control system shall include one or more of the following:

- 15.3.2.1 A vapor-tight line from the storage tank to delivery vessel (i.e. an approved control system).

- 15.3.2.2 A refrigerator-condensation system or equivalent designed to recover at least 90 percent by weight of the organic compounds in the displaced vapor.

- 15.3.3 The permittee shall ensure that operating procedures are used so that gasoline cannot be transferred into the tank unless the vapor control system is in use.

- 15.3.4 The vapor balance system shall meet the specifications of Appendix A of Colorado Regulation No. 7.

- 15.3.5 The vapor balance system and the vapor control system shall meet the requirements of Section XV of Colorado Regulation No. 7.

Cemex gasoline storage tank is equipped with a submerged fill pipe and a vapor return hose that is connected to the tanker truck to capture the vapors displaced by the tank filling procedure. Fuel delivery was not observed during the inspection, however, previous inspections reviewed the standard operating procedure for Cemex employees is to observe the fuel delivery and verify the use of the vapor return lines to minimize emissions. Source is in compliance with this permit condition.

16. Opacity - Daily Plant Walk Through

A daily plant walk through shall be performed to look for visible emissions. During the walk through, an observer will survey the plant from 5 observation points. From these 5 locations, all of the facility's baghouses and material transfer points can be observed. The results of these observations shall be logged on a daily observations sheet.

If visible emissions are observed from any stack, the permittee shall undertake the appropriate corrective process and/or maintenance actions as soon as practicable. When these actions are completed, that stack will be observed again. If visible emissions persist, the permittee shall perform a Method 9 test of that stack. Subject to the provisions of C.R.S. 15-7-123 and in the absence of credible evidence to the contrary, exceedance of the limit shall be considered to exist from the time a Method 9 reading is taken that shows an exceedance of the opacity limit until a Method 9 reading is taken that shows the opacity is less than the opacity limit.

The corrective actions taken and the follow-up observations shall be noted on the daily observation sheet. The daily observation sheets shall be maintained and made available for inspection upon request.

If visible emissions from fugitive sources are noted, the permittee shall investigate to insure that the provisions of the appropriate fugitive dust control plan are being implemented, and shall note on the observation sheet, along with additional actions taken.

The daily walk through applies to remote locations of the facility, i.e. Dowe Flats Quarry and the conveyor. Daily visible emission observations are not required at these remote locations on days when operations are not occurring at these locations.

These procedures do not apply to the kiln and clinker cooler, which are equipped with opacity monitors.

Sources subject to CAM requirements (see Condition 24) are subject to visible emission observations as specified in the CAM Plan in Appendix H of this permit.

Cemex maintains records of daily inspections for fugitive dust emissions and documentation was reviewed onsite indicating when emissions are observed and corrective actions implemented. Source is in compliance with this condition.

17. Insignificant Activities

The permittee shall at least annually inspect and determine whether the categorically insignificant activities are in compliance with the opacity standard and are exempt from the current regulatory APEN reporting requirements. The permittee shall record in a log the results of this inspection.

Cemex conducts annual evaluations of the insignificant activities to determine compliance with opacity standards and verify APEN status. Documentation of the insignificant activity evaluation was provided. Source is in compliance with this permit condition.

18. Reporting

All reports shall be postmarked within the first thirty (30) days immediately following the end of the reporting period, unless a different response time is identified elsewhere in this permit. The monitoring and permit deviation report shall be in the format identified in Appendix B of this permit.

Cemex submitted all required reports since the last inspection to the Division within the thirty-day time period, as required. Source is in compliance with this permit condition.

19. Good Operating Practices

At all times, including periods of startup, shutdown, and malfunction, owners and operators shall, to the extent practicable, maintain and operate any affected facility including associated air pollution control equipment in a manner consistent with good air pollution control practice for minimizing emissions. Determination of whether acceptable operating and maintenance procedures are being used will be based on information available to the Division which may include, but is not limited to, monitoring results, opacity observations, review of operating and maintenance procedures, and inspection of the source. (40 CFR Part 60, 60.11(d)).

No article, machine, equipment or process shall be used to conceal an emission which would otherwise constitute a violation of an applicable standard. Such concealment includes, but is not limited to, the use of gaseous diluents to achieve compliance with an opacity standard or with a standard which is based on the concentration of a pollutant in the gases discharged to the atmosphere. (40 CFR Part 60, Subpart A, 60.12)

Cemex is operating and maintaining equipment consistent with good air pollution control practices. Cemex has undertaken several operational and equipment improvements to the facility since 2006. Source is in compliance with this condition.

20. Opacity and PM Limits for Subpart OOO Sources

40 CFR Part 60, Subpart OOO, Standards of Performance for Non-Metallic Mineral Processing Plants, as adopted by reference in Colorado Regulation No. 6, Part A, limits emissions as follows:

- 20.1 The permittee shall not cause to be discharged into the atmosphere any stack emissions which exhibit greater than 7 percent opacity.

Monitoring

A six (6) minute EPA Method 9 opacity observation shall be conducted semi-annually for the primary crusher bagfilter and a representative bagfilter for the conveyor. Method 9 observations shall be conducted in accordance with Appendix A of 40 CFR Part 60. Subject to the provisions of C.R.S. 15-7-123 and in the absence of credible evidence to the contrary, exceedance of the limit shall be considered to exist from the time a Method 9 reading is taken that shows an exceedance of the opacity limit until a Method 9 reading is taken that shows the opacity is less than the opacity limit.

This semi-annual test is in addition to the daily observations set forth under Condition 16 of this permit.

The results of the semi-annual Method 9 readings and a copy of the Method 9 reader's certification shall be kept on site and made available to the Division upon request. Copies of any observations exceeding the applicable standard shall be submitted with the next scheduled monitoring and permit deviation report required by Appendix B of this permit.

Cemex has performed the semi-annual Method 9 readings for the primary crusher and representative bagfilter on the conveyor as required. The testing was most recently conducted on 11/14/11 (crusher), 12/1/12 (conveyor) and 6/6-7/12 and demonstrates compliance with the opacity limit. Source is in compliance with this condition.

- 20.2 The permittee shall not cause to be discharged into the atmosphere any stack emissions which contain particulate matter in excess of 0.05 gram per dry standard cubic meter.

Stack Testing

Stack testing for PM shall be performed as set forth in Condition 22 of this permit. Frequency of testing shall be every five years. Performance test shall be conducted on the primary crusher bagfilter and one representative bagfilter stack for the conveyor.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

This PM and opacity standard apply at all times except during periods of startup, shutdown, and malfunction (40 CFR Part 60, 60.11 (c)), however, at all times, the permittee shall use good operating practices as required by Condition 19 of this permit. The permittee shall maintain records of the occurrence and duration of any startup, shutdown, or malfunction in operation of this source; and any malfunction of the air pollution control equipment (40 CFR Part 60, Subpart A, 60.7(b), as adopted by reference in Colorado Regulation No. 6, Part A).

Cemex performed PM stack testing for the primary crusher and representative baghouse for the conveyor on June 18-19, 2008 demonstrating compliance with the 0.05 grams per dry standard cubic meter (0.022 gr/dscf). The results of the 2008 test indicate 0.0008 gr/dscf for the primary crusher (BH 125-1) and 0.0007 gr/dscf for a representative conveyor baghouse (BH 125-3). Source is in compliance with this condition.

21. Opacity Limits

These limits apply only to those sources, which are referred to this Condition throughout this permit.

- 21.1 Except as provided in Condition 21.2, below, no owner or operator of a source shall allow or cause the emission into the atmosphere of any air pollutant which is in excess of 20% opacity. This standard is based on 24 consecutive opacity readings taken at 15-second intervals for six minutes. The approved reference test method for visible emissions measurement is EPA Method 9 (40 CFR Part 60, Appendix A (July, 1992)) in all subsections of Section II.A of Regulation No. 1. (Colorado Regulation No. 1, II.A.1).

During the inspection, the facility and Dowe Flats quarry were operating normally. Upsets reported on 10/31/11, 11/12/11, 11/13-14/11, 12/10/11, 3/16/12, and 6/8/12 where opacity was observed were reviewed and

appear to be valid malfunctions based on information received from the source. Source is in compliance with this condition.

21.2 No owner or operator of a source shall allow or cause to be emitted into the atmosphere any air pollutant resulting from the building of a new fire, cleaning of fire boxes, soot blowing, start-up, any process modification, or adjustment or occasional cleaning of control equipment, which is in excess of 30% opacity for a period or periods aggregating more than six minutes in any sixty consecutive minutes (Colorado Regulation No. 1, Section II.A.4).

Complaints of dust on the patio from the conveyor from Dowe Flats have prompted changes to the way the system is operated during shutdown. The 02 system (Dowe Flats conveyor system) has been significantly repaired/rebuilt during previous shutdowns. The SOP was rewritten to ensure rock is run-out or quarry personnel informed of material on the belt so that quarry employee can be positioned at stacker with start-up to adjust foam. The conveyor system and dust collector baghouses were reprogrammed so that they will dump on a fully loaded belt and run out before the system is shut down for the day. During the inspection, a startup of the 02 system occurred with new hi cal limestone being delivered to the patio. Approximately 30 seconds of opacity was observed before the plant side foam system caught up controlling emissions from the material dropped on the pile. No Method 9 opacity observation was attempted. Source is in compliance with this condition.

Compliance with these limits shall be monitored by meeting the requirements of Condition No. 16 of this permit, and by meeting the Bagfilter Operation and Maintenance requirements specified in this Condition 21.

Compliance with the 30% limit set forth in Condition 21.2 shall be monitored by conducting visual emission observations in accordance with EPA Method 9, if any of the activities listed in Condition 21.2 occurs continuously for one hour or more. A reading shall be conducted within one hour and ten minutes of commencement of any of the above activities and every 1 hour thereafter during the activity.

If any observation indicates an exceedance of the limit, additional observations shall be performed. Subject to the provisions of C.R.S. 25-7-123.1 and in the absence of credible evidence to the contrary, exceedance of the limit shall be considered to exist from the time a Method 9 reading is taken that shows an exceedance of the opacity limit until a Method 9 reading is taken that shows the opacity is less than the opacity limit.

The permittee shall maintain records of the day, time and length for which any listed activity occurs, and the type of activity. These records, results of Method 9 readings, and a copy of the Method 9 reader's certification, shall be maintained and made available to the Division for inspection upon request. Copies of observations exceeding the applicable standard shall be submitted with the next scheduled report.

Cemex maintains records of daily inspections for visible emissions and documentation was reviewed onsite indicating when emissions are observed and corrective actions implemented. Source is in compliance with this condition.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and/or good engineering practices. These procedures

shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

Cemex tracks all maintenance activities with a preventative maintenance schedule, daily walk through inspection reports, and work orders issued to correct any deficiencies noted.

S066 – Cement Silo A5 Pressure Drop Monitoring

The pressure drop across the inlet and outlet of the baghouse shall be recorded daily, when the silo is operating. Results of the daily reading will be recorded in a log book and made available for Division inspection upon request. A reading outside of the manufacturer's recommendation shall trigger the source to investigate the baghouse performance and make any repairs or adjustments necessary. A log of any repairs shall be maintained and made available upon request. The manufacturer's recommended pressure drop shall be maintained for Division inspection upon request. Note that the recording of the pressure drop readings is not required on days when the cement silo is not operating.

Differential pressures are recorded daily when the source is operating to determine proper operation and ensure compliance with the emissions limit. Source is in compliance with this condition.

22. PM Stack Testing

This requirement applies only to those sources, which are referred to this condition throughout this permit. Performance testing for particulate emissions shall be performed in accordance with the requirements and procedures set forth in EPA Test Method 5 as set forth in 40 CFR Part 60, Appendix A. Frequency of testing shall be as specified for those sources which are referred to this condition.

A stack testing protocol shall be submitted for Division approval at least thirty (30) calendar days prior to performance of the test required under this condition. No stack test required herein shall be performed without prior written approval of the protocol by the Division. The Division reserves the right to witness the test. In order to facilitate the Division's ability to make plans to witness the test, notice of the date(s) for the stack tests shall be submitted to the Division at least thirty (30) calendar days prior to the test. The Division may for good cause shown, waive this thirty (30) day notice requirement. In instances when a scheduling conflict is presented, the division shall immediately contact the permittee in order to explore the possibility of making modifications to the stack test schedule. The required number of copies of the compliance test results shall be submitted to the Division within forty-five (45) calendar days of the completion of the test unless a longer period is approved by the Division.

Emission points P002, P007, P008, P009, P010, P011, P012, P013, P007A are subject to this condition. Cemex has submitted testing protocols to the Division prior to the testing and stack tests have been conducted for each emission point as required. Source is in compliance with this condition.

23. National Emission Standards for Hazardous Air Pollutants From the Portland Cement Manufacturing Industry

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Compliance Date	23.1	See Condition 23.1			
Circumvention	23.2	See Condition 23.2			

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
O&M Requirements	23.3	See Condition 23.3			
Startup, Shutdown, & Malfunction Plan	23.4	See Condition 23.4			
General Monitoring Requirements	23.5	See Condition 23.5			
Recordkeeping Requirements	23.6	See Condition 23.6			
Reporting Requirements	23.7	See Condition 23.7			
Emission standards and operating limits	23.8.1.1	Kiln: 0.30 lb PM/ton feed (dry basis)		Performance test O&M Plan	Once every five years
	23.8.1.2	Kiln D/F emission limit - See Condition 23.8.1.2		Performance test Temperature monitoring Inspection O&M Plan	Once every 30 months* Annually
	Kiln operating limits - See Conditions 23.8.1.3 thru 23.8.1.7				
	23.8.2	Clinker Cooler: 0.10 lb/ton feed (dry basis)		Performance test O&M Plan	Once every five years
Opacity	23.9.1	Kiln: Shall not exceed 20%		COM	Continuous
	23.9.2	Clinker cooler: Shall not exceed 10%			
	23.9.3	Raw mill: Shall not exceed 10%		Method 9 Method 22 O&M Plan	Once every five years Daily
	23.9.4	Other sources: Shall not exceed 10%		Method 9 Method 22 O&M Plan	Once every five years Monthly to Annually

*In addition, for periodic monitoring purposes (Colorado Regulation No. 3, Part C, Section V.C.5.b), testing for D/F shall be conducted 45 days after burning tire derived fuel.

This source is subject to the requirements of 40 CFR Part 63, Subpart LLL - National Emission Standards for Hazardous Air Pollutants From the Portland Cement Manufacturing Industry, as follows.

23.1 Compliance Date

The compliance date for an existing source to meet the requirements of good combustion practices (GCP) for total hydrocarbons (THC) is December 20, 2007. (63.1351(c))

Cemex submitted certification of compliance with the requirements of Subpart LLL in a Semi-annual Certification report received by the Division on 07/31/02. Cemex is meeting the requirements of good

combustion practices for total hydrocarbons in Subpart LLL. Source is in compliance with this permit condition.

23.2 Circumvention

No owner or operator subject to the provisions of this part shall build, erect, install, or use any article, machine, equipment, or process to conceal an emission that would otherwise constitute noncompliance with a relevant standard. Such concealment includes, but is not limited to: the use of diluents to achieve compliance with a relevant standard based on the concentration of a pollutant in the effluent discharged to the atmosphere; the use of gaseous diluents to achieve compliance with a relevant standard for visible emissions; and the fragmentation of an operation such that the operation avoids regulation by a relevant standard. (63.4(b))

Source is in compliance with this condition.

23.3 Operation and maintenance requirements

23.3.1 At all times, including periods of startup, shutdown, and malfunction, the owner or operator must operate and maintain any affected source, including associated air pollution control equipment and monitoring equipment, in a manner consistent with safety and good air pollution control practices for minimizing emissions. During a period of startup, shutdown, or malfunction, this general duty to minimize emissions requires that the owner or operator reduce emissions from the affected source to the greatest extent which is consistent with safety and good air pollution control practices. The general duty to minimize emissions during a period of startup, shutdown, or malfunction does not require the owner or operator to achieve emission levels that would be required by the applicable standard at other times if this is not consistent with safety and good air pollution control practices, nor does it require the owner or operator to make any further efforts to reduce emissions if levels required by the applicable standard have been achieved. Determination of whether such operation and maintenance procedures are being used will be based on information available to the Administrator which may include, but is not limited to, monitoring results, review of operation and maintenance procedures (including the startup, shutdown, and malfunction plan required in paragraph (e)(3) of this section), review of operation and maintenance records, and inspection of the source. (63.6(e)(1)(i))

Cemex appears to be operating the facility and control equipment in accordance with good air pollution practices. Source is in compliance with this condition.

23.3.2 Malfunctions must be corrected as soon as practicable after their occurrence. To the extent that an unexpected event arises during a startup, shutdown, or malfunction, an owner or operator must comply by minimizing emissions during such a startup, shutdown, and malfunction event consistent with safety and good air pollution control practices. (63.6(e)(1)(ii))

Records reviewed onsite indicate Cemex has promptly responded to events in accordance with good air pollution practices. Source appears to be in compliance with this condition.

- 23.3.3 Operation and maintenance requirements established pursuant to section 112 of the Act are enforceable independent of emissions limitations or other requirements in relevant standards. (63.6(e)(1)(iii))
- 23.3.4 The permittee shall prepare for each affected source subject to this subpart, a written operations and maintenance plan. The plan shall be submitted for Division review and approval, and shall include the information set forth in 63.1350. (63.1350(a)) Failure to comply with any provision of the operations and maintenance plan developed in accordance with 63.1350(a) shall be a violation of the standard. (63.1350(b))

Cemex submitted a revised operations and maintenance plan to the Division on 02/09/05 that incorporated changes required by previous inspections. Source is in compliance with this condition.

23.4 Startup, shutdown, and malfunction plan

The owner or operator of an affected source must develop a written startup, shutdown, and malfunction plan that describes, in detail, procedures for operating and maintaining the source during periods of startup, shutdown, and malfunction; and a program of corrective action for malfunctioning process, air pollution control, and monitoring equipment used to comply with the relevant standard. (63.6(e)(3)) The plan shall be developed, maintained and made available to the Division in accordance with the requirements set forth in 63.6(e)(3) and the permittee shall follow the recordkeeping and reporting requirements set forth in 63.6(e)(3).

Cemex has developed and implemented a startup, shutdown, and malfunction (SSM) plan. A revised SSM plan dated 12/12/06 was submitted to address minimizing emissions from air pollution control equipment identified in previous inspections. Cemex has submitted semiannual reports indicating all actions have been consistent with the SSM plan. Source is in compliance with this condition.

23.5 General monitoring requirements

Monitoring shall be conducted as set forth in 63.8 and the relevant standards. The owner or operator of an affected source shall maintain and operate each continuous monitoring system (CMS) as specified in 63.8, or in the relevant standard, and in a manner consistent with good air pollution control practices. The owner or operator of an affected source that is required to use a CMS and is subject to the monitoring requirements of 63.8 and a relevant standard shall develop and implement a CMS quality control program, as set forth in 63.8(d). When required by a relevant standard, and at any other time the Administrator may require under section 114 of the Act, the owner or operator of an affected source being monitored shall conduct a performance evaluation of the CMS, as set forth in 63.8(e). The owner or operator of each CMS shall reduce the monitoring data as specified in 63.8(g), and in addition, as set forth in Subpart LLL. (63.8(a-e) and (g))

Cemex has developed and implemented a CMS quality control plan for the continuous monitoring system. A completely revised CMS quality control plan dated 4/15/09 was submitted to incorporate the new Nexus DAS system installation. Source is in compliance with this condition.

23.6 Recordkeeping requirements

The permittee shall maintain files of all information (including all reports and notifications) required by 63.1355 recorded in a form suitable and readily available for inspection and review

as required by 63.10(b)(1). The files shall be retained for at least five years following the date of each occurrence, measurement, maintenance, corrective action, report, or record. At a minimum, the most recent two years of data shall be retained on site. The remaining three years of data may be retained off site. The files may be maintained on microfilm, on a computer, on floppy disks, on magnetic tape, or on microfiche. (63.1355(a))

The owner or operator shall maintain records for each affected source as required by 63.10(b)(2) and (3), and the items listed in 63.1355(b). (63.1355(b))

In addition, to the recordkeeping requirements in 63.1355(b), the owner or operator of an affected source equipped with a CMS shall maintain all records required by 63.10(c). (63.1355(c))

You must keep annual records of the amount of CKD which is removed from the kiln system and either disposed of as solid waste or otherwise recycled for a beneficial use outside of the kiln system. (63.1355(d))

You must keep records of all fly ash supplier certifications as required by §63.1350(o). (63.1355(f))

Cemex has maintained records and information as required since the last inspection. Source is in compliance with this condition.

23.7 Reporting requirements

The permittee shall submit reports of the results of performance tests, results of opacity or visible emission observations, progress reports, and periodic startup, shutdown, and malfunction reports in accordance with the procedures set forth in 63.10(d) and 63.1354.

Additional reporting requirements for sources with CMS

For each CMS, the permittee shall report the results of CMS performance evaluations, excess emissions and CMS performance report and summary report, and continuous opacity monitoring system data produced during a performance test in accordance with the procedures set forth in 63.10(e) and 63.1354, except that the excess emissions and CMS performance summary reports shall be submitted on a quarterly basis, based on the operating permit reporting dates.

Cemex has submitted the results of all performance tests, opacity readings, startup, shutdown, malfunctions, and semi-annual excess emission reports as required. Source is in compliance with this condition.

23.8 Non-opacity emissions standards and operating limits

Standards for kilns and in-line kiln/raw mills(P007 - Kiln/Calcliner)

23.8.1 No owner or operator of an existing kiln or an existing kiln/raw mill located at a facility that is a major source subject to the provisions of this subpart shall cause to be discharged into the atmosphere from these affected sources, any gases which:

23.8.1.1 Contain particulate matter (PM) in excess of 0.15 kg per Mg (0.30 lb per

ton) of feed (dry basis) to the kiln. When there is an alkali bypass associated with a kiln or in-line kiln/raw mill, the combined particulate matter emissions from the kiln or in-line kiln/raw mill and the alkali bypass are subject to this emission limit. (63.1343(b)(1))

Cemex conducted a stack test in May 2006 and demonstrated compliance with the PM emission limit of 0.30 lbs/ton feed into the kiln, the results of the stack test were 0.142 lbs PM per ton feed. A stack test conducted on June 19, 2008 also confirmed compliance with the 0.30 lbs/ton of feed rate limit, with a result of 0.069 lb/ton of feed. Source is in compliance with this condition.

23.8.1.2 Contain D/F in excess of:

- a. 0.20 ng per dscm (8.7×10^{-11} gr per dscf) (TEQ); or
- b. 0.40 ng per dscm (1.7×10^{-10} gr per dscf) (TEQ) when the average of the performance test run average temperatures at the inlet to the particulate matter control device is 204 °C (400 °F) or less. (63.1343(b)(3))

Operating limits for kilns and in-line kiln/raw mills

23.8.1.3 The owner or operator of a kiln subject to a D/F emission limitation under §63.1343 (Condition 23.8.1.2) must operate the kiln such that the temperature of the gas at the inlet to the kiln particulate matter control device (PMCD) and alkali bypass PMCD, if applicable, does not exceed the applicable temperature limit specified in paragraph (b) of this section. The owner or operator of an in-line kiln/raw mill subject to a D/F emission limitation under §63.1343 must operate the in-line kiln/raw mill, such that:

- a. When the raw mill of the in-line kiln/raw mill is operating, the applicable temperature limit for the main in-line kiln/raw mill exhaust, specified in paragraph (b) of this section and established during the performance test when the raw mill was operating is not exceeded.
- b. When the raw mill of the in-line kiln/raw mill is not operating, the applicable temperature limit for the main in-line kiln/raw mill exhaust, specified in paragraph (b) of this section and established during the performance test when the raw mill was not operating, is not exceeded.
- c. If the in-line kiln/raw mill is equipped with an alkali bypass, the applicable temperature limit for the alkali bypass specified in paragraph (b) of this section and established during the performance test, with or without the raw mill operating, is not exceeded. (63.1344(a))

23.8.1.4 The temperature limit for affected sources meeting the limits of paragraph (a) of this section or paragraphs (a)(1) through (a)(3) of this section is determined in accordance with §63.1349(b)(3)(iv). (63.1344(b))

23.8.1.5 Existing kilns and in-line kilns/raw mills must implement good

combustion practices (GCP) designed to minimize THC from fuel combustion. GCP include training all operators and supervisors to operate and maintain the kiln and calciner, and the pollution control systems in accordance with good engineering practices. The training shall include methods for minimizing excess emissions. (63.1344(f))

23.8.1.6 No kiln and in-line kiln/raw mill may use as a raw material or fuel any fly ash where the mercury content of the fly ash has been increased through the use of activated carbon, or any other sorbent unless the facility can demonstrate that the use of that fly ash will not result in an increase in mercury emissions over baseline emissions (i.e. emissions not using the fly ash). The facility has the burden of proving there has been no emissions increase over baseline. (63.1344(g))

23.8.1.7 All kilns and in-line kilns/raw mills must remove (i.e. not recycle to the kiln) from the kiln system sufficient cement kiln dust to maintain the desired product quality. (63.1344(h))

Cemex has conducted several stack tests to demonstrate compliance with the D/F emissions requirements at specific temperatures. The test dates, temperatures, and D/F results are summarized in the table below. The most recent D/F stack test was performed and passed on 1/29/10. Cemex is performing quarterly calibrations of each thermocouple as required. Dates of thermocouple calibrations since the last inspection occurred on 3/1/12 and 6/1/12. Cemex has programmed into the FLS control system an automatic shutdown of the kiln when temperatures are approaching the temperature limit at each baghouse to prevent temperature exceedances. Kiln operators receive annual training as required. No fly ash has been used as raw material or fuel inside the kiln since the last inspection. The alkali bypass side of the Hart baghouse is no longer used in the process. Source is in compliance with this condition.

Date	Kiln Baghouse Inlet Temp °F	Hart/Alkali By-Pass Baghouse Inlet Temp °F		Dioxin & Furan Emissions TEQ ng/dscm @ 7% O ₂ Limit = 0.20 TEQ ng/dscm
6/4-5/2002 Coal Fired Only	496	476		0.00740
11/13-14/2002 Coal Fired Only	517	509		0.01200
11/20-21/2002 Coal & Tire Derived Fuels	512	513		0.00019
3/29-30/2005 Coal Fired Only	517	505		0.00001
12/19-21/2005 Coal Fired Only	556	516		0.00330
7/25-31/2007 Coal Fired Only	523	485	526 - Hart baghouse	0.00038
1/28-29/2010 Coal Fired Only	545	N/A	551 – Hart baghouse	0.00
6/20-21/2012	554	N/A	547 – Hart baghouse	0.00046

Cemex has not reported any upsets regarding D/F temperature exceedances since the last inspection. Source is in compliance with this condition.

Standards for clinker coolers (P008 - Clinker Cooler)

23.8.2 No owner or operator of a new or existing clinker cooler at a facility which is a major source subject to the provisions of this subpart shall cause to be discharged into the atmosphere from the clinker cooler any gases which contain particulate matter in excess of 0.050 kg per Mg (0.10 lb per ton) of feed (dry basis) to the kiln. (63.1345(a)(1))

Cemex conducted a stack test on April 6, 2011 and demonstrated compliance with the PM emission limit of 0.10 lbs/ton feed into the clinker cooler, the results of the stack test were 0.03 lbs PM per ton feed. Source is in compliance with this permit condition.

Compliance with nonopacity emission standards

Compliance with nonopacity emission standards will be determined as set forth in 63.6(f).

The non-opacity emission standards set forth in this part shall apply at all times except during periods of startup, shutdown, and malfunction, and as otherwise specified in Subpart LLL. If a startup, shutdown, or malfunction of one portion of an affected source does not affect the ability of particular emission points within other portions of the affected source to comply with the non-opacity emission standards set forth in this part, then that emission point must still be required to comply with the non-opacity emission standards and other applicable requirements. (63.6(f)(1)).

Cemex is demonstrating compliance with this condition based on the results of performance test conducted for particulate matter and the operation of continuous opacity monitor. Source is in compliance with this condition.

Performance testing requirements

The permittee shall demonstrate initial compliance with these emission limits using the test methods and procedures in paragraph (b) of 63.1349 and 40 CFR Part 63, Subpart A, 63.7. Performance test results shall be documented in complete test reports that contain the information required by 63.1349(a)(1) through (a)(10). The plan to be followed during testing shall be made available to the Division prior to testing. (63.1349(a))

The permittee shall perform such tests within 180 days after the compliance date. (63.7(a)(2))

The owner or operator of an affected source must notify the Division in writing of his or her intention to conduct a performance test at least 60 calendar days before the performance test is initially scheduled to begin to allow the Division to review and approve the site-specific test plan required under 63.7(c) and to have an observer present during the test. In the event the owner or operator is unable to conduct the performance test on the date specified in this notification due to unforeseeable circumstances beyond his or her control, the owner or operator must notify the Division as soon as practicable and without delay prior to the scheduled performance test date and specify the date when the performance test is rescheduled. (63.7(b)(1) and (2), and 63.9(e))

Performance tests for PM shall be conducted in accordance with the requirements in 63.1349(b)(1) and shall be repeated every five years (except as provided for below), except that the owner or operator of a kiln, in-line kiln/raw mill or clinker cooler is not required to repeat the initial performance test of opacity for the kiln, in-line kiln/raw mill or clinker cooler. (63.1349(c))

Performance tests for D/F shall be conducted in accordance with the requirements in 63.1349(b)(3) and shall be repeated every 30 months (except as provided for below). (63.1349(d))

If a source plans to undertake a change in operations that may adversely affect compliance with an applicable D/F standard under this subpart, the source must conduct a performance test and establish new temperature limit(s) as specified in 63.1349(b)(3). (63.1349(e)(1))

If a source plans to undertake a change in operations that may adversely affect compliance with an applicable PM standard under §63.1343 (Conditions 23.8.1.1 and 23.8.2), the source must conduct a performance test as specified in 63.1349(b)(1). (63.1349(e)(2))

In preparation for and while conducting a performance test required in 63.1349(e)(1), a source may operate under the planned operational change conditions for a period not to exceed 360 hours, provided that the conditions in 63.1349(e)(3)(i) through (iv) are met. The source shall submit temperature and other monitoring data that are recorded during the pretest operations. (63.1349(e)(3))

When tire derived fuel is to be burned in the kiln, a performance test shall be conducted within 45 days of initiating the change to tires. (Colorado Regulation No. 3, Part C, Section V.C.5.b)

Stack tests performed to demonstrate compliance with the MACT standard may be used to comply with any other stack tests requirements in this permit for the same pollutant from the same stack that would otherwise be required that year or permit term.

Cemex has conducted the performance tests as required and demonstrated compliance with the emission limit. D/F performance tests have been completed every 30 months as required (see above table); however, the Division granted an extension for the test originally scheduled for October 2009 and conducted in January 2010 because of the kiln being down due to market conditions. The most recent D/F stack test was performed and passed on 6/21/12. No significant changes to feed or fuel type have changed since the performance tests were conducted. Source is in compliance with this permit condition.

Quality assurance program

Submission of site-specific test plan: Before conducting a required performance test, the owner or operator of an affected source shall develop and submit a site-specific test plan to the Division for approval, as set forth in 63.7(c)(1), (2) and (3). The plan shall be submitted at least 60 days before the performance test is scheduled to take place. (63.7(c)(1-3))

Performance test method audit program: The owner or operator shall analyze performance audit (PA) samples during each performance test, as set forth in 63.7(c)(4). (63.7(c)(4))

Performance testing facilities: At the request of the Division, the owner or operator shall provide performance testing facilities as set forth in 64.7(d). (63.7(d))

Conduct of performance tests: Performance tests shall be conducted as set forth in 63.7(e). (63.7(e))

Data analysis, recordkeeping and reporting: Unless otherwise specified in a relevant standard or test method, or as otherwise approved by the Administrator in writing, results of a performance

test shall include the analysis of samples, determination of emissions, and raw data. A performance test is “completed” when field sample collection is terminated. The owner or operator of an affected source shall report the results of the performance test to the Division before the close of business on the 60th day following the completion of the performance test, unless specified otherwise in a relevant standard. The results of the performance test shall be submitted as part of the notification of compliance status required under 63.9(h). For a minimum of 5 years after a performance test is conducted the owner or operator shall retain and make available, upon request, for inspection by the Division the records of results of such performance test and other data needed to determine emissions from an affected source. (63.7(g))

Cemex has submitted the required test protocols to the Division for approval as required. Cemex conducts performance audits, analyzes the results of the audits, maintains and submits records to the Division as required. Source is in compliance with this condition.

Monitoring requirements

Kiln and in-line kiln/raw mill - D/F emission limit (P007 - Kiln/Calciner)

The owner or operator shall install, calibrate, maintain, and continuously operate a continuous monitor to record the temperature of the exhaust gases from the kiln, in-line kiln/raw mill and alkali bypass, if applicable, at the inlet to, or upstream of, the kiln, in-line kiln/raw mill and/or alkali bypass PM control devices in accordance with 63.1350(f)(1) through (6). (63.1350(f))

In addition, the owner or operator of any kiln or in-line kiln/raw mill subject to a D/F emission limit shall conduct an inspection of the components of the combustion system of each kiln or in-line kiln/raw mill at least once per year. (63.1350(i))

For kilns and in-line kilns/raw mills complying with the requirements in Section 63.1344(g) (Condition 23.8.1.6), each owner or operator must obtain a certification from the supplier for each shipment of fly ash received to demonstrate that the fly ash was not derived from a source in which the use of activated carbon, or any other sorbent, is used as a method of mercury emissions control. The certification shall include the name of the supplier and a signed statement from the supplier confirming that the fly ash was not derived from a source in which the use of activated carbon, or any other sorbent, is used as a method of emission control. (63.1350(o))

If the facility opts to use a fly ash derived from a source in which the use of activated carbon, or any other sorbent, is used as a method of mercury emissions control and demonstrate that the use of this fly ash does not increase mercury emissions, they must obtain daily fly ash samples, composites monthly, and analyze the samples for mercury. (63.1350(p))

Records reviewed during the inspection indicate that Cemex is recording the temperature of the kiln exhaust gas continuously and calibrated the thermocouples on a quarterly basis, as required. The dates of certification for the calibration equipment were up to date and are valid for one-year terms. Thermocouple calibrations since the last inspection were performed on 3/1/12 and 6/1/12. Source is in compliance with this condition.

23.9 Opacity standards

Standards for kilns and in-line kiln/raw mills (P007 - Kiln/Calciner)

- 23.9.1 No owner or operator of an existing kiln or an existing kiln/raw mill located at a facility that is a major source subject to the provisions of this subpart shall cause to be discharged into the atmosphere from these affected sources, any gases which exhibit opacity greater than 20 percent. (63.1343(b)(2))

Standards for clinker coolers (P008 - Clinker Cooler)

- 23.9.2 No owner or operator of a new or existing clinker cooler at a facility which is a major source subject to the provisions of this subpart shall cause to be discharged into the atmosphere from the clinker cooler any gases which exhibit opacity greater than ten percent. (63.1345(a)(2))

Standards for raw and finish mills (P005 - Raw Mill Grinding, P011 - Finish Mill P012 - Separator)

- 23.9.3 The owner or operator of each new or existing raw mill or finish mill at a facility which is a major source subject to the provisions of this subpart shall not cause to be discharged from the mill sweep or air separator air pollution control devices of these affected sources any gases which exhibit opacity in excess of ten percent. (63.1347)

Standards for affected sources other than kilns, in-line kiln/raw mills, clinker coolers, new and reconstructed raw material dryers, and raw and finish mills (P002 - Raw Materials Drying, P004 – Raw Materials Storage, P006 – Homogenizing and Blending, P009 - Clinker/Gypsum Silos, P010 – Sheltered Clinker Storage, P015 – Outdoor Clinker Storage, P013 - Cement Silos/Packhouse/Loadout, P014 – Clinker Hopper, P007A – Handling and Processing of CKD and Raw material Waste Dust, P050 – Cement Rail Car Unloading and Handling)

- 23.9.4 The owner or operator of each new or existing raw material, clinker, or finished product storage bin; conveying system transfer point; bagging system; and bulk loading or unloading system; and each existing raw material dryer, at a facility which is a major source subject to the provisions of this subpart shall not cause to be discharged any gases from these affected sources which exhibit opacity in excess of ten percent. (63.1348)

Cemex is monitoring opacity continuously using a COM for both the kiln and clinker cooler and performs visible emission observations during daily walkthroughs of the facility. An upset reported on 10/31/11 was related to an instrument technician working on the control system inadvertently bringing down the kiln ID fan by choosing to download changes resulting in the kiln system going positive. Fugitive emissions escaped through cracks and openings. The software has been reprogrammed to limit the option to download changes to the pLC to prevent reoccurrence. Upset events on 11/12/11, 11/13-14/11, and 12/10/11 were caused by blown off bags in individual compartments in the kiln baghouse creating opacity exceedances. The process kept running, although at lower feed and fuel, to isolate the baghouse compartments. The malfunctions occurred after aggressive rebagging in the kiln baghouse despite average opacities still being below the standard. According to Cemex, a third check has been added to verify bag installations by a party not involved in the installation. Cooler opacity events on 11/19/11 and 11/25/11 occurred after kiln burning zone temperature cooled allowing unclinkered fine raw material to enter the clinker cooler overwhelming the cooler baghouse. Cooler vent fan speed was immediately dropped along with kiln feed rate. Cemex reported deviations in the semiannual reports related to an upset event on 3/16/12 causing opacity from cracks and openings within the system upstream of the baghouses lasting approximately 22 minutes. The event was related to an input shaft on the kiln ID fan reducer

shearing causing the equipment to cease operation and the system to go positive. The console operator immediately shut off fuel but continued feed to safely stop the kiln system and avoid any temperature exceedances. Based on the responses to the malfunctions and actions to prevent reoccurrence the source appears to be in compliance with this condition.

Compliance with opacity and visible emission standards

Compliance with the opacity and visible emission standards in this part will be determined as set forth in 63.6(h), except for (h)(5)(ii-iv). (Subpart LLL specifies the test duration.)

The opacity and visible emission standards set forth in this part must apply at all times except during periods of startup, shutdown, and malfunction, and as otherwise specified in Subpart LLL. If a startup, shutdown, or malfunction of one portion of an affected source does not affect the ability of particular emission points within other portions of the affected source to comply with the opacity and visible emission standards set forth in this part, then that emission point shall still be required to comply with the opacity and visible emission standards and other applicable requirements. (63.6(h)(1))

Source is in compliance with this condition.

Performance testing requirements

The permittee shall demonstrate initial compliance with these emission limits using the test methods and procedures in paragraph (b) of 63.1349 and 40 CFR Part 63, Subpart A, 63.7. Performance test results shall be documented in complete test reports that contain the information required by 63.1349(a)(1) through (a)(10). The plan to be followed during testing shall be made available to the Division prior to testing. (63.1349(a))

The permittee shall perform such tests within 180 days after the compliance date. (63.7(a)(2))

The owner or operator of an affected source must notify the Division in writing of his or her intention to conduct a performance test at least 60 calendar days before the performance test is initially scheduled to begin to allow the Division to review and approve the site-specific test plan required under 63.7(c) and to have an observer present during the test. In the event the owner or operator is unable to conduct the performance test on the date specified in this notification due to unforeseeable circumstances beyond his or her control, the owner or operator must notify the Division as soon as practicable and without delay prior to the scheduled performance test date and specify the date when the performance test is rescheduled. (63.7(b)(1) and (2), and 63.9(e))

Performance tests for opacity shall be conducted in accordance with the requirements in 63.1349(b)(2) and shall be repeated every five years, except that the owner or operator of a kiln, in-line kiln/raw mill or clinker cooler is not required to repeat the initial performance test of opacity for the kiln, in-line kiln/raw mill or clinker cooler. (63.1349(c))

As of the date of the inspection, Cemex had demonstrated initial compliance with the requirements of conducting the performance testing specified above. Source is in compliance with this condition.

Quality assurance program

The permittee shall follow the quality assurance program requirements set forth above for the nonopacity emission standards.

Cemex appears to be following the QA/QC plan as required. Source is in compliance with this condition.

Notification of opacity and visible emission observations

The owner or operator of an affected source shall notify the Division in writing of the anticipated date for conducting the opacity or visible emission observations specified in 63.6(h)(5) and Subpart LLL, if such observations are required for the source by the relevant standard. The notification shall be submitted with the notification of the performance test date, as specified in 63.9(e), or if no performance test is required or visibility or other conditions prevent the opacity or visible emission observations from being conducted concurrently with the initial performance test required under 63.7, the owner or operator shall deliver or postmark the notification not less than 30 days before the opacity or visible emission observations are scheduled to take place. (63.9(f))

Source is in compliance with this permit condition.

Monitoring requirements

Kiln and in-line kiln/raw mill (P007 - Kiln/Calcliner)

The owner or operator of a kiln or in-line kiln/raw mill shall monitor opacity at each point where emissions are vented from these affected sources including alkali bypasses as follows. The owner or operator shall install, calibrate, maintain, and continuously operate a COM located at the outlet of the PM control device to continuously monitor the opacity. The COM shall be installed, maintained, calibrated, and operated as required by 40 CFR Part 63, Subpart A, and according to PS-1 of 40 CFR Part 60, Appendix B. To remain in compliance, the opacity must be maintained such that the 6-minute average opacity for any 6-minute block period does not exceed 20 percent. If the average opacity for any 6-minute block period exceeds 20 percent, this shall constitute a violation of the standard. (63.1350(c)(1) and (3))

Clinker cooler (P008 - Clinker Cooler)

The owner or operator of a clinker cooler shall monitor opacity at each point where emissions are vented from the clinker cooler as follows. The owner or operator shall install, calibrate, maintain, and continuously operate a COM located at the outlet of the PM control device to continuously monitor the opacity. The COM shall be installed, maintained, calibrated, and operated as required by 40 CFR Part 63, Subpart A, and according to PS-1 of 40 CFR Part 60, Appendix B. To remain in compliance, the opacity must be maintained such that the 6-minute average opacity for any 6-minute block period does not exceed 10 percent. If the average opacity for any 6-minute block period exceeds 10 percent, this shall constitute a violation of the standard. (63.1350(d)(1) and (3))

Cemex maintains a COM for both the kiln and clinker cooler and performs visible emission observations during daily walkthroughs of the facility. Source is in compliance with this permit condition.

Raw mill or finish mill (P005 - Raw Mill Grinding, P011 - Finish Mill P012 - Separator)

The owner or operator of a raw mill or finish mill shall monitor opacity by conducting daily visual emissions observations of the mill sweep and air separator PMCD of these affected sources in accordance with the procedures of Method 22 of appendix A to part 60 of this chapter. The Method 22 test shall be conducted while the affected source is operating at the

representative performance conditions. The duration of the Method 22 test shall be 6 minutes. If visible emissions are observed during any Method 22 visible emissions test, the owner or operator must:

- (1) Initiate, within one-hour, the corrective actions specified in the site specific operating and maintenance plan developed in accordance with 63.1350(a) (Condition 23.2.4 of this permit); and
- (2) Within 24 hours of the end of the Method 22 test in which visible emissions were observed, conduct a followup Method 22 test of each stack from which visible emissions were observed during the previous Method 22 test. If visible emissions are observed during the followup Method 22 test from any stack from which visible emissions were observed during the previous Method 22 test, conduct a visual opacity test of each stack from which emissions were observed during the follow up Method 22 test in accordance with Method 9 of appendix A to part 60 of this chapter. The duration of the Method 9 test shall be 30 minutes. (63.1350(e))

Affected sources other than kilns, in-line kiln/raw mills, clinker coolers, new and reconstructed raw material dryers, and raw and finish mills (P002 - Raw Materials Drying, P004 - Raw Materials Storage, P006 - Homogenizing and Blending, P009 - Clinker/Gypsum Silos, P010 - Sheltered Clinker Storage, P015 - Outdoor Clinker Storage, P013 - Cement Silos/Packhouse/Loadout, P014 - Clinker Hopper, P007A - Handling and Processing of CKD and Raw material Waste Dust, P050 - Cement Rail Car Unloading and Handling

For these sources, the permittee shall monitor opacity in accordance with the operation and maintenance plan developed in accordance with 63.1350(a) (Condition 23.2.4 of this permit). (63.1350(j))

Cemex is conducting daily and monthly Method 22 visible emission tests and conducting thirty minute Method 9 tests for each emission point where opacity was observed. Records reviewed during the inspection indicate that Cemex is monitoring opacity for other affected sources in accordance with the O&M Plan developed. Source is in compliance with this condition.

Additional notification requirements for sources with Continuous Monitoring Systems

The owner or operator of an affected source required to use a CMS by a relevant standard shall furnish the Division written notification as follows:

- (1) notification of the date the CMS performance evaluation is scheduled to begin, submitted simultaneously with the notification of the performance test date required under 63.7(b). If no performance test is required, or if the requirement to conduct a performance test has been waived for an affected source under 63.7(h), the owner or operator shall notify the Division in writing of the date of the performance evaluation at least 60 calendar days before the evaluation is scheduled to begin;
- (2) a notification that COMS data results will be used to determine compliance with the applicable opacity emission standard during a performance test required by 63.7 in lieu of Method 9 or other opacity emissions test method data, as allowed by 63.6(h)(7)(ii), if compliance with an opacity emission standard is required for the source by a relevant standard. The notification shall be submitted at least 60 calendar days before the performance test is scheduled to begin;

(3) a notification that the criterion necessary to continue use of an alternative to relative accuracy testing, as provided by 63.8(f)(6), has been exceeded. The notification shall be delivered or postmarked not later than 10 days after the occurrence of such exceedance, and it shall include a description of the nature and cause of the increased emissions. (63.9(g))

Cemex has submitted written notification to the Division prior to CMS testing and performance testing as required. Source is in compliance with this permit condition.

Temporary, conditioned exemption from particulate matter and opacity standards (P007 - Kiln/Calcliner)

Subject to the limitations specified in 63.1357(b) through (f), an owner or operator conducting PM CEMS correlation tests (that is, correlation with manual stack methods) is exempt from the following:

- (1) Any particulate matter and opacity standards of 40 CFR Part 60 or 63 that are applicable to cement kilns and in-line kiln/raw mills.
- (2) Any permit or other emissions or operating parameter or other limitation on workplace practices that are applicable to cement kilns and in-line kiln raw mills to ensure compliance with any particulate matter and opacity standards of 40 CFR Part 60 or 63.

Cemex is monitoring opacity continuously using a COM, and no PM CEMS correlation tests are being used. Source is in compliance with this condition.

24. Compliance Assurance Monitoring (CAM)

24.1 The Compliance Assurance Monitoring (CAM) requirements in 40 CFR Part 64, as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV, apply to the sources listed below with respect to the PM/PM₁₀ and Pb limitations identified in the Conditions listed below, as follows:

Source	Condition/Limit
P002 – Raw Material Drying	
S005 – Raw Materials Drying	Condition 5.4 - 22.8 tons/year PM/PM ₁₀ 6.5 lbs/hour PM ₁₀ Condition 5.5 - 1.6 tons/year Pb
P007 & P008	
S016 – Precalciner Kiln	Condition 10.7 - 133 tons/year PM/PM ₁₀ (Kiln and Clinker)
S018 – Clinker Cooler	Condition 10.4 - 0.275 lb PM/PM ₁₀ /ton feed Condition 10.10 - 4.4 tons/year Pb (Kiln and Cooler)
P009 – Glinker/gypsum silos and weigh feeders	
S032 – Bottom of A Frame Transfer	Condition 11.3 - 7.82 tons/year PM
S024 - #2 Clinker Silo	
P010 – Sheltered Clinker Storage	
S024A – Top of A Frame	Condition 11.3 - 21.96 tons/year PM
S0??- #6 Reclaim Feeder	10.98 tons/year, 201 lb/day PM ₁₀
P011 – Cement Finish Mill and Auxiliaries and Lime Handling	
S036 – Finish Mill	Condition 11.3 – 17.05 ton PM/year

Source	Condition/Limit
S037 – Finish Mill Auxillary Dust Collector	8.65 ton PM ₁₀ /year 48 lbs PM ₁₀ /day
P012 – Separator	
S065 – Separator	Condition 11.3 - 21.52 ton PM/year 10.76 ton PM ₁₀ /year 107 lbs PM ₁₀ /day
P013 – Cement Storage Silos/Packhouse/Loadout	
S043 – Cement Silos A10 and A13	Condition 11.3 – 12.3 ton PM/year 6.2 ton PM ₁₀ /year 43 lbs PM ₁₀ /day
S044 – Cement Silo A7	
S046 – East Load Spout	
S047 – West Load Spout	
S049 & S050 – Packhouse	
P007A – Handling & Processing of CKD & Raw Material Waste Dust	
S001 – Waste Dust Silo	Condition 13.2 - 15.39 tpy PM 7.7 tpy, 69.5lbs/day PM ₁₀

24.1.1 The permittee shall follow the CAM Plan provided in Appendix H of this permit. Excursions for purposes of reporting are as follows

24.1.1.1 For Visible Emissions:

- a. All stacks except kiln: any calendar day (midnight to midnight) in which visible emissions are observed, or
- b. All stacks except kiln: any calendar day (midnight to midnight) in which the equipment was operating but no visible emission observation was conducted.
- c. Kiln stack: Any six (6) minute average opacity reading from the COMS at or above 15% opacity, or
- d. Kiln stack: Any instance where the six (6) minute average opacity readings from the COMS increases and remains consistently equal to or greater than 10% opacity for a period exceeding eight (8) hours.

24.1.1.2 For Pressure Differential:

- a. All stacks: any calendar day (midnight to midnight) in which the pressure drop is at or below 0 or above 7 inches of water.
- b. All stacks: any calendar day (midnight to midnight) in which the equipment was operating but a pressure differential reading was not recorded.

Excursions shall be reported as required by Section IV, Conditions 21 and 22.d of this permit.

24.1.2 Operation of Approved Monitoring

- 24.1.2.1 At all times, the owner or operator shall maintain the monitoring, including but not limited to, maintaining necessary parts for routine repairs of the monitoring equipment (40 CFR Part 64 § 64.7(b), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- 24.1.2.2 Except for, as applicable, monitoring malfunctions, associated repairs, and required quality assurance or control activities (including, as applicable, calibration checks and required zero and span adjustments), the owner or operator shall conduct all monitoring in continuous operation (or shall collect data at all required intervals) at all times that the pollutant-specific emissions unit is operating. Data recorded during monitoring malfunctions, associated repairs, and required quality assurance or control activities shall not be used for purposes of these CAM requirements, including data averages and calculations, or fulfilling a minimum data availability requirement, if applicable. The owner or operator shall use all the data collected during all other periods in assessing the operation of the control device and associated control system. A monitoring malfunction is any sudden, infrequent, not reasonably preventable failure of the monitoring to provide valid data. Monitoring failures that are caused in part by poor maintenance or careless operation are not malfunctions (40 CFR Part 64 § 64.7(c), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- 24.1.2.3 Response to excursions or exceedances
- a. Upon detecting an excursion or exceedance, the owner or operator shall restore operation of the pollutant-specific emissions unit (including the control device and associated capture system) to its normal or usual manner of operation as expeditiously as practicable in accordance with good air pollution control practices for minimizing emissions. The response shall include minimizing the period of any startup, shutdown or malfunction and taking any necessary corrective actions to restore normal operation and prevent the likely recurrence of the cause of an excursion or exceedance (other than those caused by excused startup or shutdown conditions). Such actions may include initial inspection and evaluation, recording that operations returned to normal without operator action (such as through response by a computerized distribution control system), or any necessary follow-up actions to return operation to within the indicator range, designated condition, or below the applicable emission limitation or standard, as applicable (40 CFR Part 64 § 64.7(d)(1), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
 - b. Determination of whether the owner or operator has used acceptable procedures in response to an excursion or exceedance will be based on information available, which may include but is not limited to, monitoring results, review of operation and maintenance procedures and records, and inspection of the control device, associated capture system, and the process (40 CFR Part 64 § 64.7(d)(2), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.2.4 After approval of the monitoring required under the CAM requirements, if the owner or operator identifies a failure to achieve compliance with an emission limitation or standard for which the approved monitoring did not provide an indication of an excursion or exceedance while providing valid data, or the results of compliance or performance testing document a need to modify the existing indicator ranges or designated conditions, the owner or operator shall promptly notify the Division and, if necessary submit a proposed modification for this permit to address the necessary monitoring changes. Such a modification may include, but is not limited to, reestablishing indicator ranges or designated conditions, modifying the frequency of conducting monitoring and collecting data, or the monitoring of additional parameters (40 CFR Part 64 § 64.7(e), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.3 Quality Improvement Plan (QIP) Requirements

24.1.3.1 Based on the results of a determination made under the provisions of Condition 24.1.2.3.b, the Division may required the owner or operator to develop and implement a QIP (40 CFR Part 64 § 64.8(a), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.3.2 The owner or operator shall maintain a written QIP, if required, and have it available for inspection (40 CFR Part 64 § 64.8(b)(1), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.3.3 The QIP initially shall include procedures for evaluating the control performance problems and, based on the results of the evaluation procedures, the owner or operator shall modify the plan to include procedures for conducting one or more of the following actions, as appropriate:

- a. Improved preventative maintenance practices (40 CFR Part 64 § 64.8(b)(2)(i), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- b. Process operation changes (40 CFR Part 64 § 64.8(b)(2)(ii), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- c. Appropriate improvements to control methods (40 CFR Part 64 § 64.8(b)(2)(iii), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- d. Other steps appropriate to correct control performance (40 CFR Part 64 § 64.8(b)(2)(iv), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- e. More frequent or improved monitoring (only in conjunction with one or more steps under Conditions 24.1.3.3.a through d above) (40 CFR Part 64 § 64.8(b)(2)(v), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.3.4 If a QIP is required, the owner or operator shall develop and implement a QIP as expeditiously as practicable and shall notify the Division if the period for

completing the improvements contained in the QIP exceeds 180 days from the date on which the need to implement the QIP was determined (40 CFR Part 64 § 64.8(c), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.3.5 Following implementation of a QIP, upon any subsequent determination pursuant to Condition 24.1.2.3.b, the Division or the U.S. EPA may require that an owner or operator make reasonable changes to the QIP if the QIP is found to have:

- a. Failed to address the cause of the control device performance problems (40 CFR Part 64 § 64.8(d)(1), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV); or
- b. Failed to provide adequate procedures for correcting control device performance problems as expeditiously as practicable in accordance with good air pollution control practices for minimizing emissions (40 CFR Part 64 § 64.8(d)(2), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.3.6 Implementation of a QIP shall not excuse the owner or operator of a source from compliance with any existing emission limitation or standard, or any existing monitoring, testing, reporting or recordkeeping requirement that may apply under federal, state, or local law, or any other applicable requirements under the federal clean air act (40 CFR Part 64 § 64.8(e), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.4 Reporting and Recordkeeping Requirements

24.1.4.1 Reporting Requirements: The reports required by Section IV, Condition 22.d, shall contain the information specified in Appendix B of the permit and the following information, as applicable:

- a. Summary information on the number, duration and cause (including unknown cause, if applicable), for monitor downtime incidents (other than downtime associated with zero and span or other daily calibration checks, if applicable) ((40 CFR Part 64 § 64.9(a)(2)(ii), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV); and
- b. The owner or operator shall submit, if necessary, a description of the actions taken to implement a QIP during the reporting period as specified in Condition 24.1.3 of this permit. Upon completion of a QIP, the owner or operator shall include in the next summary report documentation that the implementation of the plan has been completed and reduced the likelihood of similar levels of excursions or exceedances occurring (40 CFR Part 64 § 64.9(a)(2)(iii), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.4.2 General Recordkeeping Requirements: In addition to the recordkeeping requirements in Section IV, Condition 22.a through c.

- a. The owner or operator shall maintain records of any written QIP required pursuant to Condition 24.1.3 and any activities undertaken to implement a QIP, and any supporting information required to be maintained under these CAM requirements (such as data used to document the adequacy of monitoring, or records of monitoring maintenance or corrective actions) (40 CFR Part 64 § 64.9(b)(1), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- b. Instead of paper records, the owner or operator may maintain records on alternative media, such as microfilm, computer files, magnetic tape disks, or microfiche, provided that the use of such alternative media allows for expeditious inspection and review, and does not conflict with other applicable recordkeeping requirements (40 CFR Part 64 § 64.9(b)(2), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

24.1.5 Savings Provisions

- 24.1.5.1 Nothing in these CAM requirements shall excuse the owner or operator of a source from compliance with any existing emission limitation or standard, or any existing monitoring, testing, reporting or recordkeeping requirement that may apply under federal, state, or local law, or any other applicable requirements under the federal clean air act. These CAM requirements shall not be used to justify the approval of monitoring less stringent than the monitoring which is required under separate legal authority and are not intended to establish minimum requirements for the purposes of determining the monitoring to be imposed under separate authority under the federal clean air act, including monitoring in permits issued pursuant to title I of the federal clean air act. The purpose of the CAM requirements is to require, as part of the issuance of this Title V operating permit, improved or new monitoring at those emissions units where monitoring requirements do not exist or are inadequate to meet the requirements of CAM (40 CFR Part 64 § 64.10(a)(1), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).
- 24.1.5.2 Nothing in these CAM requirements shall restrict or abrogate the authority of the U.S. EPA or the Division to impose additional or more stringent monitoring, recordkeeping, testing or reporting requirements on any owner or operator of a source under any provision of the federal clean air act, including but not limited to sections 114(a)(1) and 504(b), or state law, as applicable (40 CFR Part 64 § 64.10(a)(2), as adopted by reference in Colorado Regulation No. 3, Part C, Section XIV).

Cemex is following the CAM procedures identified above. Cemex performs daily visible emission observations and pressure differential readings on the CAM listed sources and records the results in a daily log. For the kiln and clinker cooler, daily COM reports with 6-minute averages are printed and added to the daily records. Cemex has reported excursions from the CAM requirements in the semi-annual deviation reports submitted on 1/31/12 and 7/30/12. The excursions reported relate to pressure drop readings on baghouses being out of range where no visible emissions are observed, equipment malfunctions causing opacity, and bag failures inside baghouses. Other excursions involved preventative maintenance activities that inadvertently caused kiln opacities. These excursions represent less than 1.0%

of total monitoring required and were not indicative of control equipment performance. To date, the Division has not requested Cemex develop and implement a Quality Improvement Plan (QIP) based upon the corrective action response and information available for each excursion. The Division will continue to monitor excursions from the CAM rule and may require a QIP be developed in the future. Source is considered to be in compliance with this condition.

25. Fugitive Particulate Emissions

The permittee shall use a water truck and truck wash to minimize fugitive particulate emissions from roads and other traffic areas, loading areas, the edges of clinker piles, and other sources of fugitive particulate emissions. At a minimum:

- 25.1 During the day shift, the permittee will operate the plant based water truck on a full time basis, 12 hours a day, 7 days a week. Operation of the water truck will be the sole assignment of the individual assigned such duty. Watering will occur according to this schedule except under the following circumstances: Freezing conditions, snow/ice covered roads, rain, or during a shutdown of the crushing/drying system and the kiln system for greater than 24 hours. As used here, the term “freezing conditions” means weather conditions severe enough to clog the water truck due to freezing. The permittee shall take reasonable precautions, including but not limited to storing the water truck in a heated garage at night, to prevent such freezing conditions. (Compliance Order on Consent 2002-124, Paragraph 41.a)
- 25.2 The water truck shall be operated during nights as necessary to water such areas adequately to control particulate emissions. (Compliance Order on Consent 2002-124, Paragraph 41.b)
- 25.3 The permittee shall install and operate a truck wash system to minimize tracking out of any materials. The truck wash system shall be used to wash cement spillage off of cement transport trucks before the trucks leave the facility. Operation of the truck wash is not required when ambient temperatures are such that use of the truck wash creates a safety hazard due to ice formation and when the truck wash is non-operational. When the truck wash is not in use, the permittee shall use alternate methods of removing cement spillage from the trucks before they leave the facility. The permittee shall keep records of the time periods when the truck wash is not used and the reason it is not being used and shall make such records available to the Division upon request. The truck wash shall be repaired as soon as practicable after break-downs.

Cemex employs a water truck that operates continuously throughout the day. Records were reviewed onsite and indicate the water truck operates for 12 hours each day except during periods of precipitation or freezing conditions with the following exceptions. Cemex has reported deviations related to days when the plant water truck was run for less than 12 hours on 6/9/12 and 6/22/12. On 6/9/12, the plant water truck was down for 5.5 hours that day because the water truck operator had a family emergency while on 6/22/12 both water trucks were down for repairs since the quarry water wagon is often used to cover the plant water truck when it is down. No complaints received by the Division have corresponded with these events. The water truck was observed onsite during the inspection and no visible emissions were observed from the paved areas. Cemex is operating a truck wash, as required, and trucks are channeled through the truck wash with barricades after loading with cement. Trucks were observed using the truck wash during the inspection and no evidence of vehicle carry out was observed. Source is in compliance with this condition.

26. Fugitive or Excess Emissions Observations or Complaints

The permittee shall document all reported observations or complaints from citizens, inspectors, contractors, or employees of fugitive or excess emissions. The permittee will investigate each occurrence and will document its findings and any corrective action taken or implemented. (Compliance Order on Consent 2002-234, Paragraph 44)

Cemex performs daily visible emission observations as part of their daily walk through and CAM requirements and maintains records of all documented observations. Cemex also maintains a record of complaints received by the Division and Boulder County with their response and corrective action. Source is in compliance with this condition.

27. P050 - Cement Rail Car Unloading and Handling Facility

Parameter	Permit Condition Number	Limitations	Emission Factors	Monitoring	
				Method	Interval
Process Rate	27.1	50,000 tons/yr of imported cement		Recordkeeping	Monthly
PM &PM ₁₀	27.2	PM - 0.6 tons/yr PM ₁₀ – 0.4 tons/yr	See Condition 27.2	Recordkeeping and Calculation	Monthly
Control Device and Operating Requirements	27.3	See Condition 27.3		Control Equipment Maintenance	Annual Certification
Opacity	27.4	Shall not exceed 20%, except as provided for below		Inspection Method 9 Baghouse Maintenance and Operation	Daily As Needed
		Certain Operating Conditions -Shall not exceed 30%, for a period or periods aggregating more than six (6) minutes in any 60 consecutive minutes			
MACT Standard - Opacity	27.5	Shall not exceed 10%		Method 9 Method 22 O&M Plan	Every five (5) years Monthly to Annually

Compliance Status: P050 – Cement Rail Car Unloading and Handling Facility

Parameter	Limitations	Reported Data
Process Rates	50,000 tons/yr	0 tons
PM	0.6 ton/yr	0.0 tons
PM ₁₀	0.4 ton/yr	0.0 tons

Cemex provided the reported data above for the rolling 12-month period June 2012. No railcar unloading has occurred since 2007. Source is in compliance with the above permit limits.

- 27.1 The amount of cement processed through the rail car unloading system shall not exceed the limitation listed in the table above (Construction Permit 05BO0703). The quantity of cement unloaded shall be monitored and recorded monthly. Monthly quantities of cement unloaded shall be used in a twelve month rolling total to monitor compliance with the annual limitation. Each month a new twelve month total shall be calculated using the previous twelve months' data.

Cemex is maintaining monthly and rolling 12-month totals of quantities of cement processed through the rail car unloading system. No railcar unloading has occurred since 2007. Source is in compliance with this condition.

- 27.2 PM and PM₁₀ emissions from the rail car unloading system shall not exceed the limitations listed in the table above (Construction Permit 05BO0703, as modified under the provisions of Section I, Condition 1.3 to increase the PM₁₀ emission limitation). Compliance with the PM and PM₁₀ emission limitations shall be monitored by calculating emissions monthly using the emission factors specified in the table below and the monthly quantity of cement unloaded. Monthly emissions shall be used in a rolling twelve month total to monitor compliance with the annual limitations. Each month a new twelve month total shall be calculated using the previous twelve months' data.

Pollutant	Activity	Emission Factor	Control Efficiency	Source
PM	Rail car to hopper	0.60	98 %	From Division's Preliminary Analysis for Construction Permit – AP-42, cement handling portion of concrete batching, section 11.12, corrected for site differences.
	hopper to pneumatic pump (screw conveyor)	0.45	99 %	
	Pneumatic trans to silo	0.27	98 %	
PM ₁₀	Rail car to hopper	0.40	98 %	
	hopper to pneumatic pump (screw conveyor)	0.29	99 %	
	Pneumatic trans to silo	0.17	98 %	

Note that the control efficiencies listed in the above table may be applied to the emission calculations provided the requirements in Condition 27.3 have been met.

Cemex is calculating monthly and rolling 12-month totals of particulate emissions from the rail car unloading system using the above emission factors and control efficiencies. No railcar unloading has occurred since 2007. Source is in compliance with this condition.

- 27.3 The rail car unloading system shall is subject to the following control device and operational requirements:

- 27.3.1 This source shall be equipped with a pulse jet fabric filter baghouse capable of limiting particulate matter emissions to 0.02 grains per dry standard cubic feet. (Construction Permit 05BO0703)

In the absence of credible evidence to the contrary, compliance with the grain loding limitation is presumed provided the baghouse is operated and maintain in accordance with the following requirements.

Bagfilter Operation and Maintenance

Routine maintenance of and operational procedures performed on the baghouses shall be conducted in accordance with manufacturer's specifications and good engineering practices. These procedures shall be in written format. Any maintenance work performed shall be documented and maintained to be made available to the Division upon request.

- 27.3.2 Prior to initiating the discharge from the railcar into the unloading hopper, the seals between the railcar and hopper shall be firmly engaged and the exhaust fan started to maintain a negative pressure of at least 3 inch water gauge in the hopper. After the railcar is emptied and the hopper is also emptied, the negative pressure shall be maintained for at least an additional five minutes to ensure all particulate matter is vented. A gauge showing the negative pressure shall be readily visible of the operations. (Construction Permit 05BO0703)

No railcar unloading has occurred since 2007 and therefore was not observed during the inspection. However, the system is equipped with a pulse jet fabric filter baghouse, as required, and has a pressure gauge to monitor negative pressure during railcar unloading. The O&M Plan dated 8/1/06 states that Cemex will perform monthly Method 22 emission observations, as well as operational maintenance (i.e. check for leaks, evaluate equipment operation, check differential pressure, and check for dust in control equipment exhaust) at least every 240 hours. Source is in compliance with this condition.

- 27.4 These sources are subject to the opacity limits set forth in Condition 21 of this permit.

Refer to Condition 21.

- 27.5 These sources are subject to the MACT requirements, as set forth in Conditions 23 of this permit.

Refer to Condition 23.

SECTION III – Permit Shield

Cemex submitted an application requesting renewal of Operating Permit 95OPBO082 at least twelve months before the permit expiration date of February 1, 2005 and received the Application Shield provision allowed by Colorado regulation Number 3, Part C, Section II.B, to operate without a valid Operating Permit while the Division or the Air Quality Control Commission makes a final determination on the permit application. Cemex operated under this provision from the date of expiration, February 1, 2005, until the date of the renewal permit issuance March 1, 2008. **Cemex is in compliance with all the conditions outlined in the Permit Shield.**

SECTION IV – General Permit Conditions

Cemex is in compliance with the General Permit Conditions with the exception of the item listed below.

Reporting Requirements:

Semi-Annual Monitoring Report: Due on August 1 and February 1 (beginning 3/1/08)

Annual Compliance Certification: Due on February 1 (beginning 3/1/08)

Semi-Annual Monitoring Reports

Time Period	Due Date	Date Received
1/1/10 – 6/30/10	8/1/10	8/13/10*
7/1/10 – 12/31/10	2/1/11	1/26/11
1/1/11 – 6/30/11	8/1/11	8/1/11
7/1/11 – 12/31/11	2/1/12	1/31/12
1/1/12 – 6/30/12	8/1/12	7/30/12

Semi-annual Monitoring and Deviation Reports (SAR) have been submitted as required since the last inspection.

*The SAR received 8/13/10 was granted an extension by the Division to be received by 8/15/10 due to information requests by the Division for Regional Haze/BART. Deviations reported in the SARs received have been evaluated and any findings are summarized following the permit condition that specifically addresses the emissions source/point. Source is in compliance with this requirement.

Annual Compliance Certifications

Time Period	Due Date	Date Received
1/1/09 – 12/31/09	2/1/10	2/1/10
1/1/10 – 12/31/10	2/1/11	1/26/11
1/1/11 – 12/31/11	2/1/12	1/31/12

The annual compliance certifications (ACC) have been submitted as indicated in the table above. Each of the ACC's received beyond the due date since 2007 prior to the permit renewal on 3/1/08 have been addressed in the NOV dated 6/17/08. *The ACC (with SAR) received 2/2/09 was received in the mail room on 1/30/09 but not stamped by the Division until 2/2/09. Source is in compliance with this requirement.

- 22.e The permittee shall file an Air Pollutant Emissions Notice ("APEN") prior to constructing, modifying, or altering any facility, process, activity which constitutes a stationary source from which air pollutants are or are to be emitted, unless such source is exempt from the APEN filing requirements of Regulation No. 3, Part A, § II.D. A revised APEN shall be filed annually whenever a significant change in emissions, as defined in Regulation No. 3, Part A, § II.C.2. occurs; whenever there is a change in owner or operator of any facility, process, or activity; whenever new control equipment is installed; whenever a different type of control equipment replaces an existing type of control equipment; whenever a permit limitation must be modified; or before the APEN expires. An APEN is valid for a period of five years. The five-year period recommences when a revised APEN is received by the Air Pollution Control Division. Revised APENs shall be submitted no later than 30 days before the five-year term expires. Permittees submitting revised APENs to inform the Division of a change in actual emission rates must do so by April 30 of the following year. Where a permit revision is required, the revised APEN must be filed along with a request for permit revision. APENs for changes in control equipment must be submitted before the change occurs. Annual fees are based on the most recent APEN on file with the Division.

Air Pollutant Emission Notices (APENs) for each point have been submitted and are still valid. Cemex is currently working with the Division to update the inventory for the source. Several points that either no longer exist or are misrepresented will be corrected in a future administrative modification to Operating Permit 95OPBO082.

This page intentionally left blank